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VISION-BASED ANALYSIS OF MEDICAL IMAGES FOR ASSESSING BONE LOSS IN DENTAL IMPLANT PATIENTS

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Abstract

Long-term dental implant success can be compromised by peri-implantitis, a condition characterized by progressive bone loss. Traditional radiographic assessments are often subjective and time-consuming. This study developed an automated artificial intelligence (AI) pipeline for objective quantification of peri-implant bone loss from periapical radiographs. The proposed method integrates implant detection models, multiple segmentation strategies—including a novel hybrid technique combining classical computer vision and advanced AI—and a dual-metric quantification system (vertical and area-based measurements). An active learning approach was also implemented to enhance model adaptability to new clinical data.

The results demonstrated robust implant detection and accurate segmentation, particularly highlighting the clinical advantages of the hybrid method in precisely isolating relevant anatomical regions. The dual quantification provided complementary information, improving defect characterization, and active learning enhanced model generalization to diverse clinical scenarios.

In conclusion, this integrated AI-based system offers a viable, automated, and objective tool for assessing peri-implant bone loss, laying the groundwork for improved early detection, personalized patient monitoring, and optimized clinical management.

Resum

L'èxit a llarg termini dels implants dentals pot veure's compromès per la periimplantitis, una condició caracteritzada per una pèrdua òssia progressiva. Les avaluacions radiogràfiques tradicionals sovint són subjectives i requereixen molt de temps. Aquest estudi desenvolupa un sistema automatitzat basat en intel · ligència artificial (IA) per a la quantificació objectiva de la pèrdua òssia periimplantària a partir de radiografies periapicals. El mètode proposat integra models de detecció d'implants, múltiples estratègies de segmentació —inclosa una nova tècnica híbrida que combina visió per computador clàssica amb IA avançada— i un sistema de quantificació bimètrica (mesuraments verticals i basats en àrea). A més, s'implementa un enfocament d'*active learning* per millorar l'adaptabilitat del model a noves dades clíniques.

Els resultats mostren una detecció d'implants robusta i una segmentació precisa, destacant especialment els avantatges clínics del mètode híbrid a l'hora d'aïllar amb precisió les regions anatòmiques rellevants. La quantificació dual proporciona informació complementària que millora la caracterització dels defectes, i l'*active learning* potencia la capacitat de generalització del model en escenaris clínics diversos.

En conclusió, aquest sistema integrat basat en IA ofereix una eina automatitzada, objectiva i viable per avaluar la pèrdua òssia periimplantària, establint les bases per a una millor detecció precoç, un seguiment personalitzat del pacient i una gestió clínica optimitzada.

Resumen

El éxito a largo plazo de los implantes dentales puede verse comprometido por la periimplantitis, una condición caracterizada por una pérdida ósea progresiva. Las evaluaciones radiográficas tradicionales suelen ser subjetivas y requerir mucho tiempo. Este estudio desarrolla un sistema automatizado basado en inteligencia artificial (IA) para la cuantificación objetiva de la pérdida ósea periimplantaria a partir de radiografías periapicales. El método propuesto integra modelos de detección de implantes, múltiples estrategias de segmentación —incluida una novedosa técnica híbrida que combina visión por computador clásica con IA avanzada— y un sistema de cuantificación bimétrica (mediciones verticales y basadas en área). Además, se implementa un enfoque de *active learning* para mejorar la adaptabilidad del modelo a nuevos datos clínicos.

Los resultados muestran una detección de implantes robusta y una segmentación precisa, destacando especialmente las ventajas clínicas del método híbrido al aislar con precisión las regiones anatómicas relevantes. La cuantificación dual proporciona información complementaria que mejora la caracterización de los defectos, y el *active learning* potencia la capacidad de generalización del modelo en escenarios clínicos diversos.

En conclusión, este sistema integrado basado en IA ofrece una herramienta automatizada, objetiva y viable para evaluar la pérdida ósea periimplantaria, sentando las bases para una mejor detección precoz, un seguimiento personalizado del paciente y una gestión clínica optimizada.

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1. Introduction

Dental implants have become a crucial solution for patients suffering from severe tooth loss, significantly improving their quality of life. However, post-implantation monitoring is essential to ensure the long-term success of the procedure. One of the most common complications that arise over time is bone loss, which can jeopardize the stability of the implant and lead to further complications such as peri-implantitis. This inflammatory condition affects the soft and hard tissues surrounding the implant, potentially leading to implant failure if not properly managed. Accurate assessment and early detection of both bone loss and peri-implantitis are therefore critical for timely intervention and improved patient outcomes.

Recent advancements in computer vision and artificial intelligence (AI) have opened new possibilities for automating the analysis of medical images, enhancing diagnostic precision, and supporting clinical decision-making. Traditionally, implantologists manually evaluate X-ray images to determine bone degradation over time. However, this process is subjective, time-consuming, and prone to variability among experts. Additionally, in many cases, a face-to-face patient assessment is required, further complicating timely diagnosis. The integration of AI-driven vision-based analysis offers a promising approach to enhance accuracy and efficiency while also incorporating the patient's clinical history to evaluate risk factors that may contribute to bone loss.

This research aims to develop an automated model to estimate the risk of peri-implantitis by combining vision-based bone loss detection with clinical history analysis. The study's primary objectives include:

- 1. **Data Collection and Preprocessing** – Gathering and preparing radiographic images and clinical data from patients, ensuring compliance with ethical and confidentiality guidelines.
- 2. **Algorithm Development** – Implementing a vision-based AI model capable of detecting bone and gum levels in dental X-rays.
- 3. **Machine Learning Model** – Training a regression model to predict bone loss progression at different time intervals (2, 5, and 10 years), considering patient health factors.
- 4. **Validation and Performance Evaluation** – Comparing the AI model's predictions with expert assessments to ensure reliability and clinical applicability.
- 5. **Data Privacy and Ethical Considerations** – Addressing privacy concerns, ethical guidelines, and responsible AI deployment in clinical settings.

This study explores the use of X-ray image analysis to detect periodontal bone loss, which is closely linked to peri-implantitis. It comprehensively examines various risk factors, including oral hygiene practices, pre-existing medical conditions such as diabetes, and lifestyle habits like smoking, which can significantly impact the success of dental implants. By harnessing the power of AI-driven image analysis and clinical data interpretation, this research aims to develop a standardised, reproducible, and objective method for assessing bone loss. This advancement will ultimately enhance clinical decision-making and provide improved patient care.

This project contributes to the rapidly growing field of AI applications in dentistry, demonstrating the potential of automated diagnostic tools in modern healthcare. If successful, the proposed model could aid dentists and implantologists in monitoring implant stability, predicting peri-implantitis risk, and optimising treatment strategies for long-term success.

2. Theoretical framework

This chapter presents the conceptual basis and the state-of-the-art concerning peri-implant bone loss, measurement methods, and the application of computer vision and artificial intelligence in the field of dentistry.

2.1. Medical background and clinical relevance

Peri-implantitis is a chronic inflammatory condition that affects dental implants, characterized by an inflammation of the soft tissues that produces a progressive loss of the supporting bone where an implant is placed, posing significant threat to implant longevity [1]. Soft tissues in this context refer to gum and oral mucosa surrounding the dental implant. These are crucial to protect and maintain its stability. When they become inflamed due to a series of potential risk factors, which will be explained shortly, they can lead to the destruction of the supporting bone, which compromises implant integration and can result in its loss if not treated properly.

The prevalence of this disease varies, impacting approximately 10 % to 30 % of patients with dental implants [2]. Clinically, it results in severe consequences including as mentioned bone loss and increased risk of implant failure, affecting both patient quality of life and healthcare resources.

Risk factors for peri-implantitis are multifactorial and involve patient-specific factors, implant characteristics and prosthetic aspects. It has been shown that factors such as smoking significantly increase the risk by 2 to 3 times. Other risk factors such as systemic conditions like diabetes impair bone healing and exacerbate inflammatory responses and a poor oral hygiene facilitates biofilm triggering an inflammatory response. Additional systemic conditions such as osteoporosis, as well as genetic predispositions and life habits (e.g., bruxism and stress), further compound the risk [2, 3, 4, 5].

Clinically, peri-implantitis is characterized by increased probing depths and bleeding or suppuration upon probing. Radiographic diagnosis is essential for assessing marginal bone loss, typically using periapical and panoramic radiographs. In this project, only periapical radiographs will be utilized due to their enhanced precision for monitoring implant-bone interfaces and the detailed visualization necessary for accurate longitudinal assessment. Despite their utility, radiographic methods have limitations related to subjective interpretation and inter-observer variability[5]. Periapical radiographs offer useful visualization of hard tissues, including the implant and surrounding bone, allowing the detection of changes in bone levels that serve as indirect indicators of soft tissue health. However, they are limited in capturing soft tissue details directly, and their quality is often lower compared to other imaging modalities, there can be several artifacts on the produced images as well as noise and dispertion caused by the physical process of scattering of the x-rays used to form that image, known as *Compton Scattering*, making the actual contrast of the ima-

ge decrease [6]. Overall, despite their high utility, radiographic methods have limitations related to subjective interpretation and inter-observer variability[7].

Early diagnosis of peri-implantitis can significantly improve clinical outcomes by enabling timely interventions, which can prevent advanced bone loss, reduce the complexity of required treatments, and decrease the associated social and economic impact on patients and healthcare systems.

2.2. Computer vision and deep learning in dentistry

Recent advancements in artificial intelligence (AI) have led to transformative changes in the field of healthcare, with computer vision and deep learning emerging as key technologies for medical image analysis. Computer vision involves machines analyzing and interpreting visual data from images or videos without human intervention, while deep learning enhances its capabilities using neural networks, particularly Convolutional Neural Networks (CNNs). These models excel in analyzing medical images for tasks like image classification and segmentation of areas in them. These techniques enable the automated extraction of clinically relevant features from complex visual data, which can improve diagnostic efficiency and consistency.

At the core of these advances lie Convolutional Neural Networks (CNNs), a class of deep learning models specifically designed for processing visual inputs. CNNs have demonstrated remarkable performance in a wide range of medical imaging tasks, including classification, detection, and segmentation. Their ability to learn hierarchical features directly from pixel data makes them particularly well-suited for interpreting radiographic and tomographic scans, especially in structured environments with ample annotated data.

However, despite their success, CNNs also present limitations. Their performance can degrade when faced with limited training data, high inter-patient variability, or subtle anatomical features that require fine-grained interpretation. Moreover, their black-box nature and reliance on large labeled datasets pose challenges in clinical adoption, where transparency and generalizability are essential.

For these reasons, it is crucial to explore a broader range of computer vision techniques—both classical and modern—that can complement or even surpass CNN-based approaches in specific contexts. These alternatives offer diverse strengths, such as interpretability, robustness to noise, or reduced dependency on labeled data, and are particularly valuable in domains like dentistry, where image quality and annotation scarcity are common concerns.

One prominent example is the detection of dental caries, where CNNs can identify early signs of decay in radiographic images with a level of sensitivity that supports timely diagnosis and treatment. In addition to static tasks, deep models have also been applied to predictive problems. For instance, they can analyze radiographic or facial images to detect

features associated with temporomandibular joint dysfunction and other maxillofacial pathologies, enabling earlier clinical intervention.

Traditional methods such as edge detection help identify structural boundaries—like tooth contours or bone edges—enhancing visibility and supporting downstream analysis. Feature extraction techniques, which focus on measurable properties like shape or texture, remain useful for identifying anomalies in tooth morphology or bone structure. Classical segmentation approaches, including thresholding or region-growing, can be particularly effective when data is limited or consistent, while morphological operations like erosion and dilation refine image features by reducing noise or sharpening contours.

Moreover, multi-scale analysis—processing images at different resolutions—allows for the simultaneous capture of fine details and broader anatomical context. This is especially beneficial in longitudinal dental studies, where subtle changes in bone or tooth structure over time must be reliably detected. Rather than viewing these approaches in isolation, combining classical techniques with deep learning often yields the most robust and accurate results. These hybrid strategies draw on the strengths of both paradigms: the adaptability and power of deep learning with the clarity and efficiency of traditional vision methods.

2.3. Detection and Segmentation Models in Radiographic Images

Radiographic image analysis often requires identifying and delineating anatomical structures or anomalies within medical images. Modern computer vision approaches address these tasks through object detection (finding objects and drawing bounding boxes) and segmentation (classifying each pixel, or each object's shape) at high accuracy.

We progressively build from core computer vision concepts into advanced model architectures that are state-of-the-art for object detection and segmentation, with an emphasis on models applicable to radiographic images (e.g. dental X-rays, chest radiographs). We cover foundational concepts – convolutional feature extraction, receptive fields, anchor boxes, loss functions – and then dive into four notable architectures: YOLOv8 (You Only Look Once v8, for detection and instance segmentation), Faster R-CNN (two-stage detector with a Region Proposal Network), U-Net with a ResNet-34 encoder (fully convolutional segmentation network), and Mask R-CNN (extension of Faster R-CNN for instance segmentation, here discussed with Detectron2 implementation). Throughout, we highlight each model's internal design, training pipeline insights, theoretical motivations, and evaluation metrics (mAP, IoU, Dice) used to quantify performance.

2.3.1. Foundational Concepts in Classical Computer Vision

Classical computer vision and mathematical morphology techniques, while distinct from deep learning, are vital for tasks like image preprocessing, robust segmentation, and feature enhancement in medical imaging. They offer interpretable, computationally efficient solutions, often complementing learning-based approaches, as in the pipeline detailed in Section 3.4.4. This section outlines the core theory of the classical methods employed.

1. Median Filtering : Radiographic images often contain impulse noise (e.g., "salt-and-pepper"). The median filter, a non-linear technique, effectively removes this noise while preserving edge sharpness better than linear filters. It replaces each pixel's value with the median of intensities in its local neighborhood (e.g., a 3×3 kernel), crucial for subsequent segmentation.

2. Image Segmentation Techniques Segmentation partitions an image into meaningful regions, typically corresponding to different structures.

2.1 Adaptive Thresholding : Unlike global thresholding, which struggles with non-uniform illumination (common in radiographs), adaptive thresholding calculates a local threshold $T(x, y)$ for each pixel:

$$T(x, y) \;=\; \frac{1}{|\Omega_{xy}|} \sum_{(u,v)\in\Omega_{xy}} I(u, v) \;-\; C \tag{1}$$

where Ω_{xy} is the local neighborhood around (x, y), and C is a constant. This local adaptation, used in Section 3.4.4, enables effective segmentation despite varying background illumination.

2.2 Multi-Otsu Thresholding : Otsu's method automatically finds optimal global thresholds by maximizing inter-class variance. The Multi-Otsu method (explained on Section 2.4.1) extends this to find multiple thresholds $(k_1, \ldots, k_{M-1})$ to segment an image into M classes, ideal for multimodal histograms like those in dental radiographs (e.g., air/soft tissue, bone, metal).

3. Mathematical Morphology : Mathematical morphology analyzes and processes geometric structures in images using a structuring element (SE), a small pattern that probes the image.

3.1 Fundamental Operators: Dilation and Erosion : For a binary image A and SE B:

- **Dilation** ($A \oplus B = \{z \mid (\hat{B})_z \cap A \neq \emptyset\}$) expands objects by adding pixels to boundaries.
- **Erosion** ($A \ominus B = \{z \mid (B)_z \subseteq A\}$) shrinks objects by removing pixels from boundaries.

In grayscale images, dilation uses local maxima and erosion local minima.

3.2 Compound Operators: Opening and Closing : These combine dilation and erosion for complex shape manipulation:

- **Opening** ($A \circ B = (A \ominus B) \oplus B$) smooths contours, breaks narrow connections, and removes small objects or protrusions. The area-based filtering in Section 3.4.4 (step 5) is analogous, eliminating small pixel "islands".
- **Closing** ($A \bullet B = (A \oplus B) \ominus B$) smooths contours, fills small holes, and fuses narrow gaps. This relates to the optional closing before hole filling in the pipeline.

3.3 Morphological Reconstruction and Hole Filling : Morphological reconstruction iteratively modifies a "marker" image based on a "mask" image using conditional operations. Hole filling is a key application: for a binary image I, a marker F (zero except at borders matching I^c) is conditionally dilated within I^c; the complement of the result fills holes in I. The pipeline's hole filling (Section 3.4.4, step 6) uses a similar principle of label propagation to ensure topological consistency.

3.4 Morphological Gradients for Edge Detection : The standard morphological gradient, $G(A) = (A \oplus B) - (A \ominus B)$, highlights intensity transitions by subtracting the eroded image from the dilated one, enhancing edges. The "conditioned morphological gradient" (Section 3.4.4, step 8) is a specialized variant. It uses semantic information from Multi-Otsu segmentation and the `adaptive_mask`: an edge is marked only at transitions between different semantic classes within a "valid" (high local contrast, non-implant) region. This targeted approach significantly reduces spurious edges, focusing on clinically relevant boundaries like the peri-implant bone line.

The strategic combination of these classical computer vision and mathematical morphology techniques provides a robust, interpretable framework for complex image analysis, effectively preparing data and extracting key features for direct assessment or as input for advanced models.

2.3.2. Foundational Concepts in Vision Models

Convolutional Feature Extraction and Receptive Field Deep vision models are centered around Convolutional Neural Networks (CNNs), which learn hierarchical features from images [8]. CNNs use convolution filters to detect patterns, from low-level

(edges, textures) to complex structures (e.g., organs in medical images), preserving spatial hierarchy through convolutional and pooling layers to build global representations [8]. The receptive field—the input region influencing a feature—expands with network depth, enabling broader contextual understanding crucial for tasks like tumor detection in radiographs. While deep networks can suffer from vanishing gradients, skip connections (e.g., in ResNet) mitigate this, facilitating the training of very deep CNNs. These CNN backbones transform pixel data into rich feature representations for detection/segmentation models.

Multi-scale Feature Representations Radiographic objects vary significantly in size, necessitating multi-scale feature maps for effective detection. Modern models often use feature pyramids, processing feature maps at different resolutions. For instance, YOLOv8 employs a backbone with pyramidal features and a neck (e.g., PANet, FPN) to merge these, allowing detection heads to predict at multiple scales for both small and large objects. Similarly, U-Net's encoder-decoder structure inherently generates and combines multi-scale features for segmentation [9]. This multi-scale learning is crucial for high accuracy across diverse object sizes.

Anchor Boxes and Region Proposals Region-based detectors often use anchor boxes—predefined bounding boxes serving as priors for object location and scale on a feature map [10]. Faster R-CNN's Region Proposal Network (RPN) pioneered this, predicting objectness scores and regression offsets relative to multiple anchors at each feature map location [11, 10]. Anchors are labeled positive/negative during training based on IoU with ground-truth objects, providing samples for classification and regression. This learnable, multi-scale sliding-window approach generates candidate regions for further evaluation [10]. YOLOv8, however, uses an anchor-free detection head, directly predicting object centers and sizes [12, 13]. This avoids anchor tuning, reduces duplicate predictions, and simplifies post-processing (e.g., NMS), enhancing efficiency and robustness across datasets. Both anchor-based and anchor-free methods address the fundamental challenge of object localization [13].

Loss Functions for Detection and Segmentation Training vision models involves minimizing composite loss functions. For detection, classification loss (often cross-entropy, e.g., in Faster R-CNN's RPN [11]) ensures correct class prediction, while localization loss (e.g., Smooth L1 loss [10] or IoU-based losses) refines bounding box accuracy. For segmentation, semantic segmentation typically uses per-pixel cross-entropy. Instance segmentation (e.g., Mask R-CNN) often employs per-pixel binary cross-entropy for each class mask, combined in a multi-task loss (classification, box regression, mask loss) for joint optimization [14]. In medical image segmentation, Intersection over Union (IoU) and the

Dice coefficient are common overlap-based metrics for loss or evaluation:

$$\text{IoU} = \frac{|\text{Prediction} \cap \text{GroundTruth}|}{|\text{Prediction} \cup \text{GroundTruth}|} \tag{2}$$

$$\text{Dice} = \frac{2|\text{Prediction} \cap \text{GroundTruth}|}{|\text{Prediction}| + |\text{GroundTruth}|} \tag{3}$$

These metrics quantify mask overlap, penalizing false positives and negatives. The Dice coefficient, equivalent to the F1-score, is widely used, especially with imbalanced data (e.g., small tumors) [15].

2.4. Multi-Otsu Thresholding

Segmentation of radiographic images is a fundamental step for the automatic detection and quantification of relevant anatomical structures, such as the peri-implant bone line. A classical and effective technique for this purpose is thresholding based on Otsu's method, extended to multiple thresholds, commonly known as *Multi-Otsu* thresholding. This method allows the division of an image into several classes or regions, facilitating the distinction between different tissues or areas of interest in medical images.

2.4.1. Theory of Otsu's Method and Multi-Otsu Extension

The original Otsu method, introduced by Otsu in 1979, is an automatic thresholding algorithm that selects the optimal threshold to binarize an image by maximizing the between-class variance, separating the foreground from the background. Given a grayscale image with intensity levels L (typically 256), and the normalized histogram p_i representing the probability of intensity level i, the goal is to find the threshold t that maximizes the between-class variance defined as:

$$\sigma_B^2(t) = \omega_0(t)\,\omega_1(t)\,[\mu_0(t) - \mu_1(t)]^2 \tag{4}$$

where

$$\omega_0(t) = \sum_{i=0}^{t} p_i, \quad \omega_1(t) = \sum_{i=t+1}^{L-1} p_i \tag{5}$$

are the cumulative class probabilities for the two classes, and

$$\mu_0(t) = \frac{1}{\omega_0(t)} \sum_{i=0}^{t} i\,p_i, \quad \mu_1(t) = \frac{1}{\omega_1(t)} \sum_{i=t+1}^{L-1} i\,p_i \tag{6}$$

are the class means. The optimal threshold t^* is thus given by

$$t^* = \arg\max_t \sigma_B^2(t) \tag{7}$$

The *Multi-Otsu* method generalizes this concept to segment an image into k classes by finding a set of thresholds $\mathbf{T} = \{t_1, t_2, \ldots, t_{k-1}\}$ that maximizes the sum of the between-class variances:

$$\sigma_B^2(\mathbf{T}) = \sum_{j=0}^{k-1} \omega_j(\mathbf{T}) \left[\mu_j(\mathbf{T}) - \mu_T\right]^2 \tag{8}$$

where

$$\omega_j(\mathbf{T}) = \sum_{i=t_j+1}^{t_{j+1}} p_i, \quad \mu_j(\mathbf{T}) = \frac{1}{\omega_j(\mathbf{T})} \sum_{i=t_j+1}^{t_{j+1}} i\, p_i \tag{9}$$

with $t_0 = -1$, $t_k = L - 1$, and $\mu_T = \sum_{i=0}^{L-1} i\, p_i$ being the global mean intensity of the image.

The algorithm searches over all possible combinations of thresholds to maximize $\sigma_B^2(\mathbf{T})$, resulting in an optimal multi-level segmentation based on the histogram statistics.

2.4.2. Application in Medical Imaging

In dental radiographic imaging, the Multi-Otsu thresholding method is particularly useful for distinguishing regions with varying bone densities and soft tissues. By segmenting the image into multiple intensity classes, this technique enables robust and automated delineation of the bone line relative to areas of lower density, which correspond to bone loss or pathological changes.

Due to its non-parametric and histogram-based nature, Multi-Otsu is robust to variations in illumination and noise, common in clinical radiographs. This makes it an effective preprocessing step in pipelines for automated assessment of peri-implant bone loss, facilitating subsequent feature extraction and quantitative analysis.

Multi-Otsu thresholding extends the classic Otsu method to multiple classes, optimizing inter-class variance to segment images into several meaningful regions. Its theoretical foundation in histogram analysis and variance maximization makes it a computationally efficient and robust tool for medical image segmentation, particularly in the evaluation of peri-implant bone levels from radiographic images. This method requires no prior training data and serves as a valuable component in computer-aided diagnostic systems.

2.5. Comparison of State-of-the-Art Techniques

Several deep-learning models have been proposed for automated bone-loss detection, each with different levels of performance and clinical reliability. To facilitate the analysis of the various contributions and their limitations, this section presents each of the most

significant contributions separately. For each one, the core technological innovation and its technical details are explained, followed by a discussion highlighting its limitations.

1. Periodontal bone-loss detection with Faster R-CNN

Thanathornwong et al., 2020 [16] published one of the first studies of the decade to employ *Faster R-CNN* for the automatic detection of periodontal bone loss on digital panoramic radiographs. Using a ResNet-101 backbone pre-trained on natural images, the model generates region proposals and classifies each one to localise teeth with compromised bone support. The main features of the work are summarised below:

- **Image type and data:** 100 retrospectively collected orthopantomographs, carefully annotated by experts to mark the teeth with bone loss. Because of the limited data size, the authors applied geometric and photometric augmentation to increase variability during training.
- **Performance:** the detector reached an *Average Precision* (AP) of **0.81**; sensitivity **0.84**; specificity **0.88**; recall **0.80**; and F1-score **0.81**. These figures show a reasonable balance between false positives and false negatives, which is notable given the data scarcity.
- **Clinical strengths:** the methodology demonstrated the feasibility of using two-stage CNNs on panoramic images to perform rapid patient screening, reducing diagnostic time and standardising periodontitis detection.

Limitations: Despite its promising results, the study has several limitations. The relatively small sample size, composed of only 100 annotated panoramic images, raises concerns about the model's ability to generalize to broader populations or to radiographs acquired using different imaging settings. Moreover, the detection approach adopted is inherently dichotomous—it identifies the presence of bone loss but does not quantify its severity or localize its exact extent along the tooth root, which are clinically relevant aspects. Another limitation is the computational complexity of the Faster R-CNN model; while it offers high accuracy, its heavier processing requirements may hinder real-time deployment in clinical environments. Additionally, the reported slight imbalance between sensitivity and specificity suggests a tendency to over-detect certain complex anatomical regions, which could lead to an increased rate of false positives, particularly in cases with overlapping structures or radiographic noise.

Relevance and contribution to the state of the art

Despite its restrictions, the work of Thanathornwong et al. is a key *proof of concept*: it showed that a generic detection architecture, transferred to the dental domain with minimal tuning, could achieve competitive results (AP = 0.81) and motivated later research

with larger datasets and more efficient networks—e.g. YOLOv7/v8—to improve inference speed, bone-loss quantification and generalisation.

2. Hybrid segmentation and geometric calculation to quantify bone loss

Chang et al., 2020 [17] proposed a hybrid system that goes beyond simple periodontitis detection: it segments key structures on panoramic radiographs and computes the *Radiographic Bone Loss* (RBL) percentage to stage the disease. Their pipeline combines deep neural networks with classical geometric algorithms, providing quantitative outputs and explanatory images useful in clinical practice.

- **Architecture:** a CNN segmentation network with a *Feature Pyramid Network* (similar to Mask R-CNN) that produces binary masks of (i) the alveolar bone crest, (ii) the cemento–enamel junction (CEJ) and (iii) teeth/implants. From these masks, an algorithm calculates each tooth's long axis and intersects that axis with both the bone crest and the CEJ to derive the RBL; this value is then mapped to Stage I, II or III periodontitis.
- **Image type and data:** 340 orthopantomographs from a single South-Korean centre, meticulously annotated by specialists. The dataset includes both natural teeth and implants, increasing anatomical diversity.
- **Performance:**
  - *Segmentation:* Jaccard 0.92 and Dice 0.88 for the bone crest; Jaccard 0.87 and Dice 0.84 for the CEJ; similar results for teeth/implants.
  - *Staging classification:* mean absolute deviation (MAD) of 0.25 stages versus three radiologists (no statistically significant differences). High correlations confirm the method's robustness.
- **Clinical strengths:**
  1. Provides objective measures (RBL %) and automatic staging, not just binary detection.
  2. Generates intuitive visual outputs (superimposed masks), enhancing interpretability and dentist acceptance.

Limitations: The approach, while innovative, comes with several limitations. First, the pipeline is highly dependent on the precision of its segmentation outputs—errors in identifying structures such as the bone crest, CEJ, or teeth can propagate through the system, compromising the accuracy of downstream calculations like RBL and staging. Additionally, the model was trained and validated on data from a single clinical centre in South

Korea, which limits its generalisability to other populations, imaging protocols, or radiographic devices. The need for detailed, expert annotations further complicates its broader adoption. Lastly, the use of a complex architecture combining Mask R-CNN with a Feature Pyramid Network, along with the subsequent geometric analysis for staging, introduces a notable computational cost, potentially reducing the feasibility of real-time integration into clinical settings compared to faster one-stage detectors.

Relevance and contribution

This study marks a turning point by showing that AI can quantify bone loss accurately (Dice = 0.88) and reproduce clinical staging with minimal deviation. Its strategy—combining multi-label segmentation with geometric measurements—underpins recent developments that integrate keypoints and distance regression, and it underscores the importance of interpretability in advanced dental applications.

3. Peri-implant bone-loss detection with Faster R-CNN

Unlike studies focused on natural teeth,*Liu et al.*, 2022 [18] investigated marginal bone resorption around dental implants using *Faster R-CNN* on intraoral periapical radiographs. This pilot study, notable for its sizeable dataset, assesses AI's ability to detect peri-implantitis early and compares its results with human observers.

- **Architecture:** *Faster R-CNN* with a pre-trained CNN backbone (not specified in the article). The detector generates bounding boxes for bone defects on the mesial and distal surfaces of each implant.
- **Image type and data:** 1 670 periapical radiographs acquired using the paralleling technique (1 370 training, 150 validation, 150 test). To compensate for the low prevalence of peri-implantitis, the authors balanced the dataset and included triangular manual annotations around affected implants.
- **Performance:**
  - *Implant level:* sensitivity **67 %**, specificity **87 %**, PPV (Positive Predictive Value) **81 %**.
  - *Site level (mesial/distal):* sensitivity **75 %**, specificity **83 %**, PPV 87 %.
  - *Agreement:* 0.55–0.57 versus an expert periodontist, comparable to a resident.
  - *Average precision:* overall mAP 0.73; AP for implants 0.99 and for lesions 0.47, illustrating the difficulty of precisely delineating bone defects.
- **Clinical strengths:** the system matches the performance of a dentist with intermediate experience, works on high-resolution images and could serve as a screening tool for early peri-implantitis.

Limitations: Despite its strengths, the system presents notable limitations. Its scope is narrowly focused on dental implants, meaning the model cannot be directly applied to cases involving natural teeth. Moreover, the detection pipeline identifies the presence of bone defects without quantifying the extent of bone loss in millimetres or percentages, which restricts clinical interpretability and decision-making. The dataset was artificially balanced to address the low prevalence of peri-implantitis, which may introduce bias and could result in reduced specificity when applied to real-world populations with fewer positive cases. Additionally, although the results show moderate sensitivity, the detection of subtle defects remains a challenge, and the generalisability of the model has yet to be confirmed through multi-centre validations. Finally, while periapical radiographs are relatively small in size, the use of Faster R-CNN entails higher computational costs compared to one-stage detectors, which may limit its use in fast-paced clinical environments.

Relevance and outlook

Liu et al. confirm the applicability of *Faster R-CNN* for peri-implant monitoring, positioning AI performance at the level of an advanced trainee. The study broadens the scope of dental computer vision to implants and suggests future directions: quantify bone loss (e.g. in mm), integrate 3-D CBCT (Cone Beam Computed Tomography) volumes and improve sensitivity to very early lesions—all of which would consolidate CNN-based detectors in contemporary implantology.

4. Real-time bone-loss detection with YOLOv7

The arrival of the *You Only Look Once* (YOLO) family has enabled near-instant object detection in dentistry. *Lee et al.*, 2024 [19] used **YOLOv7** to identify periodontal and peri-implant bone loss on intraoral and panoramic radiographs, achieving one of the best speed–accuracy balances published to date.

- **Architecture:** YOLOv7 with an optimised multi-scale head and efficient convolutions. The model was trained to localise bone-loss regions around teeth and implants and classify them binarily (present / absent).
- **Image type and data:** several hundred to thousands of periapical and (presumably) panoramic radiographs, labelled as healthy implants or implants with peri-implantitis. A diverse hospital digital archive is assumed.
- **Performance:**
  - *Overall accuracy:* **94.74 %**.
  - *Implants with bone loss:* precision 96.18 %; *healthy implants:* precision 93.42 %.

- Sensitivity and specificity are estimated in the 93–96 % range; F1 = 0.95; AUC>0.95.

- **Clinical strengths:**

  1. *Inference speed:* tens of milliseconds per image on a GPU, enabling real-time integration into radiology software.
  2. *High precision* comparable to experts, facilitating automatic screening for periodontitis and peri-implantitis.
  3. *Lightweight architecture* in comparison with Faster R-CNN, requiring fewer computational resources for deployment.

Limitations: While the YOLOv7-based system delivers impressive performance and real-time capability, it has several important limitations. Its diagnostic scope is currently restricted to peri-implantitis, and its effectiveness in detecting general periodontitis or more nuanced defect patterns remains unproven. The model identifies regions of bone loss but does not quantify them in clinically meaningful terms such as millimetres or percentages—an enhancement that could potentially be achieved through regression models or keypoint-based extensions like YOLOv8-Pose. Furthermore, although accuracy figures exceed 94 %, these results need validation across multiple clinical centres and diverse image qualities to confirm generalisability. Another limitation is the model's inability to differentiate between types of bone defects, such as horizontal, vertical, or furcation loss, which are critical for treatment planning. Lastly, the exceptionally high reported reliability (AUC >0.95) could foster an overreliance on automated results; ethical considerations and regulatory oversight are essential to ensure that such systems complement rather than replace clinical expertise.

Relevance and perspectives

Lee et al. show that one-stage detectors can achieve clinical-grade precision without sacrificing speed, opening the door to radiological assistance systems that operate *live*. Future developments should add bone-loss quantification, defect typing and exhaustive external validations to confirm robustness in real-world settings.

Conclusion

In just half a decade, research on radiographic bone-loss assessment has evolved from exploratory two-stage detectors to lightning-fast one-stage networks and hybrid segmentation–geometry pipelines capable of producing clinically familiar metrics. Faster R-CNN-based studies broke ground by proving that generic object detectors can flag diseased sites, while subsequent Mask/Feature-Pyramid approaches showed that deep learning can

measure loss with sub-stage precision. Most recently, the YOLOv7 family has demonstrated that expert-level accuracy need not come at the expense of millisecond-level inference. Despite these advances, three strategic challenges remain. First, external validity: almost every model still relies on single-centre datasets or artificially balanced samples, so their robustness across scanners, exposure parameters and populations is untested. Second, quantitative and phenotypic richness: the field must move beyond binary labels to deliver millimetre-level loss, defect morphology (horizontal, vertical, furcation) and longitudinal change—outputs that truly guide treatment. Third, real-world deployment: strict latency budgets, cyber-security safeguards, medico-legal oversight and, above all, clinician trust demand as much attention as network architecture. Meeting these priorities will require collaborative, open-science initiatives. Large multi-institutional repositories, coupled with federated or self-supervised learning, can ease annotation burdens while boosting model robustness. Architecturally, lean one-stage detectors augmented with auxiliary segmentation-and-regression heads promise both speed and quantitative depth. Equally important are prospective clinical trials and transparent reporting that foster regulatory confidence. With these elements in place, AI-driven analysis of dental radiographs can progress from promising prototypes to everyday tools that enhance early diagnosis, personalise treatment planning and ultimately improve oral-health outcomes worldwide.

3. Methodology

3.1. General Overview of the Workflow Pipeline

The main objective of this work is to develop an automated, robust, and quantifiable system for detecting and analyzing peri-implant bone loss from periapical dental radiographs. This system is subsequently compared with other techniques based on deep convolutional neural networks (DCNNs). For both approaches, the goal is to extract features that can be integrated into a regression model, which combines these radiographic indicators with patient health data. Since peri-implantitis manifests clinically as progressive bone recession around implants, the cornerstone of the pipeline is the precise detection of the "bone line", the measurement of that bone loss, and the estimation of its evolution.

Hybrid Approach: Classical Vision + Deep Learning Models

To achieve the project's goal, a hybrid strategy has been adopted, combining traditional computer vision techniques—such as thresholding, filtering, and morphological operations—with modern deep learning models for detection and segmentation, like YOLOv8, Faster R-CNN, Mask R-CNN, and SAM. This integration is motivated by three central factors. First, the limited availability of manually annotated data makes it difficult to rely solely on supervised learning approaches. Second, the inherent variability in radiographic images, in terms of lighting conditions, quality, and the presence of metallic artifacts, often challenges the consistency of neural networks; classical methods, applied with domain-specific heuristics, can effectively extract local structure that complements these models. Finally, the combination of detection models to localize implants with handcrafted techniques for contour extraction yields a pipeline that is both efficient and adaptable to the specific characteristics of dental X-rays.

Clinical and Technical Considerations

The methodological framework of this project is shaped by both clinical goals and technical constraints. A primary clinical constraint is the lack of temporal data (different images of the same patient over time). To overcome this, the methodology is built upon a medically-justified assumption: at the time of placement, the implant is positioned perfectly at the crest of the bone. Therefore, any part of the implant that should be within the bone but is exposed is considered an indicator of bone loss.

Pipeline Description

The complete processing workflow is organized into two parallel pipelines designed to detect and quantify peri-implant bone loss from a single periapical radiograph. Although

both pipelines start from the same image and converge on a shared prediction model, they differ in methodology regarding segmentation and feature extraction.

1. Common Initial Steps

- **Input Image:** A periapical radiograph is introduced as the input to the system.
- **Implant Detection:** A state-of-the-art object detection model (e.g., Faster R-CNN or YOLOv8) is employed to automatically localize the dental implant(s). This step defines the Region of Interest (ROI), which will be the basis for subsequent segmentation and analysis in both pipelines.

2. Pipeline A: Hybrid Approach (SAM + Classical Computer Vision) This pipeline represents the primary contribution of this work, combining the precision of generative segmentation with the robustness of handcrafted image processing techniques.

- **Implant Segmentation (SAM):** The Segment Anything Model (SAM) is prompted using the detected ROI to produce an accurate binary mask corresponding solely to the implant.
- **Bone Line Detection (Classical):** Classical image processing techniques are applied around the implant mask to identify the bone line. This includes adaptive filtering, multi-level Otsu thresholding, morphological noise reduction, and directional gradients conditioned on implant geometry to detect only clinically relevant bone boundaries.
- **Bone Loss Quantification:** Bone loss indicators are computed by measuring the vertical or radial distance between the segmented bone line and the implant boundary, based on domain-specific geometric criteria.

3. Pipeline B: End-to-End Deep Segmentation Approach (Benchmark) This secondary pipeline acts as a baseline, relying solely on deep learning for feature extraction.

- **Multi-Class Segmentation (Mask R-CNN / U-Net):** A deep supervised segmentation model, trained on a set of annotated images, is used to simultaneously delineate three anatomical structures: implant, boneloss, and gum.
- **Bone Loss Quantification:** Indicators of bone loss are extracted directly from the segmented regions, focusing on the spatial relationship between bone and implant labels.

4. Final Common Step

- **Peri-Implantitis Risk Prediction:** The computed bone loss indicators—regardless of the pipeline—are fused with patient-specific clinical data (e.g., oral hygiene habits, systemic conditions, smoking status). A regression model is then trained to predict long-term bone loss or classify the overall risk of peri-implantitis progression.

3.2. Data Acquisition and Preparation

The radiographic images used in this project originate from two complementary sources: a public dataset and an initial sample of clinical data obtained in collaboration with the Hospital Odontològic Universitat de Barcelona (HOUB) with which the supervisor of this project have a established collaboration.

First, in order to train implant detection models from the early stages of the project, a public dataset available on the Roboflow platform ("implant-gag01", public domain) was used. This dataset includes 200 images of dental implants annotated in YOLOv8 format. An example YOLO annotation is included in Annex A. The images were automatically preprocessed by Roboflow through resizing to 640×640 pixels and EXIF-based auto-orientation. The dataset is already split into training, validation, and test partitions. These 200 images have been also used to train the deep convolutional neural network segmentation models developed in this project.

Subsequently, after a confidentiality agreement was formalized and authorized access to clinical data was granted, a first sample of real images from HOUB was incorporated. The initial sample includes 25 periapical radiographs from 9 patients '.jpg' format and have been used during the initial segmentation and analysis phases. Some of these 25 images have been manually annotated under expert supervision during the course of the project, and they will be used for validation purposes.

Only periapical radiographs were used in this study, as they offer higher spatial resolution in regions of interest, lower incidence of overlapping anatomical structures, and superior definition of the alveolar bone line compared to panoramic radiographs (OPG). Although OPGs provide a global view of the maxilla and mandible, they can present soft tissue overexposure, geometric distortion, and artifacts due to the inherent concave projection of the imaging system. Moreover, periapical radiographic acquisition technology yields a more favorable contrast between bone and metallic structures than between soft tissues such as the gingiva, which directly impacts the ability to detect certain regions automatically unless robust segmentation models are employed.

Descriptive Image Analysis

A global descriptive analysis was conducted to characterize the grayscale distribution, segmentation behavior, and the impact of preprocessing choices across the dataset of pe-

riapical radiographs. This exploration aims to highlight intrinsic challenges for automatic bone line extraction and support the design of robust segmentation pipelines.

Grayscale Distribution The aggregated intensity histogram of all images (Figure 1) reveals a clearly multimodal pattern, with two prominent peaks around intensities 100 and 200. The first could correspond to mid-tone anatomical structures (e.g., bone and soft tissue), while the second relate to high-brightness regions such as metal artifacts or background. Additionally, small spikes near intensity 0 and 255 are evident, likely due to image boundaries, inherent artifacts to x-ray imaging or annotation overlays. These patterns reflect significant heterogeneity in exposure and anatomical visibility across the dataset, emphasizing the need for contrast normalization.

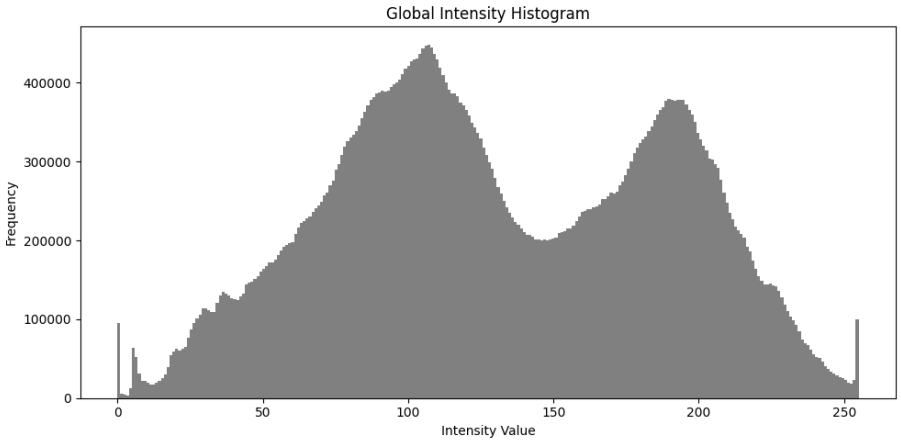
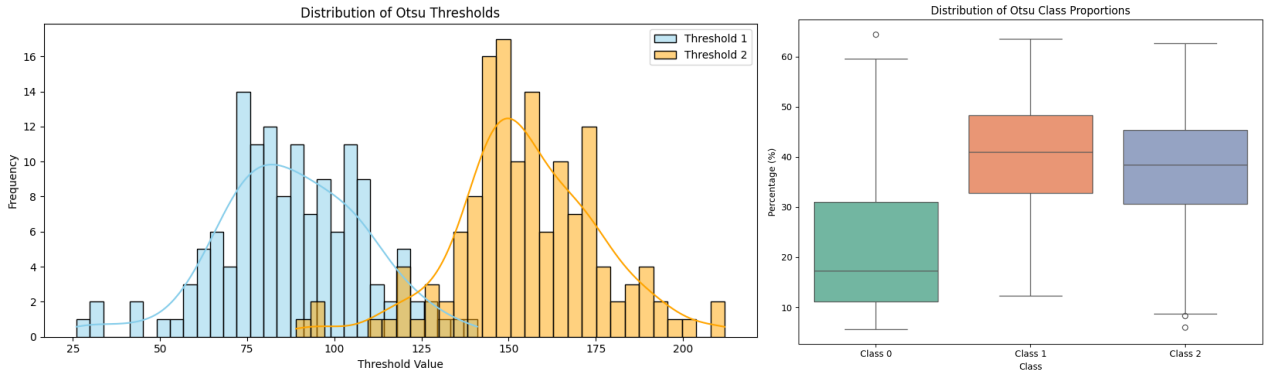


Figure 1: Global grayscale intensity histogram across all radiographs. The distribution is clearly multimodal, with major peaks around intensities 105 and 195, likely corresponding to anatomical structures and high-brightness artifacts (e.g., metal or overexposed regions). Secondary sharp spikes near 0 and 255 suggest the presence of clipping, borders or annotation artifacts. This heterogeneity reflects variability in acquisition and supports the need for intensity normalization.

To analyze structural variability, Multi-Otsu thresholding was applied to split each image into three intensity-based classes. As shown in Figure 2, the first threshold t_1 exhibits a wide distribution centered around 85 (range: 25–130), while the second threshold t_2 concentrates near 150 (range: 90–210). This separation indicates high variability in low-to-mid intensity structures (e.g., trabecular bone), with more consistency in the segmentation of brighter areas.

In terms of pixel composition, Class 1 (intermediate intensity) is the most prevalent across the dataset, with a median proportion of 41 % and values ranging from 12 % to 65 %. This class likely includes bone structures and other radiodense tissue. Class 2 (brightest regions) maintains relatively stable proportions (median: 38 %), while Class 0 (darkest) shows greater variability, spanning from 5 % to over 60 %, possibly reflecting differences in background contrast, noise, or underexposed anatomical features.

These results are summarized in Figure 2, which combines the threshold distribution and pixel class proportions derived from Multi-Otsu segmentation.



(a) Distribution of Multi-Otsu thresholds across images.

(b) Boxplot of class-wise pixel proportions.

Figure 2: Segmentation behavior analysis using Multi-Otsu. (a) Threshold distributions t_1 and t_2 show variability in mid- and high-intensity structure separation. (b) Class 1 (mid-intensity, typically bone) exhibits the highest mean and variability, potentially reflecting anatomical or exposure differences between images.

The heterogeneity observed in threshold values and class distributions indicates that adaptive strategies may be required for robust segmentation. The wide span of t_1 suggests that global thresholds may not be reliable for distinguishing darker anatomical features (e.g., bone), especially in cases of advanced bone loss or poor exposure. Conversely, the consistency in t_2 highlights the reliability of isolating metal components (implants) and bright background zones.

These findings directly inform preprocessing decisions and threshold selection in the bone line extraction pipeline. Local methods such as adaptive thresholding or CLAHE (contrast limited adaptive histogram equalization) may be needed or global strategies by accounting for the observed intra-image variability.

Image Artifacts and Dataset Heterogeneity The radiographs from both the public and the HOUB datasets exhibit a wide range of artifacts and heterogeneity, which poses a critical challenge for automated analysis. Images provided by HOUB, for instance, sometimes included patient identifiers printed directly on the radiograph, which were cropped to ensure privacy. More significantly, the dataset as a whole presents artifacts common to clinical practice, such as beam hardening, abrupt intensity shifts, and overlapping anatomical structures.

These factors introduce significant variability in contrast and texture patterns, representing a crucial divergence point for the two analytical pipelines explored in this work:

- **For the end-to-end deep learning models (Pipeline B):** These models are designed to learn robust feature representations directly from data. Therefore, only

minimal preprocessing—such as resizing and basic intensity normalization—was applied during training. The models are expected to inherently learn to manage a certain degree of this variability.

- **For the classical/hybrid pipeline (Pipeline A):** This pipeline relies on explicit, heuristic-based steps like thresholding and morphological gradients. As such, the variability introduced by the artifacts and heterogeneous acquisition conditions becomes a major obstacle. The performance of classical methods is critically dependent on consistent image characteristics.

Therefore, designing and validating a dedicated image preprocessing workflow is not merely an optimization step, but an essential prerequisite for Pipeline A to function accurately and reliably across different images. The following sections detail the evaluation and selection of this preprocessing strategy.

Image Preprocessing

Clinical image preprocessing was designed with two primary goals: to eliminate any identifying patient information and to standardize visual content for subsequent automatic analysis.

First, all metadata embedded in the ‘.jpg‘ files was removed in bulk using ExifTool. This ensured that no sensitive information—such as names, dates, or acquisition data—remained in the file attributes.

Additionally, many images contained identifying text directly printed on the radiograph (usually at the bottom). To preserve patient privacy, a Python script was developed to automatically crop the bottom 10 % of each image, maintaining the clinically relevant region (the periapical area) without significant anatomical information loss.

Preprocessing Evaluation To enhance segmentation performance, six preprocessing techniques were visually compared using a representative radiograph: histogram equalization, CLAHE, log transformation, contrast stretching, extreme clipping, and the original grayscale baseline. As shown in Figure 3, the log transformation yielded the cleanest separation between structures, particularly by suppressing mid-tone noise and enhancing boundary contrast. However, it also tended to over-amplify bright regions (e.g., metallic components), risking partial loss of anatomical detail in some images.

Linear streching and clipping offered a balanced enhancement, improving local contrast while maintaining overall structural integrity. In contrast, global histogram equalization and CLAHE produced limited or inconsistent gains and, in some cases, introduced artifacts such as exaggerated edges or washed-out transitions. Based on these findings, log transform was selected as the default preprocessing technique for most experiments.

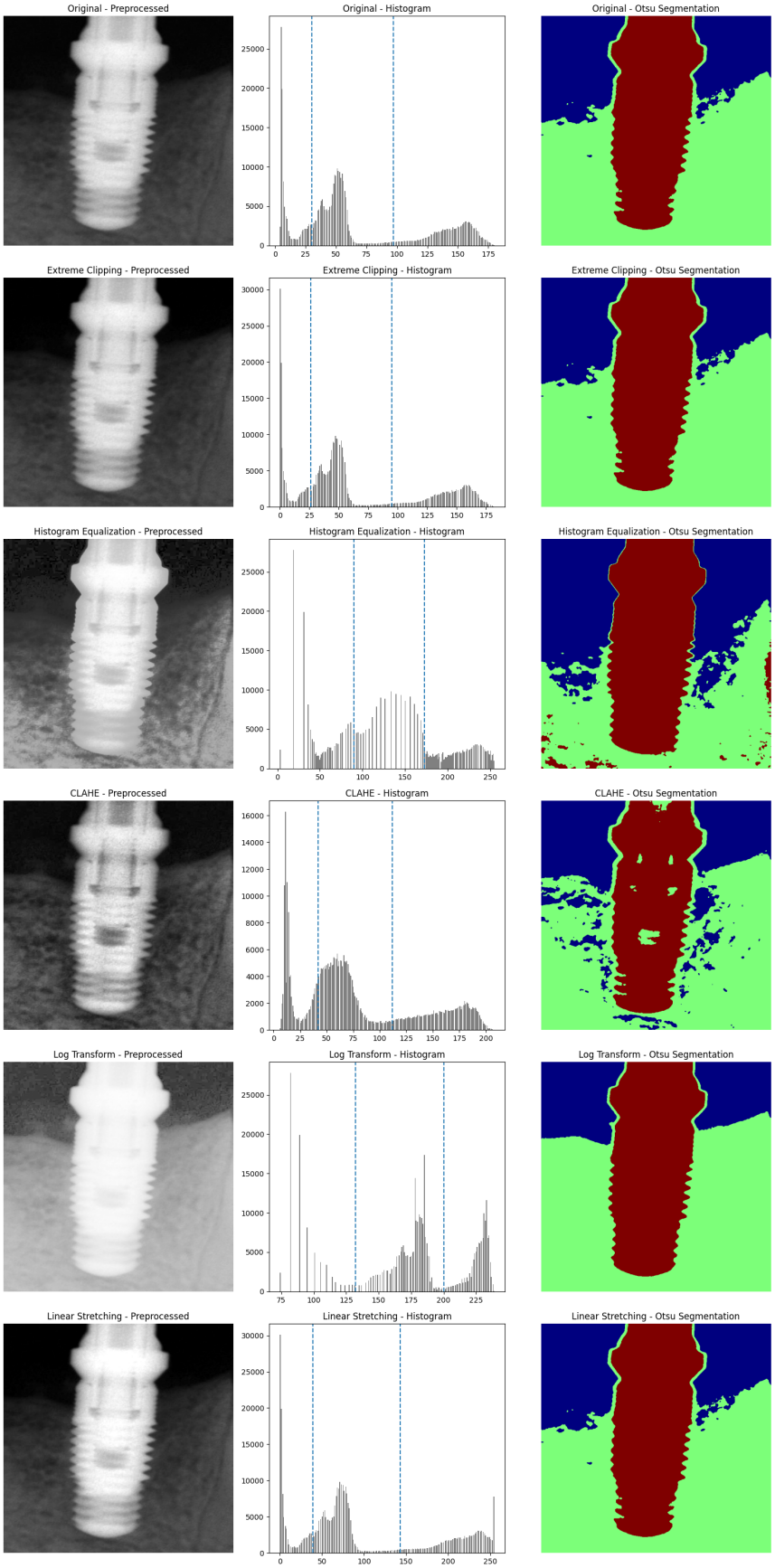


Figure 3: Comparison of preprocessing methods applied to a representative radiograph: processed image (left), histogram with thresholds (center), and Multi-Otsu segmentation (right). Log transformation yielded the most promising results in contrast enhancement and segmentation clarity.

After selecting the most promising preprocessing pipeline (median filtering followed by a log transformation), the same descriptive analysis was repeated on the preprocessed images to assess changes in grayscale distribution and segmentation characteristics.

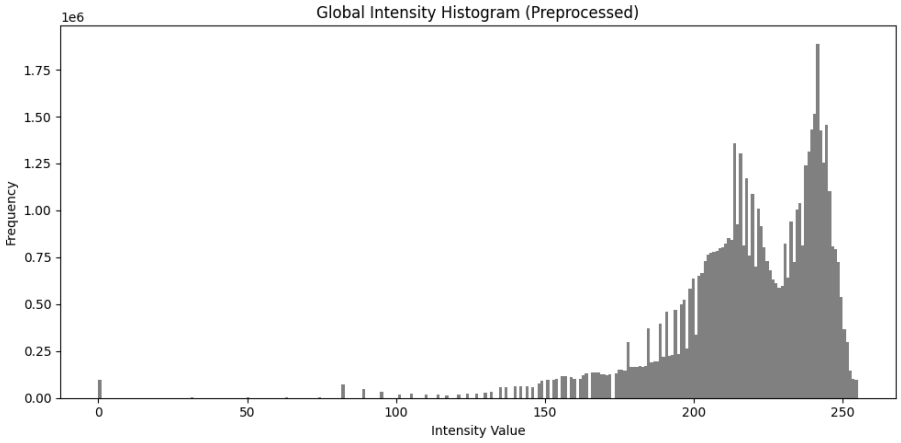
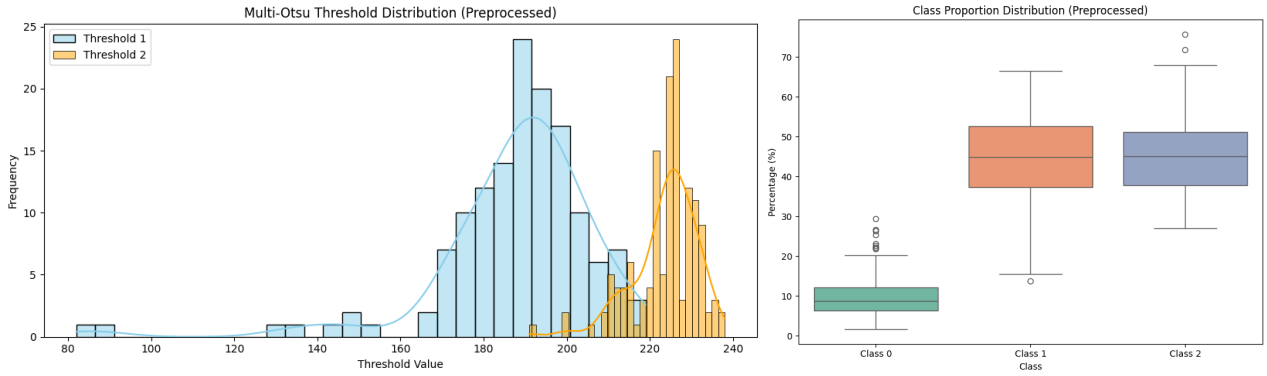


Figure 4: Global intensity histogram after applying median smoothing and logarithmic transformation. The distribution becomes more compact and skewed, emphasizing structures of interest and suppressing background noise.

The global intensity histogram (Figure 4) reveals a strong shift towards higher intensity values, with a pronounced asymmetry and dense peak structure above intensity 200. Compared to the original, this histogram is significantly more compact and skewed. The presence of multiple sharp peaks—especially near the high end of the dynamic range—suggests enhanced differentiation between highly radiodense regions (e.g., metal, cortical bone) and surrounding structures.

At the same time, low-intensity regions (below 150) are sparsely populated, indicating suppression of background noise and dim anatomical areas. This transformation boosts contrast in diagnostically relevant zones, particularly where bone and implant boundaries meet, making them more distinguishable for threshold-based and learning-based segmentation techniques. These effects align with the visual enhancements observed during qualitative assessment and justify the choice of preprocessing as a standard pipeline step.

The distributions of Multi-Otsu thresholds are presented in Figure 5a. While the overall separation between the thresholds does not increase uniformly for all cases, a clear shift in the distribution of t_1 is observed. This shift suggests improved delineation of low-intensity anatomical regions, such as trabecular bone and gingiva, which are critical for accurate segmentation in dental radiographs.



(a) Distribution of Multi-Otsu thresholds after preprocessing.

(b) Class-wise pixel proportions after preprocessing.

Figure 5: Segmentation behavior after preprocessing using Multi-Otsu thresholding. (a) Thresholds t_1 and t_2 exhibit improved separation and reduced variability. (b) Class 1 (intermediate intensity) becomes more dominant and consistent, while Class 0 (low intensity) is suppressed, suggesting reduced background interference.

In the preprocessed images, Threshold 1 (t_1) is tightly concentrated between approximately 170 and 200, while Threshold 2 (t_2) clusters between 215 and 235. This results in a more consistent and predictable thresholding behavior across the dataset, with a generally wider gap between t_1 and t_2 than observed in the raw images.

The sharper peaks and reduced spread in the distributions indicate that preprocessing improved the stratification of grayscale intensities. In particular, t_1 better separates low-intensity structures (e.g., trabecular bone, gingival regions), whereas t_2 consistently captures the transition to high-density components such as implants or cortical bone. These findings support the integration of the preprocessing pipeline as a critical step for enhancing segmentation robustness in intensity-based image analysis workflows.

Finally, Figure 5b displays the class-wise pixel proportions across the dataset after preprocessing. Class 1 (intermediate intensity), which typically represents anatomical structures such as bone, becomes the most dominant component and shows reduced variability, suggesting enhanced consistency in the segmentation of diagnostically relevant features.

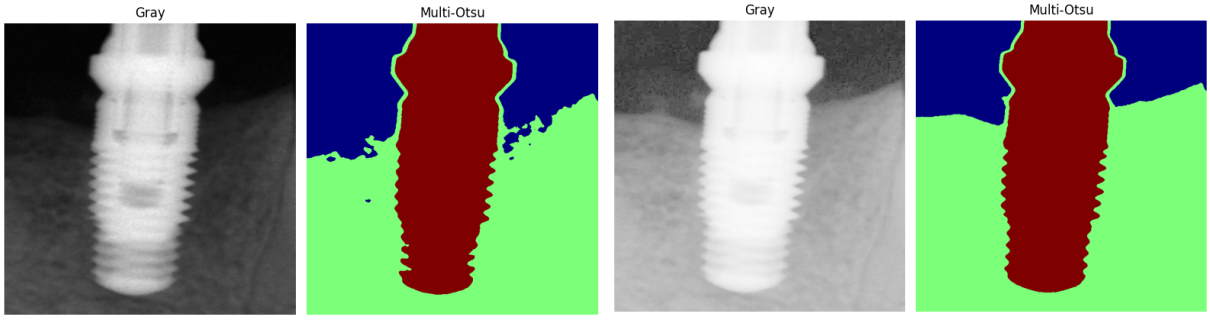
Class 2 (high intensity), often associated with radiopaque elements like implants and cortical bone, also maintains a strong and relatively stable presence. In contrast, Class 0 (low intensity), generally corresponding to background areas or poorly exposed regions, is markedly reduced in both proportion and variability. This behavior is consistent with the effect of the logarithmic transformation, which compresses lower intensity values and expands contrast in mid-to-high ranges, thereby attenuating background dominance.

Importantly, despite the suppression of low-intensity noise, background regions remain detectable thanks to the adaptive nature of Multi-Otsu thresholding, which preserves their classification while yielding clearer boundaries for the remaining structures. This balance allows the preprocessing to reduce clutter while retaining useful class separation.

These results confirm that the selected preprocessing pipeline enhances grayscale stratification, reduces background interference, and promotes a more stable intensity profile across samples. This behavior supports more robust segmentation training and motivates the use of the preprocessed images as the standard input for all subsequent model development.

To illustrate the practical effect of preprocessing, Figure 6 presents a visual comparison of Multi-Otsu segmentation applied to a representative ROI of a periapical radiograph before and after the selected preprocessing pipeline (median filtering + log transform).

Despite the suppression of low-intensity noise, background regions remain clearly identifiable in both versions, thanks to the adaptiveness of Multi-Otsu thresholding. However, the preprocessed version yields more sharply defined transitions between anatomical structures, particularly in regions involving bone and metallic components. This example highlights how preprocessing improves contrast and boundary clarity while preserving segmentation relevance.



(a) Original image and Multi-Otsu segmentation.

(b) Preprocessed image and Multi-Otsu segmentation.

Figure 6: Visual comparison of Multi-Otsu segmentation applied before and after preprocessing. The preprocessed version shows cleaner boundaries and more distinct intensity-based classes.

To objectively quantify the benefits of this preprocessing pipeline, the Multi-Otsu thresholding method was reapplied to the preprocessed dataset. A clear increase in the separation between the first and second thresholds ($t_2 - t_1$) was observed, suggesting enhanced contrast and separability between intensity classes. This is visually summarized in Figure 7.

Furthermore, statistical tests confirmed the significance of this improvement:

- Paired t-test: $p = 7{,}5 \times 10^{-62}$
- Wilcoxon signed-rank test: $p = 4{,}05 \times 10^{-23}$

These findings demonstrate that the preprocessing procedure substantially enhances the discriminability of intensity-based regions. In practical terms, this implies that anatomical features, particularly the bone crest and metallic structures, are more discernible

from the background in the preprocessed images, thereby facilitating more robust and dependable segmentations.

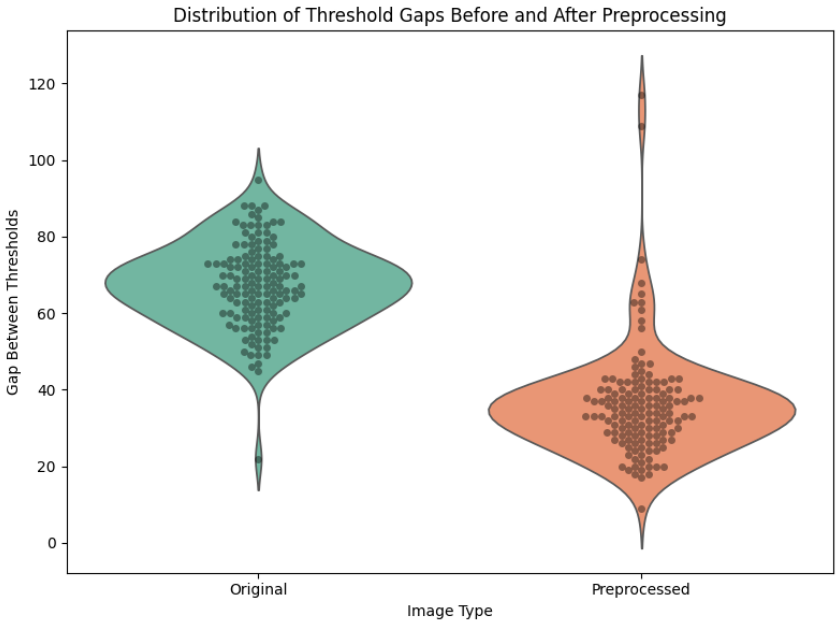


Figure 7: Effect of preprocessing on Otsu threshold separation: Distribution of $t_2 - t_1$ before and after applying log transformation and median smoothing. The increased separation after preprocessing indicates greater contrast between intensity-based classes.

Based on this evidence, the preprocessed version (log + median) was selected as the standard input for all techniques, including deep learning segmentation models (e.g., U-Net) and classical compuetr vision techniques.

Data Labeling and YOLO/COCO Format Annotations

To train deep segmentation and detection models, manual annotations were created using Label Studio, installed and run locally. Three main classes were defined: `bone`, `implant`, and `gum`, and polygon-based segmentation masks were created for a selected subset of images. The export format used was "COCO with Images", which produces a structure compatible with standard computer vision libraries [20].

This format allows direct integration with frameworks such as Detectron2 [21] and facilitates interoperability with segmentation pipelines and cross-validation systems. A simplified annotation example is provided in Annex A.

Additionally, the training process incorporates publicly available annotated datasets provided in YOLO format, widely used for object detection tasks. These annotations—originally expressed as bounding boxes or polygonal regions in YOLO format—were automatically converted to the COCO format to ensure compatibility with frameworks such as Faster R-CNN and Mask R-CNN. This transformation expanded the usable dataset and

enabled cross-architecture experimentation. An example YOLO annotation is included in Annex A.

The combined use of both native COCO annotations and converted YOLO annotations supports a flexible training setup, allowing the exploration of both segmentation and detection-based architectures under a unified annotation framework.

Synthetic Patient Data

To rigorously test the prognostic model in a controlled environment, a synthetic cohort of patient profiles ($n = 160$) was generated. This cohort was designed to reflect the demographic and clinical heterogeneity observed in real-world implantology patients by sampling variables from distributions informed by established clinical literature and epidemiological data.

Key patient characteristics were modeled as follows:

- **Age:** Patient age was sampled from a normal distribution with a mean of 55 and a standard deviation of 15 years (clipped at 20–90 years), reflecting the typical age range of patients receiving dental implants as reported in large systematic reviews [22].
- **Systemic Risk Factors:** The prevalence and impact of major systemic risk factors were explicitly modeled.
  - **Smoking status** was assigned with a prevalence reflecting general population data, acknowledging its role as a primary risk factor for increased severity of periodontal disease and subsequent bone loss [23].
  - **Diabetes** was assigned with a higher prevalence of Type 2 over Type 1, consistent with epidemiological data. Its presence is known to significantly increase the risk of peri-implant complications and was used to modulate disease progression in the simulation [24].
- **Clinical Periodontal Status:** The baseline oral health of each patient, a critical determinant of implant success, was categorized using the framework from the 2017 World Workshop on the Classification of Periodontal and Peri-Implant Diseases and Conditions. This allowed for a clinically relevant stratification from periodontal health to severe periodontitis [25].
- **Other Factors:** Additional variables such as Body Mass Index (BMI), oral hygiene level (rated on a 1–5 scale analogous to plaque indices), and implant-specific characteristics (e.g., type, material, surface) were also included to ensure a comprehensive and diverse patient profile.

This evidence-based generation process resulted in a synthetic cohort with plausible and heterogeneous clinical profiles, creating a suitable and challenging testbed for the technical validation of the prognostic regression model.

3.3. Detection Models

3.3.1. YOLOv8

YOLOv8 (You Only Look Once version 8) [12] is the latest in the one-stage YOLO family of models (released by Ultralytics in 2023). It represents a cutting-edge tradeoff between accuracy and speed for real-time object detection and also supports instance segmentation. YOLOv8's design philosophy continues YOLO's tradition of a single unified network that directly produces detections from an image in one evaluation (no separate proposal stage), making it highly efficient. Key innovations in YOLOv8 include an improved backbone, an anchor-free detection head, and the addition of flexible task-specific heads for segmentation and other tasks [12].

Architecture YOLOv8 follows the general Backbone–Neck–Head layout. The backbone is a CNN feature extractor; YOLOv8 uses an updated CSP-Darknet architecture with novel layers like the C2f (Cross Stage Partial networks with fused feature maps) that improve parameter efficiency and gradient flow [13]. This backbone yields a rich multi-scale feature pyramid. The neck further processes and combines these features (using upsampling and concatenation in a PANet/FPN style) to ensure that high-level context and low-level detail are available for the detection head [26]. Notably, YOLOv8's neck concatenates features without forcing them to same depth, reducing parameters while fusing scale information.

The detection head in YOLOv8 is anchor-free and decoupled. Unlike earlier YOLO versions which used predetermined anchor boxes, YOLOv8 directly predicts the center of objects and their width/height on the feature map. Specifically, the head outputs, for each feature map cell, a set of predictions (bounding box coordinates relative to that cell's location, objectness score, and class probabilities) without reference to preset anchors. This design simplifies training (no need to match anchors to data) and yields fewer, more meaningful candidate boxes. Additionally, YOLOv8 uses a split head: separate regression and classification sub-branches, which is known to improve optimization (each sub-task can specialize). This "anchor-free split Ultralytics head" is highlighted as a core feature that improved both accuracy and efficiency. In practice, anchor-free YOLOv8 reduces post-processing burden since far fewer overlapping boxes are predicted, and it can adapt to new domains (like dental X-rays) without recalibrating anchors [13][12].

Training pipeline YOLOv8 training follows the general YOLO recipes proven in YOLOv5. Ultralytics made training user-friendly via a PyTorch implementation that handles data loading, augmentation, and hyperparameters. Data augmentations such as mosaic augmentation (combining four images), random cropping, flipping, and color jitter are applied on the fly. Mosaic augmentation in particular helps the model learn to detect objects under varying context by stitching images together, though YOLOv8 switches off mosaic in later epochs to fine-tune on natural image distributions (as was found beneficial in YOLOv5 training) [13]. The model uses the Adam optimizer by default and a combination of losses: objectness (binary cross-entropy), classification (BCE or focal loss), and bounding box (Complete Intersection over Union(CIoU) loss, which directly optimizes IoU).

Model Configurations YOLOv8 is offered in scalable versions: `v8n`, `v8s`, `v8m`, `v8l`, and `v8x`. This flexibility allows for matching computational budgets and accuracy requirements. All configurations achieve real-time performance on GPU, making them applicable in time-sensitive medical scenarios.

Experimental Setup All YOLOv8 variants were trained on the same dataset using a uniform pipeline: 100 epochs, image size 640, batch size 16, early stopping after 20 epochs without improvement, and pretrained COCO weights [12].

Parameter	**YOLOv8n**	**YOLOv8s**	**YOLOv8m**	**YOLOv8l**	**YOLOv8x**
Parameters (M)	3.2	11.2	25.9	42.6	68.2
GFLOPs @640	8.4	28.6	78.9	131.5	195.2
Max epochs	100 (patience = 20)				
Optimizer	AdamW, LR = 1×10^{-3}, cosine decay				
Augmentations	Mosaic, H/V flip, HSV, scale jitter; mosaic off in final 10 % epochs				

Table 1: Training configuration summary across YOLOv8 variants.

Model	**Epoch**†	**mAP**$_{50}$	**mAP**$_{50\text{-}95}$	**Precision**	**Recall**	**T**$_{\text{inf.}}$ **[ms]**
YOLOv8n	55	**0.888**	0.366	0.807	0.750	17.1
YOLOv8s	71	0.849	**0.434**	0.569	**1.000**	**14.8**
YOLOv8m	21	0.373	0.182	0.311	0.500	25.3
YOLOv8l	67	0.697	0.447	0.437	0.593	56.7
YOLOv8x	73	0.736	0.355	**0.926**	0.500	76.4

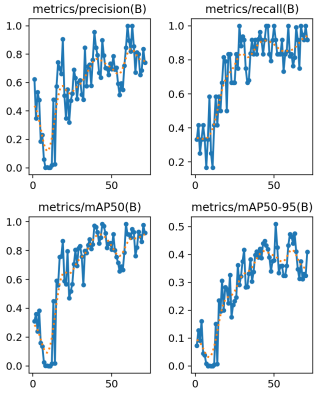
Table 2: Performance metrics on the test split. $T_{\text{inf.}}$ = average inference time per image. † Early stopping epoch.

Training Dynamics and Convergence Figure 8a displays the evolution of core metrics across training epochs. YOLOv8s exhibited a stable reduction in classification and localization loss, while key evaluation metrics—especially recall and mAP_{50}—increased steadily. Notably, recall approached 1.0 and mAP_{50} surpassed 0.9, evidencing robust learning. Despite some oscillations, the final precision metric also peaked near 0.9. The $\text{mAP}_{50:95}$ curve, while more sensitive to localization quality, stabilized around 0.45, confirming good generalization to varying IoU thresholds.

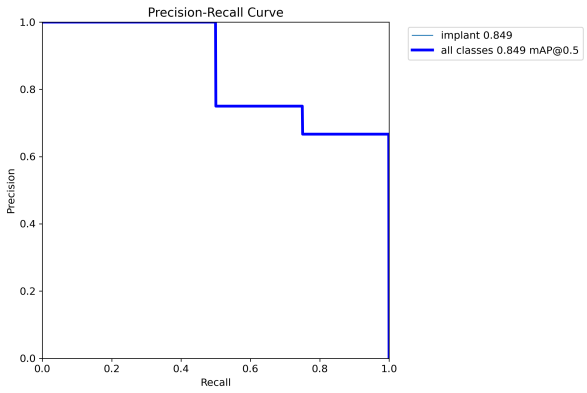
Precision–Recall Analysis As shown in Figure 8b, the Precision–Recall curve for the "implant" class yields an AP@0.50 of 0.849. The step-wise shape reflects discrete thresholds applied over a limited number of positive samples. The model exhibits a steep decline in precision as recall increases, indicating its tendency to favor sensitivity at the cost of false positives. This confirms the need for confidence threshold tuning. Based on internal threshold sweeping, the maximum F1 score is reached near $\hat{\tau}$
$approx$0,45, which is recommended for deployment scenarios.

Comparison with Faster R-CNN Compared to the best Faster R-CNN configuration (AP@0.50 = 92.9), YOLOv8s offers **comparable detection performance** (84.9 %) and **10× faster inference** (14.8 ms vs. 150 ms on ResNet-50 + FPN). While absolute precision is lower (0.57), this can be mitigated through post-processing (e.g., size filtering, adaptive thresholding), making YOLOv8s ideal for fast clinical screening pipelines.

Conclusion and Recommended Model Considering sensitivity, localization accuracy, and computational cost, **YOLOv8s** is the recommended configuration for periapical implant ROI detection. Its false positives can be reduced via confidence calibration, and its < 15 ms latency enables real-time applications.



(a) YOLOv8s precision, recall and mAP evolution during training.



(b) YOLOv8s Precision–Recall curve on the test split.

Figure 8: Training performance curves of YOLOv8s. Equal vertical scale for visual comparison.

3.3.2. Faster R-CNN

Faster R-CNN (Ren et al., 2015)[10] is a seminal two-stage object detector that significantly advanced detection accuracy while maintaining reasonable speed. It introduced the Region Proposal Network (RPN), eliminating the need for slow external proposal algorithms, and effectively combined region proposal and region classification into a single unified network[10]. Faster R-CNN has been highly influential and remains a strong baseline for detection tasks, including medical image analysis where high precision is needed.

Architecture Faster R-CNN consists of a common backbone CNN for feature extraction, followed by two sequential stages:

- **Region Proposal Network (RPN)** – This is a small network that slides over the backbone's convolutional feature map and predicts object proposals. At each sliding-window location, the RPN proposes a set of bounding boxes (anchors adjusted) and scores them for object likelihood. As discussed in foundational concepts, the RPN uses multiple anchors per location (e.g. 9 anchors) of various sizes/aspect ratios [10]. For each anchor, it outputs a binary classification (object vs. not object) and regresses a refined box. The RPN is thus a fully convolutional network that outputs a dense set of proposals. After generating these, a Non-Maximum Suppression (NMS) filters out overlapping proposals, typically keeping e.g. top 200 or 300 proposals by score.

- **Region-of-Interest (RoI) classifier + regressor (Fast R-CNN head)** – The second stage takes the proposals and classifies them into specific object categories (or background) and further refines the bounding boxes. For each proposed region, features are extracted from the shared backbone feature map using an operation called RoI Pooling (in Faster R-CNN). RoI Pooling crops the region's features and resizes them to a fixed size (e.g. 7×7) by pooling, producing a uniform feature vector which is fed into fully-connected layers [27]. These FC layers (often called the detection head) output class scores and box regressions (one for each class). In essence, this stage is equivalent to the earlier Fast R-CNN detector operating on the RPN proposals.

Faster R-CNN's innovation was mainly the RPN: by sharing the convolutional backbone between the RPN and the RoI classifier, it achieved region proposal generation at almost no extra cost. The RPN tells the network "where to look" (hence described as telling the detector where to attend), and those feature maps are reused for the second stage classification. This made the detector "nearly cost-free" in generating proposals [10].

A typical backbone for Faster R-CNN is a deep CNN like ResNet-50 or ResNet-101, often augmented with a Feature Pyramid Network (FPN) for multi-scale detection (as used

in the Detectron2 implementation). FPN adds top-down lateral connections in the backbone, enabling the RPN and head to also operate on multiple scales of feature maps, improving detection of both small and large objects. Modern implementations with ResNet-FPN greatly boost performance on challenging datasets like COCO.

Training pipeline : Training Faster R-CNN is a bit complex because it involves two sets of outputs (RPN and Fast R-CNN head). In the original paper, a 4-step training was proposed: train RPN, then train Fast R-CNN using proposals from RPN, then fine-tune RPN with the detection features, then fine-tune Fast R-CNN again. However, an approximate joint training method is often used, where both RPN and detection head are trained simultaneously, treating proposal generation as a non-differentiable step (gradient doesn't flow through NMS)[11]. Modern frameworks (like Detectron2) perform an end-to-end multi-task training: the loss is the sum of RPN losses and detector losses. The multi-task loss includes: RPN objectness (binary cross-entropy), RPN box regression (smooth L1), detection classification (cross-entropy over N+1 classes), and detection box regression (smooth L1). The RPN produces many anchors per image (e.g. 20k); during training, they are subsampled (e.g. 256 anchors per image) with a ratio of positives to negatives (e.g. 1:3) for the loss. Non-max suppression and proposal selection is also applied during training to feed a fixed number (like 128 or 256) of top proposals into the second stage.

One crucial improvement introduced later (in Mask R-CNN) is RoIAlign – it replaces RoI Pooling to avoid misalignment caused by quantization. RoIAlign uses bilinear interpolation to extract exactly aligned feature maps for each proposal, which greatly improves localization especially for segmentation masks. In Faster R-CNN implementations today, RoIAlign is often used by default as it improves box accuracy as well [14].

Faster R-CNN: Training Analysis and Results

This section delivers a complete, self-contained analysis of every Faster R-CNN configuration explored in this study. We first give a consolidated view of the training set-ups, followed by a discussion of convergence dynamics, quantitative comparisons and evidence-based recommendations.

Configuration Overview To ensure that the quantitative discussion remains anchored to reproducible hyper-parameter settings, Table 3 summarises the core choices that differentiate the five runs (learning rate, mini-batch sizes, augmentation policy and training length). All experiments share the same ResNet-50 [28]+FPN backbone and COCO-style detection head; only the scheduler and the strength of regularisation changed across runs.

Training Dynamics and Convergence Runs Run3 and Run4 —those employing warm-up plus multi-step decay—show the smoothest descent, plateauing near a minimum loss of $\approx$ 0,23. Interestingly, run HeavyAug-Large pushes the loss further down (0.144)

Model	Iterations	Init. LR	ROI Batch	Augmentations	Min. Loss	Final AP
Run1	3 000	2.5×10^{-4}	128	None	0.284	17.0
Run2	2 500	2.0×10^{-4}	128	Flip_{H}	0.343	46.8
Run3	5 000	1.0×10^{-4}	256	Flip_{H}+Resize	0.234	**49.2**
Run4	7 500	1.0×10^{-4}	256	Flip_{H}+Resize	0.231	48.4
Run5	5 500	1.0×10^{-4}	512	Bright./Contr./Flip/Rot.	**0.144**	48.1

Table 3: End-to-end configuration and outcome summary for all Faster R-CNN experiments. Lower minimum loss indicates faster or deeper convergence; higher AP reflects better localization and classification performance on the validation split.

but fails to translate that margin into higher AP. The additional colour-jitter and rotation augmentations evidently regularise the model but also introduce a distribution shift that the small validation set cannot fully absorb, a classic sign of mild over-fitting.

Accuracy Trends All configurations surpass 93 % in overall classification accuracy, yet only the two larger-batch schedules (Run3 and Run4) sustain foreground accuracies above 83 %. The marginal 1.2 percentage-point lead of Run4 over Run3 (84.32 % vs. 83.05 %) corroborates its slightly better IoU-50 precision (Table 4).

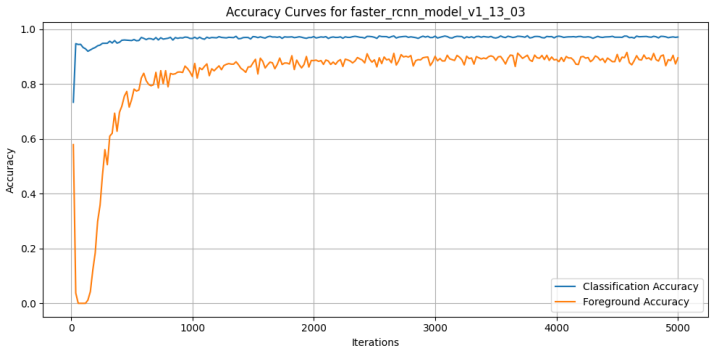
Training Curves for the Top Configurations To keep the focus on the two strongest candidates, Figure 9 juxtaposes the loss and accuracy trajectories of Run3 and Run4.

Fine-grained COCO Metrics Table 4 lists the full COCO evaluation. Despite its slightly lower overall AP, model Run4 edges out Run3 at the IoU=0.50 threshold—an important consideration when minor localisation errors are acceptable in clinical screening, prioritising sensitivity over strict IoU.

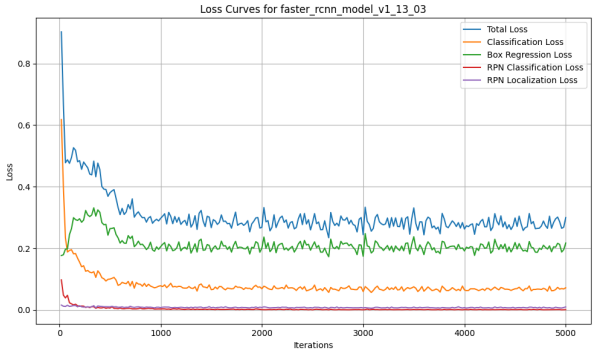
Model	AP	AP_{50}	AP_{75}	AP_{s}	AP_{m}	AP_{l}
Run1	17.0	33.3	17.2	7.6	39.1	31.2
Run2	46.8	91.7	41.9	37.4	54.3	53.9
Run3	**49.2**	92.6	**46.2**	**40.6**	**55.6**	**55.1**
Run4	48.4	**92.9**	44.7	39.9	55.3	54.2
Run5	48.1	90.9	47.2	40.3	54.5	53.6

Table 4: Detailed COCO metrics (higher is better). Bold figures highlight the leading score per column. AP_{50} is often used in the literature as a proxy for clinical sensitivity because it tolerates minor bounding-box deviations.

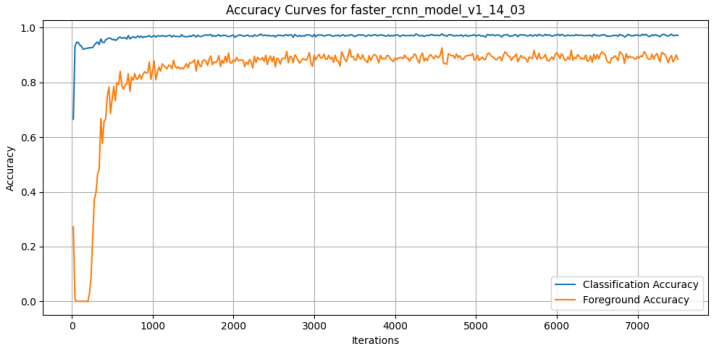
Conclusions and Recommended Baseline Balancing detection quality, convergence stability and training cost, configuration **Run4** emerges as the most dependable baseline for peri-apical implant detection. Its performance nearly matches the absolute AP of Run3 while offering better IoU-50 precision and the highest foreground classification accuracy.



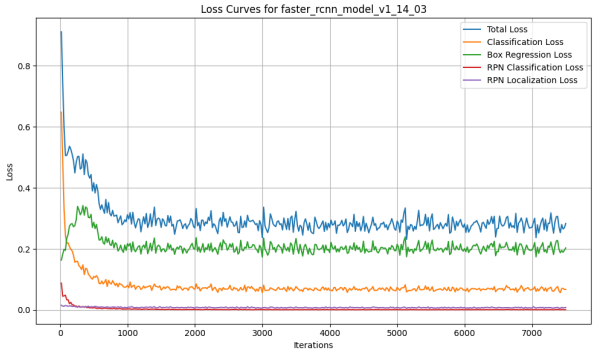
(a) Run3 – classification accuracy



(b) Run3 – total loss



(c) Run4 – classification accuracy



(d) Run4 – total loss

Figure 9: Training evolution for the two best Faster R-CNN configurations. All sub-plots share identical axis limits, allowing direct visual inspection of convergence speed and stability.

Therefore, all subsequent ablation studies and deployment prototypes in this project will be grounded on the Run4 hyper-parameter recipe.

3.3.3. Use Cases, Comparison and Justification

Role of object detection in the peri-implantitis pipeline

Periapical radiographs contain a wealth of anatomical detail—teeth, implants, restorative materials, and surrounding bone. Before any quantitative assessment of crestal bone levels can be attempted, the image area must be narrowed to the region of interest (ROI) surrounding each implant. Automatic object detectors therefore constitute the *first* stage of the pipeline described in Section 3.1: they isolate implant ROIs so that the subsequent bone-loss regression network operates on a clean and standardised context. Manual cropping is infeasible at clinical scale and introduces subjectivity; an accurate detector guarantees reproducibility and enables fully automated peri-implantitis screening.

Quantitative comparison

The main quantitative results are consolidated in Tables 2 (YOLOv8 family) and 4 (Faster R-CNN ablations). When both detectors are evaluated on the common validation split, the best one-stage variant **YOLOv8s** reaches $\mathrm{mAP}_{50} = 0{,}849$ with a latency of 14.8 ms, whereas the two-stage configuration **Faster R-CNN Run4** attains a superior $\mathrm{AP}_{50} = 0{,}929$ at 150 ms. Although YOLOv8s is an order of magnitude faster, the 9 percentage-point lead in IoU-50 accuracy is decisive given the clinical priority on sensitivity.

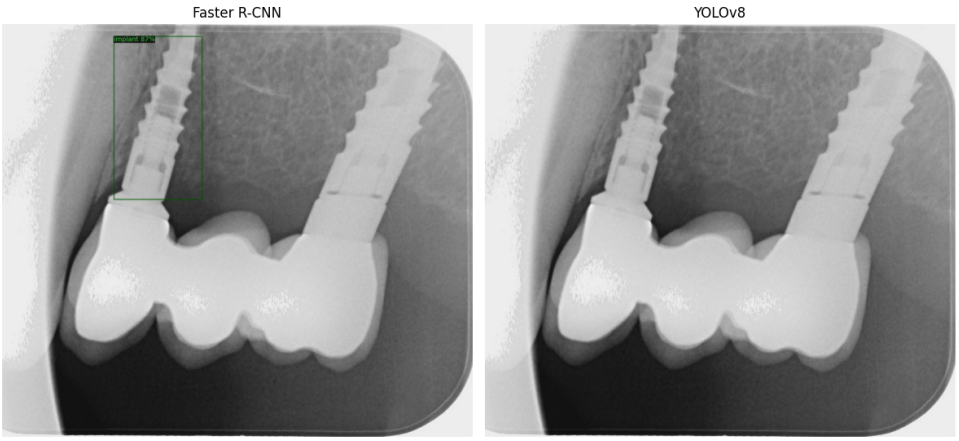
Qualitative comparison

Figure 10 juxtaposes both detectors on three representative radiographs:

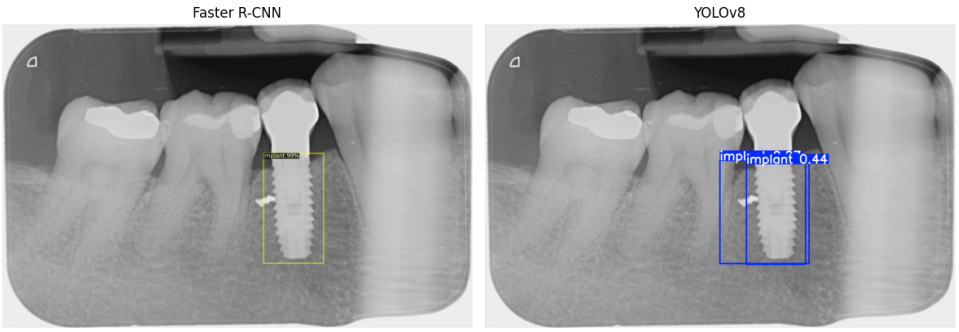
(a) *Challenging case*: two truncated implants; Faster R-CNN retrieves one, YOLOv8 none.

(b) *Easy case*: single, clearly visible implant; Faster R-CNN localises it with 99 % confidence, whereas YOLOv8 predicts two overlapping boxes at lower scores.

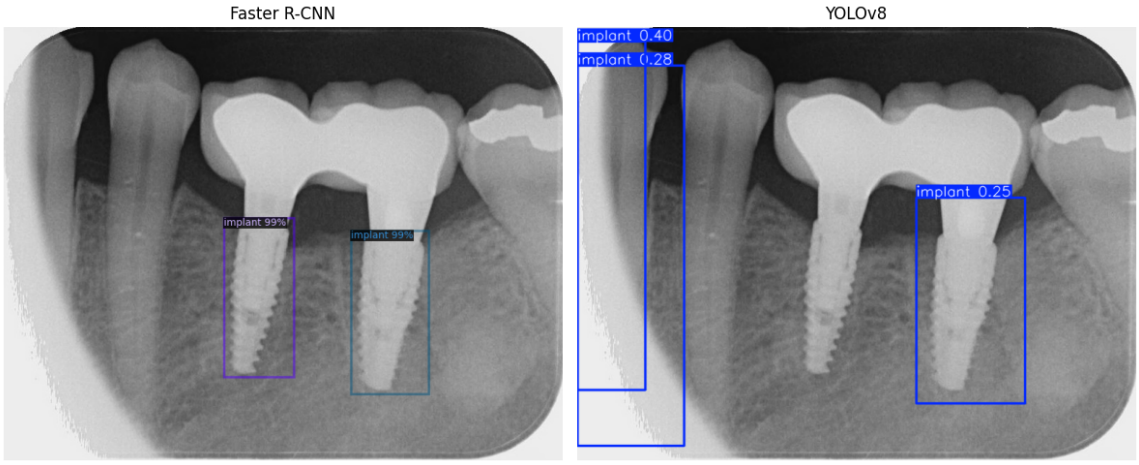
(c) *Noisy case*: double implant with scatter artefacts; Faster R-CNN detects both correctly, YOLOv8 misses one and raises two off-target boxes.



(a) (a) Challenging case



(b) (b) Easy case



(c) (c) Noisy case

Figure 10: Qualitative comparison of YOLOv8s and Faster R-CNN Run4.

Why does Faster R-CNN excel on this dataset?
Three empirical observations explain the superiority of the two-stage detector in our radiographic context:

- **Robustness to partial views**: Region Proposal Networks (RPN) generate candidate boxes at multiple scales and aspect ratios, enabling detection of implants that are truncated at the image boundary.
- **Contrast invariance**: The ResNet-50+FPN backbone learns multi-scale features that remain stable under the strong exposure variations typical of intraoral imaging.
- **Precise localisation**: RoIAlign preserves pixel-level alignment, a decisive advantage when the implant contour spans only 30–40 pixels in width.

In contrast, the anchor-free head of YOLOv8 occasionally splits the elongated implant into two adjacent predictions or suppresses it entirely when contrast is low.

Model selected for downstream analysis
Balancing the quantitative gain in AP_{50}, the qualitative robustness observed in Figure 10, and the acceptable 150 ms latency on a mid-range GPU, we select **Faster R-CNN Run4** as the default implant detector for the peri-implantitis pipeline. The higher recall at IoU= 0,50 ensures that virtually every implant is forwarded to the bone-regression stage, while occasional false positives can be pruned cheaply via post-processing. *YOLOv8s* remains a viable alternative for edge deployments where inference speed below 20 ms is mandatory.

Summary
Object detection is indispensable for localising implant ROIs in periapical radiographs. While YOLOv8 offers impressive speed, Faster R-CNN provides higher sensitivity and better geometric fidelity—both critical for subsequent quantitative bone analysis. Consequently, **Faster R-CNN Run4** is adopted as the detector of choice for the remainder of this thesis.

3.4. Models and Segmentation Techniques

The recent advances in artificial intelligence, particularly deep learning and computer vision, have introduced new opportunities for automated, objective, and quantifiable analysis of radiographic images. In the context of this project, the primary aim is to develop and validate a robust pipeline capable of segmenting anatomical structures relevant to peri-implant bone loss—including bone, gum, and dental implants—using both classical image processing techniques and state-of-the-art deep learning models. This section provides an overview of the segmentation models and techniques evaluated throughout

the project, along with the rationale for their selection and the methodologies adopted for model training and validation. Emphasis is placed on architectures with demonstrated efficacy in medical imaging tasks and on the integration of these approaches within a reproducible workflow tailored to dental radiographs.

3.4.1. U-Net (ResNet-34)

For the task of semantic segmentation of anatomical structures in dental radiographs, the U-Net architecture, specifically one utilizing a ResNet-34 backbone, is explored in this project.

The U-Net model, originally proposed by Ronneberger et al. (2015) [29], has become a reference standard in medical image segmentation due to its outstanding performance in scenarios characterized by limited datasets—a common situation in healthcare applications. U-Net's symmetrical encoder-decoder structure, complemented by skip connections, enables the network to efficiently learn both high-level contextual features and fine-grained spatial details, which are essential for the precise delineation of complex biological boundaries such as bone levels and gingival margins. The incorporation of skip connections allows the decoder to directly access high-resolution features from the encoder, facilitating the recovery of spatial information lost during downsampling.

In this work, the U-Net architecture is extended by employing a ResNet-34 network as its encoder. ResNet-34, a widely adopted convolutional neural network pre-trained on the ImageNet dataset, is leveraged for its powerful feature extraction capabilities and transfer learning benefits, which help accelerate convergence and improve generalization on small-scale medical datasets. The overall network consists of a contracting path based on ResNet-34, which progressively reduces the spatial dimensions while increasing feature richness, and an expanding path that reconstructs the segmentation map to the original image resolution. At each scale, feature maps from the encoder are concatenated with those in the decoder, ensuring optimal information flow and enhanced segmentation accuracy.

The training pipeline for U-Net (ResNet-34) in this project encompasses several essential steps: preparation of radiographic image datasets with pixel-level annotations, extensive data augmentation to improve model robustness, and the use of loss functions such as cross-entropy or Dice loss to optimize for overlap-based metrics (as detailed in Section 2.3.2, Equations (2) and (3)). Model performance is systematically evaluated using metrics like Intersection over Union (IoU) and the Dice coefficient, which quantify the agreement between the predicted segmentation and the ground truth. Iterative training with careful validation ensures reliable model selection and prevents overfitting, ultimately leading to a solution capable of supporting automated and reproducible assessment of peri-implant bone loss in dental radiographs.

Training Methodology and Experimental Setup The training logic was implemented in `PyTorch` and included encapsulated routines for both the training and validation phases. The model was trained on a workstation equipped with GPU acceleration whenever available, using a multi-class Dice loss that ignored the background class to focus the optimization process on the relevant anatomical structures. The number of segmentation classes was set to four (three anatomical structures: bone, implant, gum; plus background).

A standard supervised training loop was followed, where the model was trained over multiple epochs using the Adam optimizer and a learning rate of 1×10^{-3}, unless otherwise specified. In each epoch, the training loss was monitored, and after every epoch, the model was evaluated on a separate validation set to compute loss, IoU, and Dice coefficient. The model checkpoint with the highest validation Dice score was saved for further testing.

The following experimental configurations, were systematically explored:

- **U-Net with ResNet-34 encoder**: Experiments were conducted with 25, 50, and 100 training epochs, with a learning rate of 0,001 and a batch size of 4.
- **U-Net++ with ResNet-34 encoder**: A more complex variant, trained for 100 epochs with a learning rate of 0,001 and batch size of 4.
- **U-Net with EfficientNet-B0 encoder**: To investigate the effect of a modern and lightweight encoder, this configuration was trained for 100 epochs with a learning rate of 0,001 and batch size of 4.
- **Learning Rate variations**: U-Net (ResNet-34) was trained for 100 epochs with a reduced learning rate (1×10^{-4}) and batch size of 4, for finer convergence.
- **Batch Size variations**: U-Net (ResNet-34) was also trained for 100 epochs with an increased batch size of 8 and a learning rate of 0,001, to assess its impact on gradient stability and convergence.

During training, the loss curves for both training and validation sets were plotted and stored, allowing for a detailed inspection of convergence and overfitting behavior. Final model evaluation was performed on a hold-out test set, reporting the IoU and Dice metrics. Additionally, confusion matrices were generated to provide an in-depth analysis of class-specific performance and potential misclassifications.

Experimental Results and Discussion The various experimental configurations for the U-Net based segmentation models were evaluated based on their performance on the validation and test datasets. The key results are summarized in Table 5. This table highlights the best validation Dice score (`best_val_dice`), final test Intersection over Union (`final_test_iou`), final test Dice coefficient (`final_test_dice`), and training duration for each experimental setup.

Experiment ID	Model Type	Encoder	Epochs	LR	BS	Best Val Dice	Test IoU	Test Dice	Duration (s)
UNET_BS8_E100	Unet	resnet34	100	10^{-3}	8	0.8779	0.7973	0.8801	331.39
UNET_EFFNETB0_E100	Unet	efficientnet-b0	100	10^{-3}	4	0.8764	0.7872	0.8706	347.33
UNET_E50	Unet	resnet34	50	10^{-3}	4	0.8770	0.7850	0.8690	179.24
UNETPP_E100	UnetPlusPlus	resnet34	100	10^{-3}	4	0.8803	0.7850	0.8684	584.66
UNET_LR1E-4_E100	Unet	resnet34	100	10^{-4}	4	0.8755	0.7841	0.8677	353.73
UNET_E100	Unet	resnet34	100	10^{-3}	4	0.8797	0.7815	0.8666	356.97
UNET_E25	Unet	resnet34	25	10^{-3}	4	0.8739	0.7751	0.8633	89.85

Table 5: Summary of Segmentation Model Performance from U-Net with ResNet-34. LR denotes Learning Rate, BS denotes Batch Size.

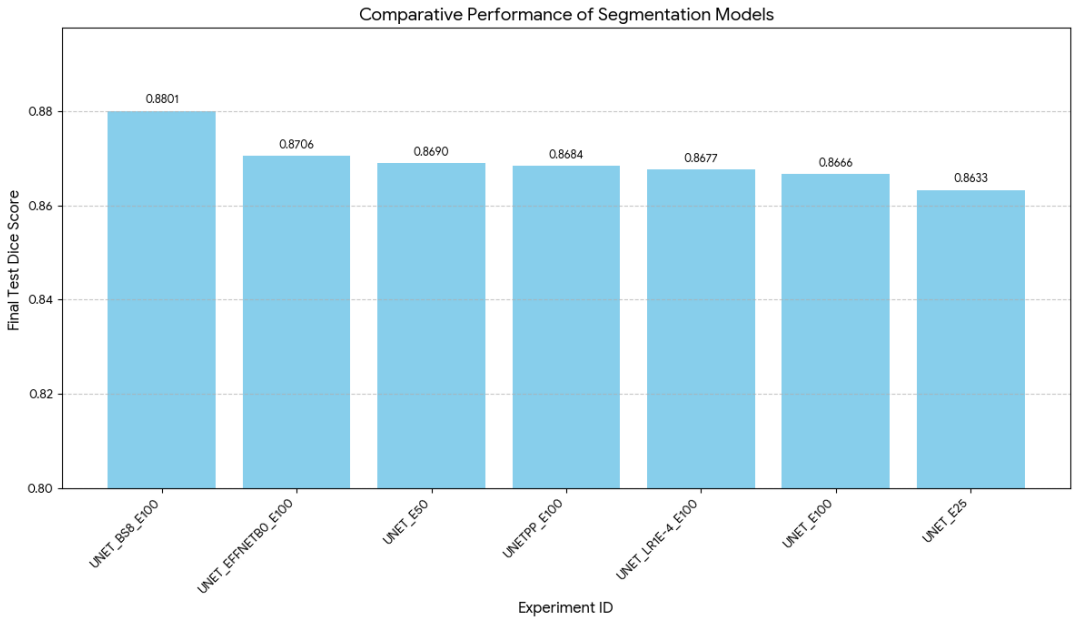


Figure 11: Comparative performance (Final Test Dice) of various segmentation model configurations from Table 5.

Several key observations can be made from these results (see Table 5):

- **Impact of Training Epochs:** For the standard U-Net (ResNet-34) with a learning rate of 0,001 and batch size of 4, the 25-epoch model (`UNET_E25`) achieved a final test Dice of 0.8633. Extending training to 50 epochs (`UNET_E50`) increased the test Dice to 0.8690. However, further extension to 100 epochs (`UNET_E100`) resulted in a slightly lower test Dice of 0.8666, suggesting that peak performance on the test set might be achieved before 100 epochs or that the model might be starting to overfit slightly. The `best_val_dice` for `UNET_E100` (0.8797) was higher than for `UNET_E50` (0.8770). (A visual representation of this can be seen in Figure 12.)
- **Effect of Batch Size:** Increasing the batch size from 4 to 8 for the U-Net (ResNet-34) trained for 100 epochs (`UNET_BS8_E100`) yielded the highest final test Dice score among all experiments (0.8801) and the highest final test IoU (0.7973). This configuration also had a competitive `best_val_dice` of 0.8779 and completed training

faster than the 100-epoch, batch size 4 ResNet-34 model. This suggests that a larger batch size might have contributed to more stable gradient estimates and better generalization. (Learning curves for this model are illustrated in Figure 13.)

- **Encoder Choice:** The U-Net with an EfficientNet-B0 encoder (`UNET_EFFNETB0_E100`) achieved a strong final test Dice score of 0.8706. This performance is notably better than the U-Net with ResNet-34 under identical epoch, learning rate, and batch size settings (`UNET_E100` test Dice: 0.8666).
- **Architectural Variants (U-Net vs. U-Net++):** The U-Net++ model with a ResNet-34 encoder (`UNETPP_E100`) achieved a `best_val_dice` of 0.8803, the highest among all experiments. However, its final test Dice score (0.8684) was slightly lower than that of the best U-Net configuration (`UNET_BS8_E100` at 0.8801) and took significantly longer to train (584.66 seconds vs. 331.39 seconds for `UNET_BS8_E100`).
- **Impact of Learning Rate:** Reducing the learning rate to 1×10^{-4} for the U-Net (ResNet-34) 100-epoch model (`UNET_LR1E-4_E100`) resulted in a test Dice score of 0.8677. This was slightly better than the standard LR (1×10^{-3}) U-Net (ResNet-34) 100-epoch model (`UNET_E100` test Dice: 0.8666).

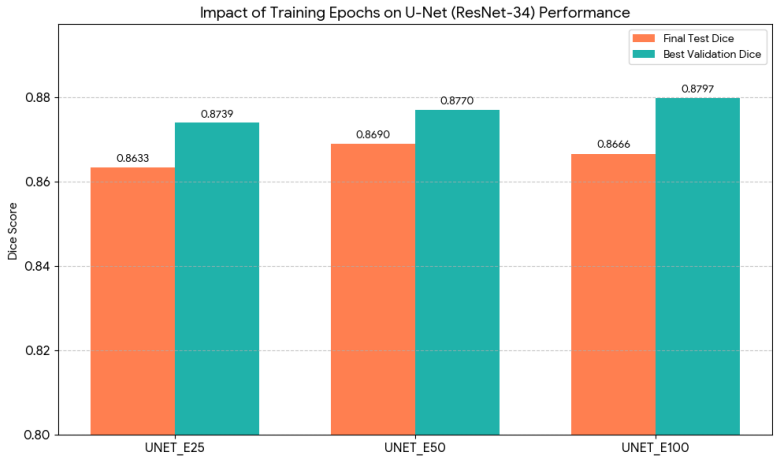


Figure 12: Impact of training epochs on the performance (Final Test Dice and Best Validation Dice) of U-Net (ResNet-34) models.

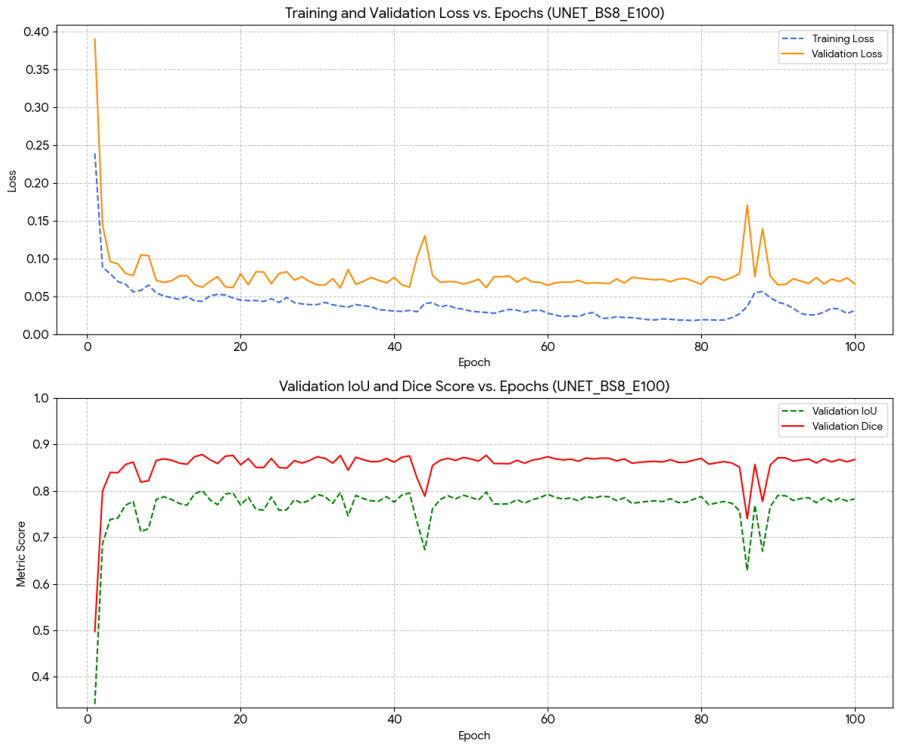
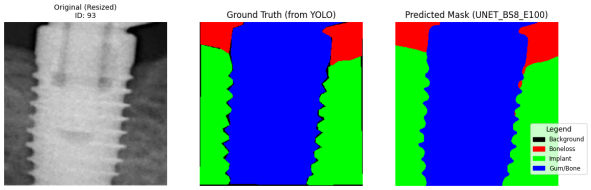
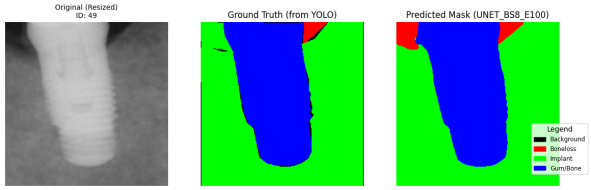


Figure 13: Learning curves (Training/Validation Loss and Validation IoU/Dice vs. epochs) for the UNET_BS8_E100 model.

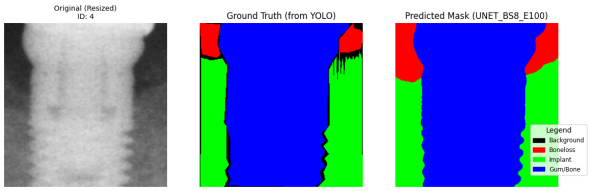
Based on the final test Dice coefficient, the `UNET_BS8_E100` (U-Net with ResNet-34, 100 epochs, batch size 8, LR 1×10^{-3}) emerged as the best performing model with a test Dice score of 0.8801. Qualitative analysis of this model's predictions on test images, as illustrated in Figure 14, provides further insight into its segmentation accuracy. Additionally, the confusion matrix presented in Figure 15 details its per-class performance.



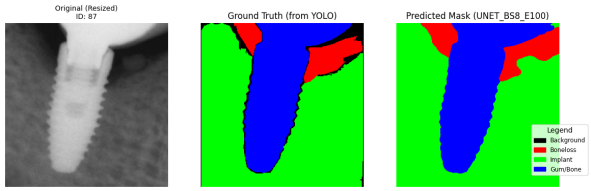
(a) Prediction Example 1



(b) Prediction Example 2



(c) Prediction Example 3



(d) Prediction Example 4

Figure 14: Example segmentation outputs from the `UNET_BS8_E100` model on test samples, showing the original image (resized), ground truth (from YOLO), and predicted mask.

The qualitative results displayed in Figure 14 corroborate the strong quantitative

performance of the `UNET_BS8_E100` model. Across the presented examples, the model consistently demonstrates a high capability in accurately delineating the different classes of interest: *Boneloss* (Red), *Implant* (Green), and *Gum/Bone* (Blue).

For instance, in Prediction Example 1 (Figure 14a), the predicted mask for the implant body (green) closely mirrors the ground truth, and the surrounding *Gum/Bone* tissue (blue) is well-defined with smooth and plausible contours. Prediction Example 4 (Figure 14d) further illustrates the model's robustness, showcasing accurate segmentation of the implant threads and adjacent bone structures, even under variations in image intensity and contrast typical of periapical radiographs.

Notably, in certain instances, such as those in Prediction Example 2 (Figure 14b) and Prediction Example 3 (Figure 14c), the model's output appears to refine the original ground truth annotations. The YOLO-based polygonal masks tend to have coarse boundaries, whereas the U-Net predictions frequently exhibit smoother and more continuous contours. This suggests the model can capture fine anatomical details that might be generalized in the original annotations.

Further supporting these qualitative observations, the confusion matrix for the `UNET_BS8_E100` model (Figure 15) provides a quantitative breakdown of its per-class performance. The diagonal values represent correctly classified pixels. The *Implant* class achieved 590,730 true positives, and *Gum* (aligned with "Gum/Bone") reached 531,467. Misclassifications of *Gum* into *Implant* (32,808) and *Boneloss* (6,650) were observed. For *Boneloss*, there were 60,699 true positives, with main confusions being with *Gum* (12,628) and *Background* (17,293).

The *Background* class was generally well-separated from anatomical classes. Misclassifications in *Boneloss* are understandable due to its diffuse appearance and proximity to other tissues. In summary, the confusion matrix affirms the model's robustness in segmenting *Implant* and *Gum/Bone*, while still offering reasonable performance for the more challenging *Boneloss* class.

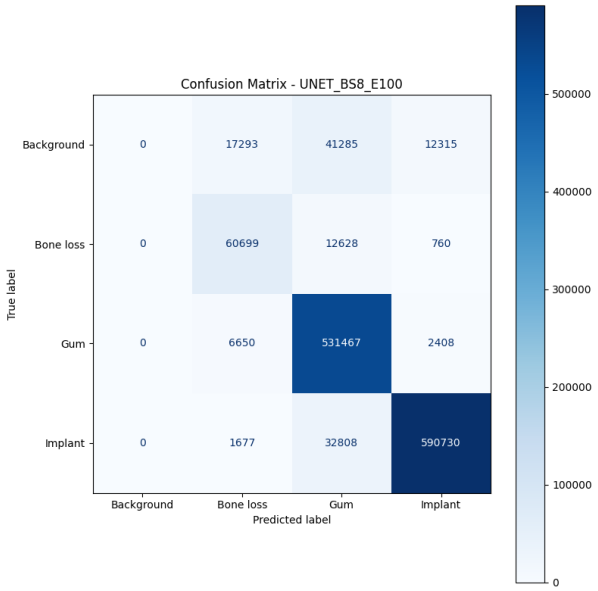


Figure 15: Confusion matrix for the UNET_BS8_E100 model, detailing per-class segmentation performance on the test set for Background, Bone loss, Gum, and Implant classes.

3.4.2. Mask R-CNN

Mask R-CNN (He et al., 2017) [14] is an extension of the Faster R-CNN model introduced in Section 3.3.2, designed specifically for *instance segmentation*. In addition to detecting objects with bounding boxes, this architecture outputs a pixel-accurate mask for every detected instance. Such precise segmentation is particularly valuable for dental radiographic analysis, where clear separation of overlapping anatomical structures is critical. The model was implemented with Facebook AI Research's `Detectron2` framework [21], a PyTorch-based platform that offers a robust, highly optimized environment for rapid and reproducible experimentation with state-of-the-art detection and segmentation models

Architecture Mask R-CNN retains Faster R-CNN's two-stage structure while adding a third branch dedicated to instance segmentation:

- **Backbone and Region Proposal Network (RPN).** A ResNet-FPN convolutional backbone produces multiscale feature maps. The RPN scans these maps to generate candidate regions of interest (RoIs) following the original Faster R-CNN methodology.
- **Detection and Segmentation Heads.** For each proposed RoI, Mask R-CNN performs three tasks in parallel:

  1. *Object classification*: predicts the object category (e.g. `Boneloss`, `Gum`, `Implant`, or `background`);

2. *Bounding-box regression*: refines the RPN's coordinates;
3. *Binary-mask generation*: employs a Fully Convolutional Network (FCN) to produce a class-specific mask, decoupled from the classification branch, enabling finer specialization.

- **RoIAlign.** Replacing `RoIPool`, Mask R-CNN introduces the `RoIAlign` layer, that applies bilinear interpolation to preserve the exact spatial correspondence between features and RoI's, essential for generating accurate masks and improving bounding-box localization.

Training Methodology and Experimental Setup Mask R-CNN was trained in `Detectron2` under controlled conditions to allow direct comparison with U-Net segmentation experiments. Training ran for the equivalent of 50, 100, and 200 epochs, using stochastic gradient descent (SGD) with momentum and a multitask loss combining classification, bounding-box regression, and binary-mask terms.

Systematic variations were explored in:

- **Encoder backbone:** ResNet-50-FPN (50 and 100 epochs) and ResNet-101-FPN (100 and 200 epochs);
- **Learning rate:** default ($2{,}5 \times 10^{-3}$) and reduced ($2{,}5 \times 10^{-4}$);
- **Batch size:** default (4 images) and enlarged (8 images).

During training, performance was monitored with the standard metrics for segmentation, such as, Average Precision (AP) or Dice/IoU scores, and the best model was saved for evaluation on the test set.

Experimental Results and Discussion The Table 6 summarizes the performance obtained by each tested configuration, ordered by the Dice coefficient in the test set to facilitate its direct comparison:

ID	Modelo	Encoder	Épocas	LR	BS	Val AP	Test IoU	Test Dice	Duración (s)
MaskRCNN_R101_E100	Mask-RCNN	ResNet101-FPN	100	0.00250	4	68.76	0.7844	**0.8540**	2090.12
MaskRCNN_R50_E100_BS8	Mask-RCNN	ResNet50-FPN	100	0.00250	8	65.45	0.7679	0.8437	1699.49
MaskRCNN_R50_E100_LR1e-4	Mask-RCNN	ResNet50-FPN	100	0.00025	4	65.49	0.7658	0.8435	1631.11
MaskRCNN_R50_E100	Mask-RCNN	ResNet50-FPN	100	0.00250	4	67.52	0.7721	0.8419	1583.60
MaskRCNN_R50_E50	Mask-RCNN	ResNet50-FPN	50	0.00250	4	65.19	0.7577	0.8327	794.38
MaskRCNN_R101_E200	Mask-RCNN	ResNet101-FPN	200	0.00250	4	68.14	0.7221	0.8255	4219.00

Table 6: Summary of instance segmentation performance with Mask R-CNN. LR: learning rate, BS: batch size.

Key observations emerging from the experimental analysis are:

- The deeper encoder (ResNet-101-FPN) trained for 100 epochs delivered the best overall results, attaining a Dice score of 0.8540 and an IoU of 0.7844, confirming that greater feature-extraction capacity benefits segmentation on complex periapical radiographs.
- Extending training to 200 epochs (`MaskRCNN_R101_E200`) did not improve performance; in fact, IoU and Dice dropped to 0.7221 and 0.8255, respectively. Despite a slight gain in validation AP, these results indicate the onset of overfitting or model saturation, ruling out this configuration.
- For ResNet-50, enlarging the batch size to 8 or lowering the learning rate yielded only marginal gains; none surpassed the ResNet-101 model trained for 100 epochs.

Consequently, `MaskRCNN_R101_E100` was selected as the optimal model, offering the best balance of segmentation accuracy (Dice and IoU), efficiency, and training-curve stability. The following figures illustrate its performance.

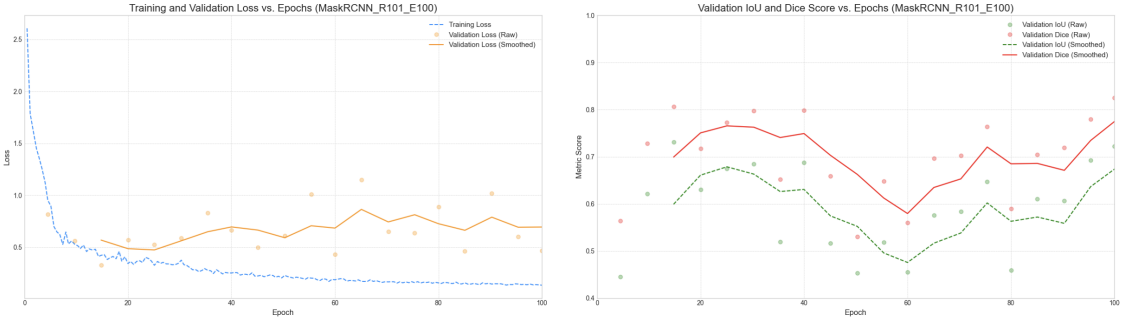
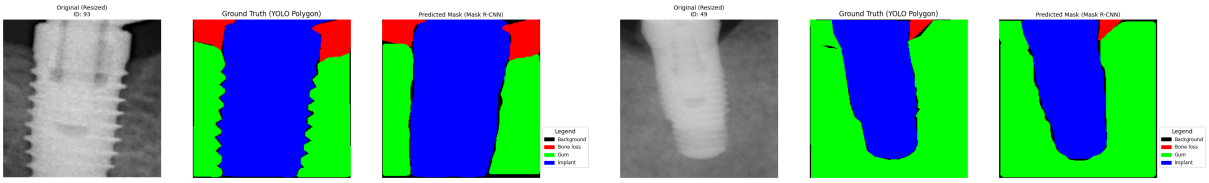


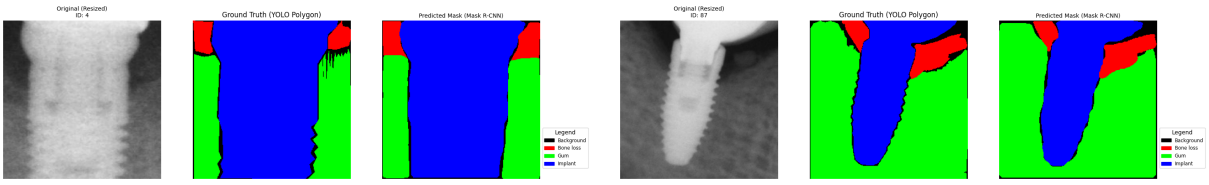
Figure 16: Learning curves for Mask R-CNN (ResNet-101-FPN, 100 epochs)

The learning curves in Figure 16 support this choice. Training loss decreases steadily to approximately 0.3 and plateaus, while validation loss fluctuates moderately without diverging. Validation IoU and Dice exhibit an intermediate dip between epochs 30 and 60, followed by a progressive recovery. This pattern suggests a brief generalization loss that the model later corrects, indicating good adaptability to validation variability. The final improvement in both metrics justifies stopping at 100 epochs, achieving a favorable equilibrium without clear signs of overfitting.



(a) Prediction Example 1

(b) Prediction Example 2



(c) Prediction Example 3

(d) Prediction Example 4

Figure 17: Segmentaciones de instancia generadas por Mask R-CNN (ResNet101-FPN).

The qualitative results in Figure 17 complement the quantitative findings, showing anatomically coherent segmentations adapted to each image's visual quality. In Prediction Example 1 (Figure 17a), the implant is segmented with high precision, faithfully tracing its serrated contour while clearly separating gingiva from bone loss even amid complex anatomy. Prediction Example 2 (Figure 17b) is particularly noteworthy: the model correctly identifies a bone-loss region in the upper right margin —as annotated— but does not detect a symmetric region on the left that is absent from the ground truth. This contrasts with U-Net, which did segment that clinically plausible yet unannotated area.

In Prediction Example 3 (Figure 17c), despite low image quality (blur and low contrast), implant segmentation remains coherent and gingival regions are located accurately, with minor errors attributable to image artefacts. Prediction Examlple 4 (Figure 17d) shows highly precise segmentation of an inclined implant and bilateral bone-loss areas, with smoother, more anatomical contours than the original, more angular ground-truth masks.

Notably, the model is conservative when segmenting the background, strictly limiting itself to annotated regions. While this behavior reduces over-segmentation, it also reveals some dependence on annotation imperfections, especially when polygonal masks generated by YOLO-style tools were used. Nevertheless, the predicted contours are smoother and morphologically plausible, suggesting that the model learns to refine anatomical structures even when training labels contain noise or imprecision.

The confusion-matrix analysis (Figure 18) details per-class performance, highlighting high precision in identifying implants and gingival tissue and anatomically justifiable confusion between bone-loss areas and adjacent tissue.

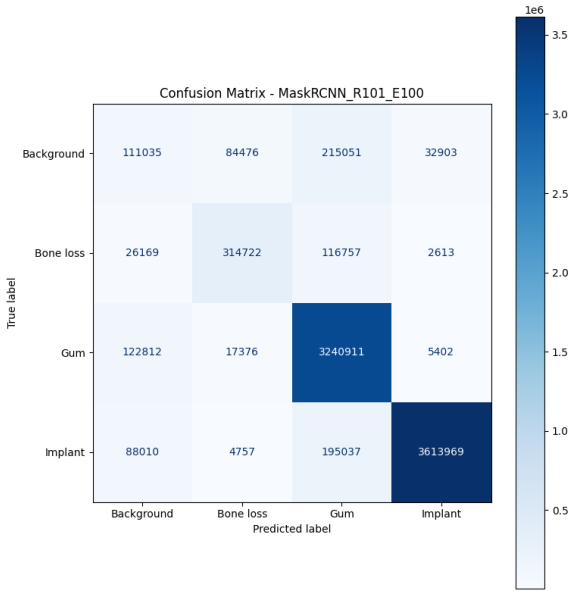


Figure 18: Confusion matrix for segmentation by Mask R-CNN (ResNet101-FPN).

In summary, Mask R-CNN with ResNet-101-FPN proves to be a robust and accurate method for instance segmentation in dental radiographic analysis, offering clear advantages over simpler approaches in clinical contexts where spatial precision is paramount.

3.4.3. Segment Anything Model (SAM)

The Segment Anything Model (SAM), introduced by Kirillov et al. (Meta AI, 2023), represents a paradigm shift in image segmentation [30]. Unlike conventional segmentation models such as U-Net or Mask R-CNN, which are trained to segment a predefined set of object classes, SAM is a **foundational model** designed for *zero-shot* segmentation of any object in an image based on a user-provided prompt. This capability makes it exceptionally valuable for tasks where annotated training data is scarce or nonexistent, as is often the case in specialized medical imaging applications.

Architecture and Functionality SAM's architecture is composed of three main components:

1. **Image Encoder:** A powerful encoder based on a Vision Transformer (ViT), pretrained at scale on the SA-1B dataset (over one billion masks across 11 million images). This encoder processes the input image once to generate a high-level vector representation (an *embedding*).

2. **Prompt Encoder:** A lightweight encoder that transforms the user's prompts—such as points, bounding boxes, text, or even rough masks—into vector embeddings.

3. **Mask Decoder:** A fast decoder that efficiently combines the image and prompt embeddings to generate one or more high-quality segmentation masks in real time.

The separation between image encoding (computationally expensive) and mask decoding (very fast) enables fluid interaction and exploration of multiple objects without reprocessing the entire image.

Integration and Mask Selection Strategy in the Pipeline In the context of this work, SAM forms the backbone of **Pipeline A: Hybrid Approach (SAM + Classical CV)** 3.1 of the pipeline, where it is used for the precise segmentation of the dental implant. The process is fully automatic and follows a robust selection strategy to ensure the best possible mask:

1. **Automatic Prompt Generation:** For each ROI, a point prompt is heuristically generated. First, a preliminary implant mask is obtained using a simplified version of the classical pipeline. The centroid of this mask is used as the point prompt for SAM. If this method fails, the geometric center of the ROI is used as a fallback.

2. **Two-Step Inference:** SAM is run twice to maximize the chance of achieving optimal segmentation. The first execution (S1) is performed on the original ROI image. The second (S2) is applied to a preprocessed version of the image (e.g., using a logarithmic transformation to enhance subtle contrasts).

3. **Consensus and Mask Selection:** A core feature of SAM is its ability to handle ambiguity. Given a single prompt, the model may return multiple candidate masks that correspond to different hierarchical interpretations of the object (e.g., a part of the implant, the full implant, or the implant with its crown). To resolve this ambiguity and obtain the most reliable segmentation, a consensus strategy is applied: the logical **intersection** of all candidate masks is computed.

   This operation produces a *consensus mask* (int_{s1} and int_{s2}), which acts as a strict confidence filter, retaining only those pixels that SAM included in *all* its segmentation hypotheses. The result is a high-certainty core representing the most unambiguous part of the object. The final choice between the consensus mask from the original image (S1) and the preprocessed one (S2) is made according to a hierarchy of criteria:

   - *Validity:* Each consensus mask must exceed a minimum area threshold to be considered valid.
   - *Solidity:* If both masks are valid, the one with greater **solidity** (the ratio between the contour area and the area of its convex hull) is selected. This favors compact and morphologically coherent segmentations, penalizing fragmented shapes. Based on qualitative analysis, preference is given to the original image mask (S1) unless the preprocessed one (S2) shows a substantial improvement in solidity.

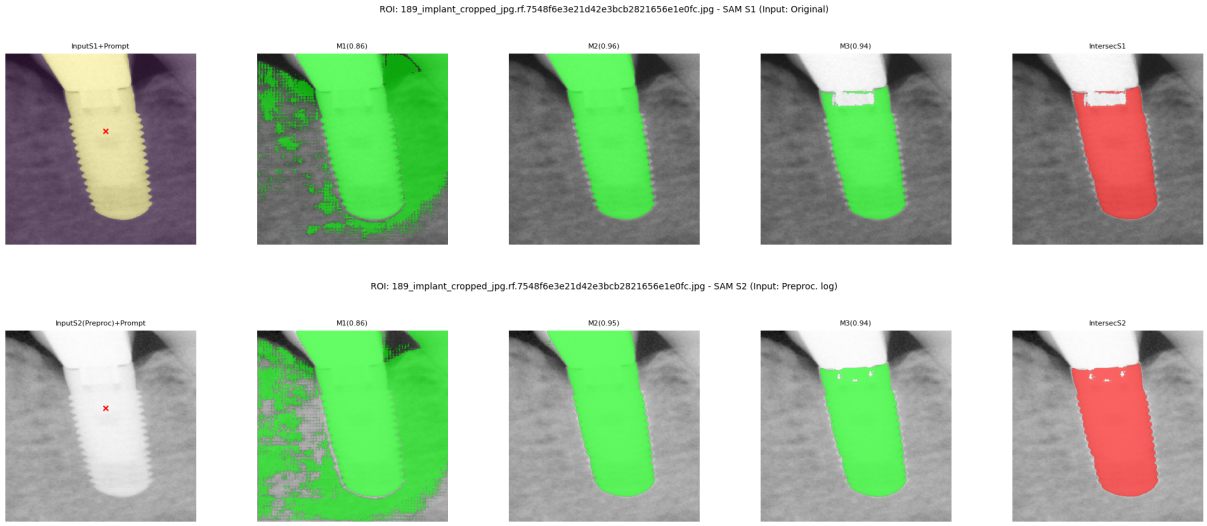


Figure 19: Inference process with SAM. **Top row (S1)**: original image, candidate masks M1–M3, and consensus mask *IntersecS1*. Note how *IntersecS1* excludes the square-shaped region corresponding to the abutment–crown joint. **Bottom row (S2)**: same sequence on the preprocessed image (logarithmic transformation). The consensus mask *IntersecS2* correctly includes this region, delineating the full endosseous portion of the implant.

Qualitative Evaluation and Observations on SAM Performance Figure 19 illustrates a representative case of SAM's behavior. In both runs (S1 and S2), the model successfully captures the threaded portion of the implant. The mask obtained from the original image omits the abutment–crown connection area, which is perceived as having homogeneous intensity with the background. In contrast, the preprocessed version improves contour definition, yielding a continuous and anatomically accurate mask, since logarithmic enhancement redistributes intensities and SAM includes the previously missing region (abutment-crown).

Once the best consensus mask is selected (in this case, IntersecS2), it is used as input for the automated extraction of bone-loss metrics, as explained in Section 3.5.

Justification and Advantage over Classical Segmentation (Multi-Otsu) The Multi-Otsu method from the classical *pipeline* (Section 3.4.4) thresholds radiographic density and thus tends to merge the implant body, crown, and other prosthetic elements into a single label. SAM, by contrast, thanks to its implicit understanding of object *atomicity*, enables *zero-shot* segmentation of only the **endosseous portion**—the threaded part integrated into the bone—while intelligently excluding the crown.

This distinction is clinically and geometrically crucial: bone-loss measurements must be taken relative to references located on the implant body, not the crown. For this reason, when the SAM-generated mask is morphologically coherent, the system prioritizes it over the classical mask to define the implant's uppermost point (see Section 3.5).

The value of this approach becomes clear when comparing both segmentations in Figu-

re 20. While Multi-Otsu, as previously explained, tends to fuse elements into a single mask, SAM separates and isolates the diagnostically relevant structure. This ability to produce semantically more accurate segmentations—excluding irrelevant regions—justifies its integration into the pipeline and represents a decisive advantage over classical methods.

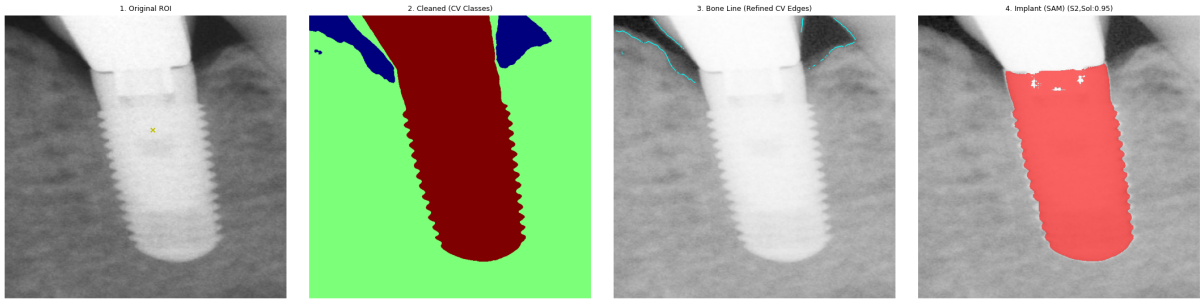


Figure 20: Final mask generated by SAM. From left to right: original ROI, segmentation by Multi-Otsu (for comparison), detected bone line, SAM-selected mask.

Limitations and Outlook Although SAM has proven useful for dental implant segmentation in periapical images, it also presents limitations. Failures have been observed in cases with low contrast, prosthetic artifacts, or radiographic saturation. In some instances, the model fails entirely to generate a coherent mask, which justifies the existence of fallback pathways in the pipeline.

On the other hand, extended versions such as SAM-HQ have shown general improvements, but at the time of this work, no stable or well-documented implementation was available for clinical integration. Moreover, the use of SAM in the dental domain remains in its early stages, with no consolidated precedent in the scientific literature. This exploration thus represents a pioneering application with relevant implications for both clinical practice and future research.

3.4.4. Classical Computer Vision Techniques and Mathematical Morphology

The proposed algorithm for the detection of the *peri-implant bone line*, while integrated into a workflow that includes *deep learning* modules to locate implants (see Section 3.3), bases its core analysis on a set of classical computer vision and mathematical morphology techniques. These techniques, which combine robust heuristics with well-founded morphological operations, are decisive in preprocessing, segmentation of regions of interest, and precise identification of the bone contour. The complete, documented source code for this pipeline is publicly available at https://github.com/amadeohm/periimplant-bone-analysis.git.

1. Grayscale Conversion. The process begins by converting the original radiographic image to a single intensity channel. This transformation reduces the data dimensionality from three channels to one, preserving diagnostic information based on luminance

variations and discarding chromatic information, which is generally non-critical in dental radiographic analysis.

2. Adaptive Local Mean Thresholding. Next, an adaptive mean thresholding algorithm is applied, which defines for each pixel a threshold $T(x, y)$ based on the mean intensity of a local neighborhood Ω_{xy} of size $N \times N$, subtracting a compensation constant C (see Section 2.3.1, Equation 1). This method is particularly effective for enhancing bone edges and sutures even in the presence of non-homogeneous illumination variations, such as vignetting, which is common in intraoral radiographs. Unlike global methods such as Otsu's (Section 2.4.1), adaptive thresholding does not assume global histogram multimodality but adjusts to the local characteristics of the image. The resulting binary mask, termed `adaptive_mask`, is fundamental in later stages for validating edge search regions.

3. Median Filtering. To attenuate impulse noise of the *salt-and-pepper* type present in radiographs, a median filter is employed (`cv2.medianBlur`) on the grayscale image. This non-linear operation replaces each pixel with the median of the intensities in its neighborhood, which allows for the elimination of atypical values (noise) while preserving the sharpness of edges and structural details with notable fidelity. This step is crucial before Multi-Otsu segmentation, as adequate smoothing prevents the generation of spurious regions due to noise, as illustrated in Figure 21.

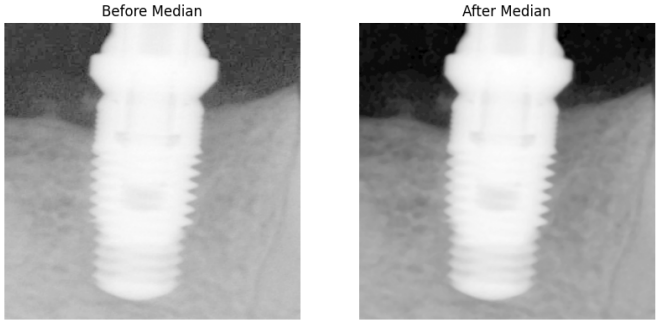
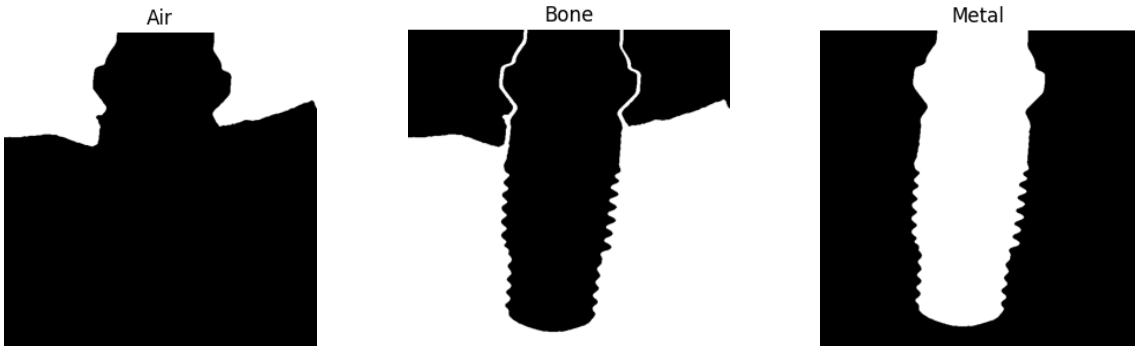


Figure 21: Effect of the median filter (kernel size 5×5) on a representative patch of a radiograph. Left: original image with impulse noise. Right: image after filtering, where a significant attenuation of noise and better preservation of structural edges are observed compared to linear filters such as the Gaussian filter.

4. Multi-Otsu Segmentation.

The main segmentation of the smoothed image is performed using Otsu's multiple thresholding method (`threshold_multiotsu` from `scikit-image`). This algorithm generalizes Otsu's original method to divide the image into $M = 3$ classes, which, in the context of dental radiographs with implants, usually correspond to: air/background and soft tissue; alveolar bone; and high-density material (metallic implant and crown). The $M - 1$ thresholds $(k_1, \ldots, k_{M-1})$ are automatically determined by maximizing the inter-class variance (or equivalently, minimizing the weighted intra-class variance), as detailed

in Section 2.4.1 (see Equation 8). This approach is especially pertinent given the typically trimodal character of the intensity histogram of these images. By convention, it is assumed that the class with the highest intensities corresponds to the implant and other metallic structures, due to the greater X-ray attenuation by these materials. The resulting binary masks for each class (Figure 22) form the basis for the subsequent refinement steps.



(a) Air / Background (b) Alveolar Bone (c) Metal / Implant

Figure 22: Representative binary masks of the three classes obtained after Multi-Otsu segmentation of a preprocessed radiograph. These initial masks are fundamental for isolating the alveolar bone and the implant before morphological refinement steps.

5. Morphological Filtering by Area. Subsequently, a morphological filter based on the area of connected components is applied to refine the masks of each class obtained in the previous step. For each class, all connected components are identified using `cv2.connectedComponentsWithStats`. Those components whose area is less than a predefined threshold (`min_area`) are considered noise or irrelevant regions and are eliminated. Only the largest connected component is preserved for each class, assuming it represents the main structure of interest (e.g., the main mass of the bone or implant). This procedure is conceptually analogous to a morphological opening operation, as it eliminates small islands without significantly affecting larger objects.

6. Hole Filling by Reconstruction. The regions eliminated in the previous step, as well as possible internal holes within the main components, are filled using an iterative morphological reconstruction process. Pixels marked as holes adopt the label of the majority class present in their immediate neighborhood (e.g., 8-connected). This diffusion process is repeated until all holes have been filled or a maximum number of iterations is reached. This filling ensures the topological coherence of the segmented regions, especially that of the alveolar bone and the implant, which is vital for a correct subsequent extraction of their contours.

7. Construction of Boolean Masks. With the refined classes, two key Boolean masks are defined for the detection of the bone line:

- `implant_mask`: A binary mask that identifies all pixels belonging to the implant (Figure 23). This is obtained directly from the highest intensity class after Multi-Otsu segmentation and morphological refinements.

- `valid_edge_mask`: This mask delimits the regions where the active search for the bone line will be performed. A pixel belongs to this mask if, and only if, (i) it is not part of the implant and (ii) it is considered foreground (white) by the adaptive thresholding from step 2. This double condition ensures that the edge search is concentrated in areas of high local contrast outside the implant body, minimizing the detection of metallic artifacts or edges in low-contrast regions.

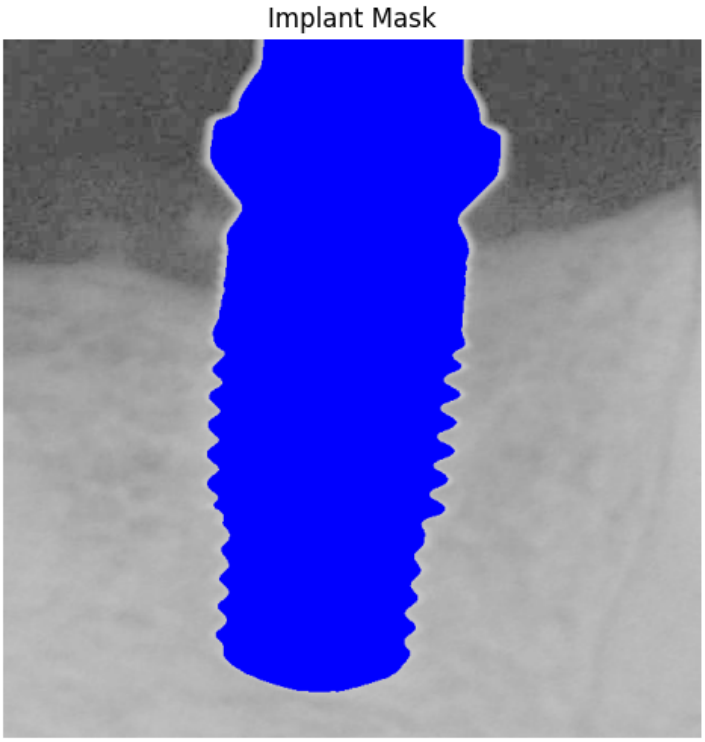


Figure 23: Visualization of the `implant_mask` (shown in blue) superimposed on the original radiograph. This mask is crucial for focusing the analysis on the bone-implant interface and for excluding the implant body during bone edge detection within the `valid_edge_mask`.

8. Edge Detection via Conditioned Morphological Gradient.

Finally, the peri-implant bone line is extracted using a conditioned and semantically filtered morphological gradient approach. Instead of applying a generic edge detector like Canny over the entire image—which could generate multiple irrelevant edges—this method inspects the neighborhood of each pixel in the refined Multi-Otsu class map.

A pixel is marked as part of the bone contour if it simultaneously meets two strict conditions: (i) there is a label transition between the central pixel and its examined

neighbor (e.g., from the “alveolar bone” class to “air/soft tissue”), indicating a change in material; and (ii) both the central pixel and its neighbor belong to the `valid_edge_mask` defined in the previous step.

This strategy ensures that only edges representing interfaces between different classes are detected, are not within the implant, and are located in regions of sufficient contrast. The result is an effective suppression of spurious edges and a high-specificity profiling of the marginal bone line, as shown in Figure 24.

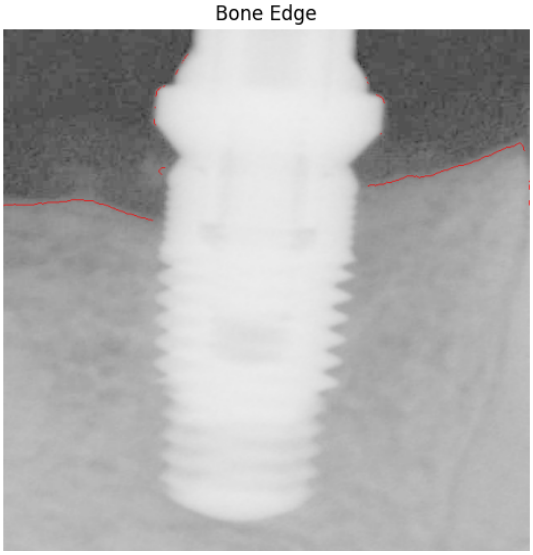


Figure 24: Peri-implant bone line detected (contour in red) using the conditioned morphological gradient, superimposed on the original radiograph. This edge map is the main result of the classical *pipeline* and is used for the identification and subsequent quantification of the bone contour.

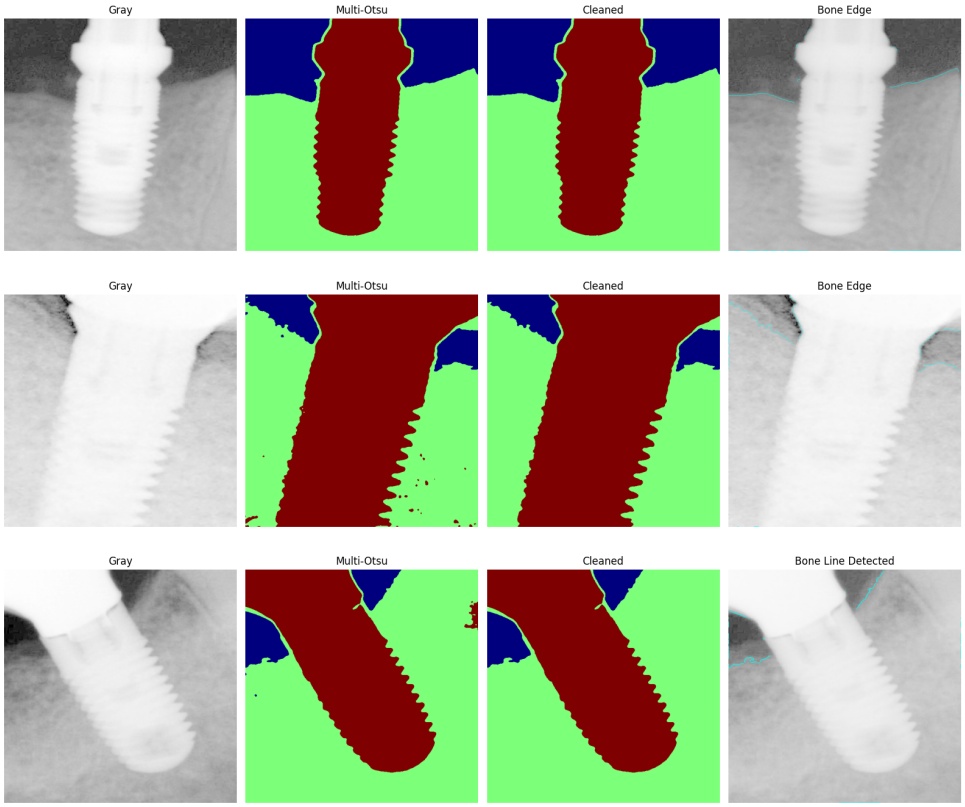


Figure 25: Mosaics of visualizations in Regions of Interest (ROI) around dental implants. Results from key stages of the classical *pipeline* are illustrated, such as the refined Multi-Otsu segmentation and the final detected bone line, demonstrating the applicability and behavior of the method in critical diagnostic zones.

In summary, this *pipeline* of classical techniques and mathematical morphology prepares the radiographic image, segments key anatomical and metallic structures, and extracts the peri-implant bone line in a robust and interpretable manner. The resulting edge map is subsequently merged with implant localization information obtained from *deep learning detection* models (Section 3.3) to perform quantitative measurement of bone loss, as described in Section 3.5. Figure 25 presents a mosaic with views of some intermediate and final results in the region of interest, illustrating the effectiveness of the process.

3.5. Comparison and Extraction Techniques

The goal of this section is to rigorously and reproducibly describe the procedures developed to quantify marginal bone loss (peri-implant bone loss, PBL) from processed periapical radiographs. The resulting values serve as predictor variables (regressors) in a comprehensive model designed to estimate the risk of peri-implantitis onset or progression. This model will integrate radiographic indicators with clinical and demographic information from the patient.

Depending on the segmentation technique used, two approaches are proposed for obtaining the indicator:

- A pipeline based on classical computer vision and hybridization with generative models (methodology developed in this work).
- A pipeline based on deep segmentation (Deep CNN) (pending implementation; mentioned here solely to contextualize future comparisons).

We detail below the first approach, which combines a pretrained generative model (SAM) for the implant with a classical pipeline for the bone line.

Method 1: Hybrid Estimation via CV and SAM

This method quantifies marginal bone loss in periapical radiographs by combining the generative Segment Anything Model (SAM) 3.4.3 with classical computer vision techniques for bone line extraction (described in Section 3.4.4).

1. Implant Segmentation (SAM) In this phase, the SAM model is used to obtain an accurate segmentation of the endosseous portion of the dental implant, isolating the relevant structure for measurement. Two versions of the Region of Interest (ROI) are used: the original image and a preprocessed version with logarithmic transformation to better enhance anatomical contrasts. SAM is guided by a single-point prompt, automatically determined from the centroid of a preliminary mask obtained by classical vision methods or, if unavailable, from the ROI center. After a consensus process based on intersection of SAM candidate masks, the final segmentation is selected based on validity (area) and morphological solidity criteria, ensuring high quality and stability of the result (see detailed strategy in Section 3.4.3). The selected mask allows extraction of the implant's upper profile, which serves as a reference for later steps. In addition, the height of the bounding box from this mask provides the total implant length, L_{imp}.

2. Bone Line Detection (Classical Vision) The automatic detection of the marginal bone line is carried out through a classical pipeline based on advanced computer vision and mathematical morphology techniques (detailed in Section 3.4.4). This pipeline includes sequential stages: grayscale conversion, median smoothing, segmentation via the Multi-Otsu algorithm, morphological cleaning to remove noise and artifacts, and final edge extraction using conditional morphological gradients. The output of this process is a robust and accurate bone line, $y_{\text{bone}}(x)$, obtained after excluding any edge contained within the previously segmented implant area.

3. Identification of Localized Vertical Defects Once the implant and bone line segmentations have been obtained, the next step is to detect and quantify specific vertical bone defects or localized bone loss pockets. This procedure is particularly important since

peri-implant bone loss does not always occur homogeneously along the implant, but may instead present as focal vertical defects.

To this end, the implant mask is dilated using a morphological kernel to create an extended region around it. All background-connected components adjacent to the dilated implant are then identified. Within each connected component, the longest continuous vertical line of background pixels is determined, and this segment is considered a robust indicator of a focal bone defect. The height of this longest line (H_{defect}) is measured from the coronal end of the implant, and is then normalized by the total implant length to compute the localized bone loss metric:

$$\text{PBL}\,\%_{\text{local}} = \frac{H_{\text{defect}}}{L_{\text{imp}}} \times 100.$$

This analysis is performed independently on the mesial and distal regions relative to the implant centroid, yielding side-specific indicators.

4. Confidence Metric To assess the reliability of each measurement, a confidence metric has been designed. This metric, denoted as c and defined on the interval $[0, 1]$, synthesizes three key aspects:

- **Solidity of the SAM-generated implant mask**: Indicates the structural robustness and absence of artifacts in the segmentation.
- **Effective coverage of the bone line beneath the implant**: Measures the proportion of columns within the implant's width that have a valid bone-line detection, strongly penalizing cases with low coverage.
- **Stability of the obtained measurements**: Assesses the variability of the vertical distances using their standard deviation (σ_d). High σ_d values indicate greater uncertainty, thereby reducing the confidence score.

Method Summary The combination of SAM for implant segmentation and classical techniques for bone line detection enables robust, detailed segmentation without requiring large volumes of annotated data (*zero-shot*). This hybrid strategy ensures granular control in detecting key anatomical structures and provides clinically interpretable indicators that are well-suited for predictive models, thereby optimizing automated risk assessment of peri-implantitis.

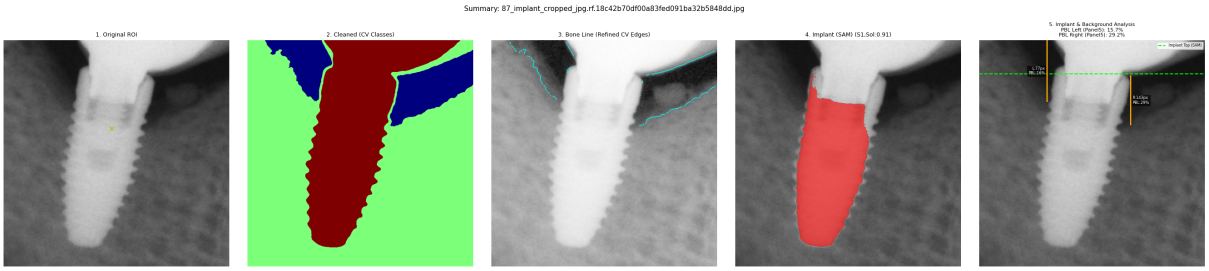


Figure 26: Visual example of the hybrid method: implant segmentation, detected bone line, and localized bone defect regions (mesial and distal).

Method 2: Estimation via Deep Segmentation (DCNN)

This second analysis pathway quantifies bone loss from a different perspective. Rather than inferring loss from edges or voids, it relies on the semantic intelligence of a deep segmentation model (DCNN)—either U-Net or Mask R-CNN—as described in Sections 3.4.1 and 3.4.2. After training, the DCNN produces a multiclass map that explicitly identifies regions of interest (implant, healthy bone/gum, and *boneloss* as a separate class). From these masks, quantitative indicators are extracted, providing a richer profile of the defect than what is possible with a single-type measurement.

1. Multiclass Semantic Segmentation The starting point is the DCNN's predictive mask for a given ROI. Each pixel is assigned a label (e.g., 0 = background, 1 = bone loss, 2 = bone/gum, 3 = implant). From this map, binary masks are isolated per class, forming the basis for all subsequent measurements.

2. Establishment of the Geometric Reference System To ensure consistent and normalized measurements, it is essential to define a stable coordinate system and reference scale for each ROI. The implant mask is chosen as the **geometric anchor** for this purpose, given its rigid structure and typically well-defined contours, in contrast to the morphological variability of surrounding biological tissues. From the `mask_implant`, the following key references are extracted:

- **Main Implant Component:** The largest connected component is isolated. This step ensures that the analysis focuses exclusively on the implant of interest, discarding potential spurious segmentations or metallic artifacts that the model may have misclassified.
- **Mesial–Distal Axis:** The centroid of the main implant component, (c_x^{implant}), is computed. Its horizontal coordinate defines a vertical axis that divides the ROI into a mesial (left) and distal (right) region, allowing localized and differentiated analysis of bone loss on each side of the implant.

- **Normalization Scale ($L_{\mathbf{imp}}$):** The height of the bounding box of the main component is defined as the apparent total length of the implant, L_{imp}. This value is crucial as it serves as the intrinsic reference scale for the image, enabling normalization of all height and area measurements so they can be expressed as percentages and thus be comparable across patients and implants of different real-world sizes.

3. Dual Quantification of Bone Loss

a) Vertical Depth Metric $\mathbf{PBL_{vertical}}$ Captures the maximum severity of the defect.

- **Primary method (crest-based).** For each side of the implant, the *boneloss* / bone-gum interface is located. The most coronal point is defined as the bone crest level y_{crest}; the most apical point of the *boneloss* region is the defect bottom y_{fondo}. The defect height is computed as:

$$H_{\text{defect}} = y_{\text{background}} - y_{\text{crest}}.$$

- **Fallback method (implant-top based).** If the above interface is unreliable, the top of the implant is used as the coronal reference; this is recorded as it may introduce higher uncertainty.

b) Area-Based Extension Metric $\mathbf{PBL_{area}}$ Robust against horizontal or "bowl-shaped" defects. It is defined as:

$$\text{PBL}_{\text{area}} = \frac{\text{Area(boneloss)}}{\text{Area(implant)}} \times 100.$$

The combination $\text{PBL}_{\text{vertical}}$ + PBL_{area} provides a more complete two-dimensional characterization of the peri-implant defect.

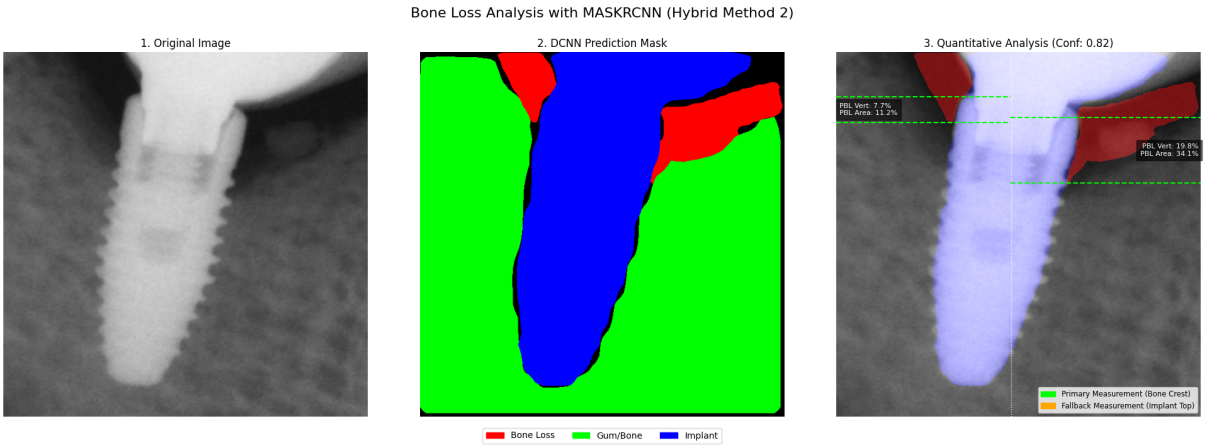


Figure 27: Workflow for dual quantification (vertical and area-based) of bone loss from semantic masks generated by a DCNN. Key reference points are illustrated: bone crest (y_{crest}), defect bottom ($y_{\text{background}}$), and the areas of *boneloss* and implant.

4. Confidence Metric The reliability of each measurement depends directly on the quality of the DCNN segmentation. For this reason, a composite confidence metric is defined as $c \in [0, 1]$, computed from the following factors:

- **Geometric coherence.** This is primarily based on the *solidity* of the implant mask: compact and morphologically plausible segmentations contribute to a high c value. Additionally, cases where the bone-loss mask is excessively fragmented are penalized.
- **Measurement method reliability.** The value of c is modulated depending on the vertical measurement method used. Measurements based on the detected bone crest (`crest_based`) are considered highly reliable; if the system resorts to the fallback method (`implant_top_backup`), the confidence score is attenuated to reflect greater uncertainty.
- **Model predictive confidence (optional).** When softmax outputs are available, c can be weighted by the average probability with which the DCNN classified pixels as implant or bone loss, thereby incorporating the model's internal certainty.

Feature Export for Predictive Analysis

Regardless of the quantification method used (hybrid or DCNN), the results of each analyzed image are consolidated into a unified feature vector. This vector contains bone-loss metrics, confidence indicators, and relevant geometric descriptors.

All these variables are exported for integration into predictive models. This standardized format facilitates direct comparison of both methods' performance and serves as input for the peri-implantitis risk prediction model.

Final remark: A systematic comparison between both methods will allow the evaluation of advantages in terms of robustness, clinical interpretability, and ease of integration into an automated pipeline. It will also help determine which method generalizes better to images of varying quality or with the presence of artifacts.

This comprehensive registration approach provides critical groundwork for integrating medical images effectively, supporting advanced diagnostic and prognostic tasks such as the assessment of bone loss in dental implants and other detailed analyses relevant to clinical practice.

3.6. Regression

The final stage of this project focuses on developing a prognostic model to predict the longitudinal progression of peri-implant bone loss (%PBL). To achieve this, a multi-stage experimental framework was designed, encompassing a hybrid real-to-synthetic data generation process and a robust validation strategy.

Experimental Design and Timeline

To standardise follow-up, three clinical milestones were defined (Fig. 28). At surgery (T_0), baseline bone loss is set to $\mathbf{PBL = 0\,\%}$. While an initial physiological bone remodelling of approximately 0.5–1.0 mm during the first year is well-documented [31, 32], establishing a common zero baseline is a necessary methodological simplification. This approach allows the model to learn any pathological deviation that occurs after this initial healing and adaptation phase.

At the one-year follow-up (T_1), peri-implant bone loss (PBL_1) is quantified from real radiographs using the computer vision techniques described previously on this section. This measurements, along with the other radiographic indicators, become the key predictors in the prognostic model. The model's target variable is the absolute PBL at future follow-up points, such as T_2 (2 years), T_5 (5 years) or T_{10} (10 years).

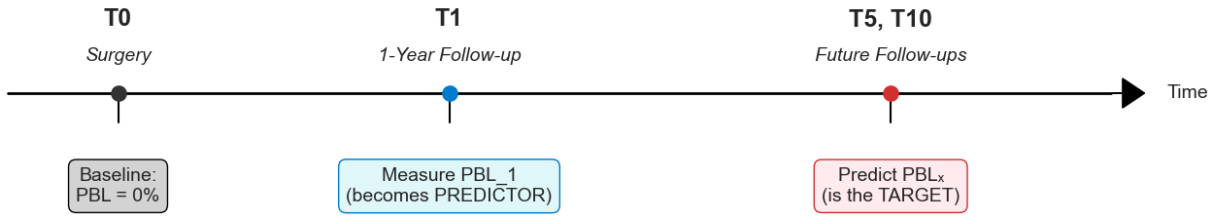


Figure 28: Chronology of the longitudinal experiment. Radiographic measurements acquired at T_1 become *predictors*, whereas the increment $\Delta\mathrm{PBL}_t = \mathrm{PBL}_t - \mathrm{PBL}_1$ at $t = \{2, 5, 10\}$ years is the *target*.

Two–stage architecture

The prognostic pipeline is built upon a two-stage architecture that systematically separates radiographic analysis from longitudinal prediction.

The first stage, Stage A (Quantification), is responsible for processing the raw radiographic image at a given time point T_x. It employs a Deep Convolutional Neural Network (DCNN) to segment the implant, surrounding bone, and any existing defect regions. The output of this stage is a quantitative metrics vector:

$$\mathbf{z} = [\mathrm{PBL}_{\mathrm{vertical}},\ \mathrm{PBL}_{\mathrm{area}},\ c],$$

where c represents the confidence score of the automated measurement.

The output of this initial stage then becomes a critical input for Stage B (Prognosis). This second stage uses a machine learning regression model. It receives the radiographic metrics vector from the first-year follow-up, $\mathbf{z}_{T_1}$, combines it with the patient's baseline clinical variables $\mathbf{x}_{\mathrm{clin}}$, and its objective is to estimate the absolute bone loss PBL_t at future horizons, such as 2, 5, or 10 years.

This architecture ensures that the complex task of image interpretation is handled separately before the data is integrated for risk prediction.

Synthetic data and ranking fusion

A fully paired longitudinal dataset is not yet available. Therefore a *synthetic cohort* was generated:

1. A cohort of clinical profiles ($n = 160$) was generated as detailed in Section 3.2. These profiles were statistically sampled to reflect published distributions of key risk factors such as age, smoking status, diabetes, and oral hygiene.

2. Radiographic metrics ($n = 90$) were taken from the automatic DCNN applied to holder radiographs.

3. Profiles were sorted by a *Risk Index* (ρ_{clin}) and radiographs by a *Severity Index* (first principal component of PBL metrics, ρ_{rad}). The ordered lists were joined pair-wise, yielding $N = 75$ synthetically coherent patient – image pairs (Spearman $\rho = 0{,}63$).[1]

With the baseline at T_1 established, the longitudinal progression was subsequently simulated. The annual bone loss slope ($\Delta\mathrm{PBL}_{\mathrm{year}}$) was modelled as a function of a baseline rate modulated by the patient's risk factors, with a stochastic component to account for biological variability:

[1]The procedure recreates the empirical association "higher systemic risk → greater baseline destruction", solely for technical validation. No clinical inference is claimed.

$$\Delta\mathrm{PBL}_{\mathrm{year}} = 0{,}8\,\% \times (1{+}0{,}20 \cdot \mathrm{smoker})(1{+}0{,}30 \cdot \mathrm{diabetic}) \; + \; \mathcal{N}\Big(0,\; 0{,}30\sqrt{t}\Big).$$

This approach, where only the slope is modulated by risk factors, prevents information leakage from the T_1 baseline measurement to the future progression rate.

Regression models and validation

The target was $\mathrm{PBL}_5\,\%$ ($t = 5$ years). The predictor set was defined as

$$\mathbf{x} = \{\texttt{PBL_area_T1}, \mathrm{confidence}, \mathrm{clinical\ factors}\}$$

Two algorithms were compared:

- **Ridge regression** (linear, ℓ_2-regularised).
- **Gradient Boosting** (GBM, 100 estimators, default learning rate).

A *GroupKFold* (5 folds, grouping by patient) was used to prevent same-subject leakage. Model performance was evaluated using MAE and RMSE. Confidence intervals at 95 % were estimated via bootstrap resampling ($B = 2000$) on a hold-out test set generated with *GroupShuffleSplit* (using 25 % of patients).

4. Results

This chapter presents the empirical and quantitative findings derived from the application of the described methodology.

4.1. Dental Implant Detection

The performance of the two primary detection architectures, YOLOv8 and Faster R-CNN, was evaluated using a comprehensive set of metrics. This section details these results, provides a comparative analysis, and examines the nature of detection errors to justify the selection of a final model for the downstream analysis pipeline.

Quantitative Performance of Detection Models

The various configurations of YOLOv8 and the ablation studies for Faster R-CNN were trained and evaluated, with their detailed performance metrics presented in the Methodology section (Table 2 and Table 4, respectively). For a direct comparative analysis, the performance of the best-performing configuration for each architecture is summarized in Table 7. The YOLOv8s model was selected from the YOLO family for its balance of high recall and efficiency, while the Run4 configuration of Faster R-CNN was chosen for achieving the highest precision among its peers.

Model	**mAP$_{50}$**	**mAP$_{50:95}$**	**Precision**	**Recall**	**Inference Time (ms)**
YOLOv8s	0.849	0.434	0.569	**1.000**	**14.8**
Faster R-CNN (Run4)	**0.929**	**0.484**	**0.843**†	0.831†	150

Table 7: Comparative performance summary of the best YOLOv8 and Faster R-CNN models on the test set. †Metrics derived from the foreground classification accuracy reported in the training analysis.

The results clearly indicate a trade-off between detection accuracy and inference speed. The premier Faster R-CNN model (Run4) achieves a significantly higher mean Average Precision at an IoU threshold of 0.50 (AP_{50} = 0,929) compared to the best YOLOv8 variant, YOLOv8s (mAP_{50} = 0,849). This 9-point lead in AP_{50} underscores its superior ability to correctly identify and localize implants with a standard degree of bounding box overlap. While YOLOv8s achieves perfect recall—identifying every ground-truth implant—it does so at the cost of lower precision (0.569), suggesting a higher rate of false positives. Conversely, YOLOv8s demonstrates a tenfold advantage in speed, with an average inference time of just 14.8 ms per image versus 150 ms for Faster R-CNN, highlighting its suitability for real-time applications.

Comparative Analysis and Error Examination

Beyond aggregate metrics, a qualitative analysis of the models' predictions reveals critical differences in their real-world performance. This examination also helps to identify the root causes of detection errors, such as false positives and false negatives.

Visual Examples of Detections To understand the practical implications of the quantitative differences, Figure 29 provides a qualitative comparison on representative test cases. The visual evidence highlights the superior robustness of Faster R-CNN in this specific clinical context, correctly identifying implants in challenging situations where YOLOv8s fails.

Analysis of False Positives and Negatives The detection failures observed visually can be attributed to the architectural differences between the models and the inherent challenges of radiographic imagery.

- **False Negatives:** YOLOv8s is more susceptible to producing false negatives, particularly in cases of low radiographic contrast or when implants are truncated at the image border (Figure 29). Faster R-CNN's Region Proposal Network (RPN), which generates candidate boxes at multiple scales and aspect ratios, provides greater robustness against these partial views. Furthermore, its ResNet-FPN backbone is more effective at learning stable features under the strong exposure variations typical of intraoral imaging.
- **False Positives and Localization Errors:** While achieving high recall, YOLOv8s occasionally produces false positives. A characteristic error is the prediction of two overlapping boxes for a single, elongated implant (Figure 29) or placing boxes on confounding structures like restorative materials in noisy images (Figure 29). This behavior may be linked to its anchor-free detection head, which can struggle with objects of unusual aspect ratios not well-represented in its training experience. In contrast, the two-stage process of Faster R-CNN—proposing regions first and then classifying them—combined with RoIAlign's precise feature extraction, results in more accurate localization and fewer spurious detections.

In summary, as seen in Section 3.3.3 the two-stage architecture of Faster R-CNN, leveraging a powerful feature extractor and a dedicated proposal mechanism, provides greater resilience to the visual variability inherent in periapical radiographs. This leads to a higher overall detection accuracy (AP_{50}) and fewer clinically significant errors compared to the faster but less robust single-stage YOLOv8s. These findings establish Faster R-CNN as the more reliable detector for the initial stage of the automated peri-implantitis analysis pipeline.

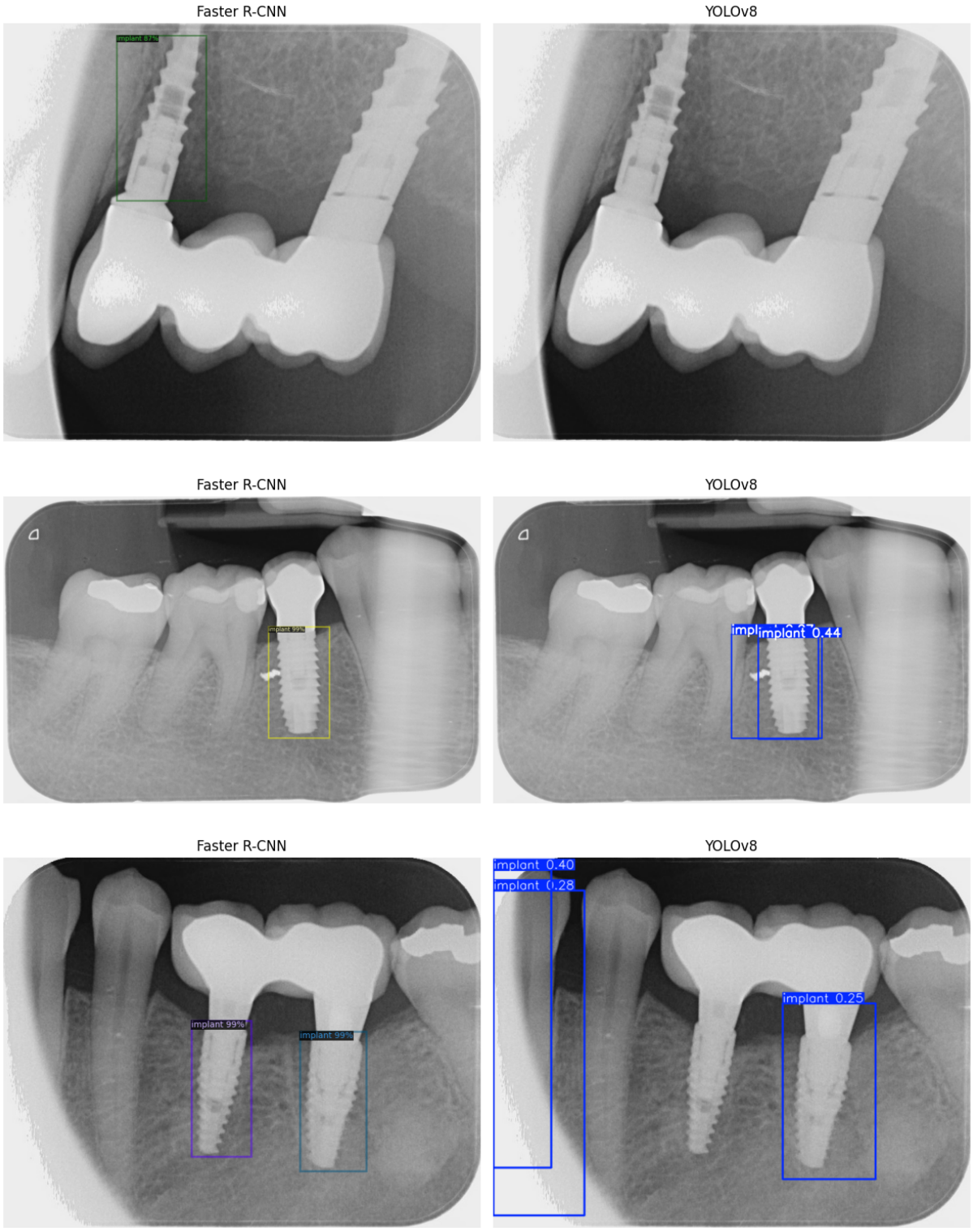


Figure 29: Qualitative comparison of YOLOv8s and Faster R-CNN Run4. Faster R-CNN demonstrates superior performance in challenging, easy, and noisy cases.

4.2. Segmentations

Following the detection of implant Regions of Interest (ROIs), the next critical stage involves the precise segmentation of key anatomical structures: the implant body, surrounding bone, and gum tissue. This section presents the results obtained from three distinct approaches: semantic segmentation using U-Net, instance segmentation with Mask R-CNN, and a hybrid pipeline combining the Segment Anything Model (SAM) with classical morphological techniques. The evaluation is based on quantitative metrics (Dice, IoU), qualitative visual assessment, and an analysis of each method's suitability for the final goal of bone loss measurement.

Performance of Deep Learning Segmentation Models: U-Net and Mask R-CNN

The primary deep learning models, U-Net and Mask R-CNN, were trained to perform multi-class segmentation. While detailed results from extensive hyperparameter tuning and ablation studies are available in the Methodology (Tables 5 and 6), a direct comparison of the best-performing configuration for each architecture is essential for evaluation.

Quantitative Results Table 8 consolidates the performance of the optimal U-Net model (UNET_BS8_E100) and the optimal Mask R-CNN model (MaskRCNN_R101_E100) on the hold-out test set.

Model Architecture	Encoder	Test IoU	Test Dice	Training Duration (s)
U-Net	ResNet-34	**0.7973**	**0.8801**	**331.39**
Mask R-CNN	ResNet-101-FPN	0.7844	0.8540	2090.12

Table 8: Comparative performance of the best U-Net and Mask R-CNN segmentation models. Bold values indicate the leading score for each metric.

The quantitative results show a clear advantage for the U-Net architecture. The UNET_BS8_E100 configuration achieved a superior Dice coefficient of 0.8801 and an IoU of 0.7973, outperforming the best Mask R-CNN model. Furthermore, the U-Net model demonstrated remarkable computational efficiency, completing its training over six times faster than its Mask R-CNN counterpart. This performance suggests that for the task of semantic segmentation on this dataset, the U-Net architecture with a ResNet-34 encoder provides a more effective and efficient solution.

Qualitative and Per-Class Performance Visual inspection of the segmentation masks generated by both models offers deeper insights into their behavior. As shown in the examples provided in the Methodology section (Figures 14 and 17), both models produce high-quality, anatomically coherent segmentations.

- **U-Net** consistently generates smooth and continuous contours that often appear to refine the original ground-truth annotations, which can have coarse, polygonal boundaries. This suggests the model learns robust representations of anatomical shapes, enabling it to generalize beyond the exact pixel-level labels.
- **Mask R-CNN**, while also highly accurate, tends to adhere more strictly to the provided ground-truth masks. This is evident in cases where it correctly ignores clinically plausible bone loss areas that were not annotated in the ground truth, showcasing its fidelity to the training data.

Analysis of the confusion matrices (Figures 15 and 18) reveals similar per-class performance profiles. Both models excel at identifying the *Implant* and *Gum/Bone* classes, which are well-defined. The majority of classification errors occur in the *Boneloss* class, which is often misclassified as adjacent *Gum* or *Background*. This confusion is clinically understandable, as bone loss regions are characterized by diffuse boundaries and subtle texture changes, making them inherently challenging to segment precisely.

Evaluation of the Hybrid Approach: SAM and Morphological Filtering

An alternative to end-to-end deep learning segmentation was evaluated through a hybrid pipeline. This approach leverages the zero-shot capabilities of the Segment Anything Model (SAM) for precise implant segmentation, followed by classical computer vision techniques to identify the bone line. This method is not evaluated with global Dice/IoU scores, as its goal is not to produce a full multi-class semantic mask but to extract specific, clinically relevant contours.

SAM for Zero-Shot Implant Segmentation The results from applying SAM were primarily qualitative, demonstrating a key strategic advantage over trained models. Using an automatically generated point prompt, SAM proved capable of segmenting the dental implant with high precision. Its most significant contribution is the ability to distinguish and isolate the **endosseous portion** of the implant (the threaded body integrated into the bone) from the prosthetic crown (Figure 19).

As illustrated in Figure 20, classical methods like Multi-Otsu tend to merge all high-density metallic components into a single mask. In contrast, SAM's inherent understanding of object atomicity allows it to deliver a semantically correct segmentation of only the clinically relevant part of the implant. This is a crucial result, as accurate bone loss measurements depend on landmarks located on the implant body, not the crown.

Classical Techniques for Bone Line Extraction The subsequent application of a robust pipeline based on adaptive thresholding, Multi-Otsu segmentation, and conditioned morphological gradients successfully identified the peri-implant bone line (Figure 24). The

result is not a segmentation mask but a precise contour representing the bone-to-soft-tissue interface. This method proved effective at suppressing spurious edges from metallic artifacts, delivering a clean and interpretable line for quantitative analysis.

Comparative Summary of Segmentation Approaches

Each evaluated technique presents a unique set of strengths and trade-offs, summarized in Table 9.

Approach	Primary Task	Key Metric	Strengths	Weaknesses
U-Net	Semantic Segmentation (Multi-class)	**Dice = 0.8801**	Highest accuracy, very fast training, produces smooth/anatomical contours.	Segments all classes jointly; may not distinguish implant substructures.
Mask R-CNN	Instance Segmentation (Multi-class)	Dice = 0.8540	High accuracy, separates individual object instances.	Computationally expensive, performance highly dependent on ground-truth quality.
Hybrid Pipeline (SAM + Classical)	Targeted Contour Extraction	Qualitative (Semantic Correctness)	Zero-shot segmentation of implant body (excludes crown), highly interpretable bone line extraction.	No single-step multi-class output, SAM can fail in low-contrast cases.

Table 9: Qualitative and quantitative comparison of the evaluated segmentation approaches.

In conclusion, the results indicate that for generating a complete, high-quality semantic segmentation of the entire ROI, U-Net is the superior method, offering the best balance of accuracy and computational efficiency. However, the hybrid pipeline provides a decisive advantage for the specific clinical task of measuring bone loss. Its ability to isolate the endosseous implant body via SAM, without requiring specific training annotations for that substructure, represents a significant practical breakthrough. This targeted approach yields a more clinically relevant segmentation for the subsequent measurement steps than what is achievable with standard semantic segmentation models.

4.3. Computation of the Percentage of Bone Loss (PBL %)

This section presents and analyzes the quantitative results obtained by applying different methodologies for estimating marginal bone loss (PBL) on a set of 50 periapical radiographs. The analysis is structured around three main axes: the operational robustness of each method, the concordance and agreement between their measurements, and the internal relationship between the dual quantification metrics. Given the limited sample size ($N = 50$), proportions are reported along with their respective confidence intervals to reflect the uncertainty of the estimates.

Robustness and Success Rate Analysis

Robustness was assessed as the success rate of each method, defined as the percentage of images for which a valid measurement was obtained. The results, shown in Table 10, indicate an overall high performance.

The methods based on deep convolutional neural networks (DCNNs) proved to be the most robust. Notably, the area-based metric (PBL area) from U-Net achieved a 100 % success rate (95 % CI: 92.9 % – 100.0 %), successfully processing all images. Close behind were the area-based metric from Mask R-CNN and the vertical metric from U-Net, both with a 98.0 % success rate. The hybrid method (CV+SAM) and the vertical metric from Mask R-CNN showed a slightly lower robustness at 96.0 %. This may be attributed to the fact that sequential pipelines based on classical vision are potentially more sensitive to contrast or noise variations than DCNN architectures, which learn semantic representations of structures.

Quantification Method	**Successes / Total**	**Success Rate (%)**	**95 % CI**
U-Net (Area)	50 / 50	100.0	92.9 % – 100.0 %
U-Net (Vertical)	49 / 50	98.0	89.5 % – 99.6 %
Mask R-CNN (Area)	49 / 50	98.0	89.5 % – 99.6 %
Hybrid (CV+SAM)	48 / 50	96.0	86.5 % – 98.9 %
Mask R-CNN (Vertical)	48 / 50	96.0	86.5 % – 98.9 %

Table 10: Success rate and 95 % Confidence Interval (Wilson method) for each methodology, evaluated on $N = 50$ radiographs.

Concordance and Agreement Between Methods

To evaluate consistency among the estimates, we analyzed the subset of 45 images in which all three methods (Hybrid, Mask R-CNN, and U-Net) produced a valid vertical PBL measurement.

The Pearson correlation analysis (Table 11) revealed a positive and statistically significant linear association ($p < 0{,}001$) among all approaches. The strongest correlation was observed between the two DCNN models (Mask R-CNN vs. U-Net, $r = 0{,}79$), suggesting that both architectures converge toward a similar semantic interpretation of the defect. The hybrid method also showed good correlation with the DCNN models (with Mask R-CNN, $r = 0{,}73$; with U-Net, $r = 0{,}64$).

Method	Hybrid	Mask R-CNN	U-Net
Hybrid	1.00	0.73	0.64
Mask R-CNN	0.73	1.00	0.79
U-Net	0.64	0.79	1.00

Table 11: Pearson correlation matrix (r) for vertical PBL measurements. All coefficients are statistically significant ($p < 0{,}001$).

This relationship is visualized in the scatter plots in Figure 30, where the point cloud between Mask R-CNN and U-Net clusters most closely around the identity line ($y = x$), confirming their high consistency.

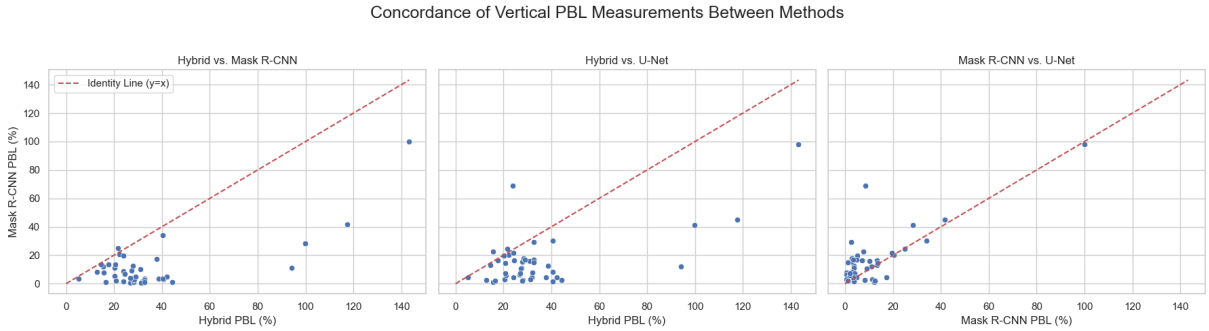


Figure 30: Scatter plots comparing vertical PBL measurements (%) between method pairs. The dashed red line represents perfect concordance ($y = x$).

However, correlation does not measure agreement. To address this, a Bland-Altman analysis was conducted (Figure 31) to quantify systematic bias and agreement limits. The analysis between Mask R-CNN and U-Net showed a mean bias close to zero, indicating the absence of systematic bias and good agreement. In contrast, the hybrid method exhibited a slight bias, confirming conceptual differences in how each technique quantifies the defect.

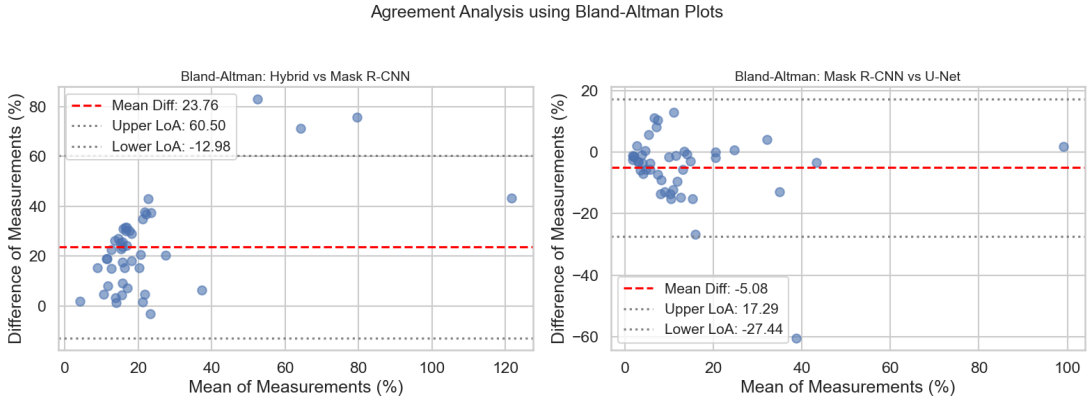


Figure 31: Bland-Altman plots assessing agreement between method pairs. The plot shows the measurement difference against their mean, with the mean bias (dashed line) and 95 % limits of agreement (dotted lines).

Validation of Dual Quantification: Correlation Between Vertical and Area-Based PBL

A key finding of this study arises from the analysis of the relationship between the two metrics proposed for DCNN models: vertical PBL and area-based PBL. For the Mask R-CNN model, the Pearson correlation coefficient was not statistically significant ($r = 0{,}078$, $p = 0{,}598$).

This value, extremely close to zero, indicates that there is no predictable linear relationship between the depth and area of the bone defect (see Figure 32). This result is highly relevant, as it empirically validates the hypothesis that PBL is a morphologically complex phenomenon that cannot be characterized by a single measure. The existence of deep but narrow defects ("pit" type) and others that are shallow but wide ("crater" type) justifies the need for dual quantification. Both metrics, being statistically independent, provide complementary and non-redundant information.

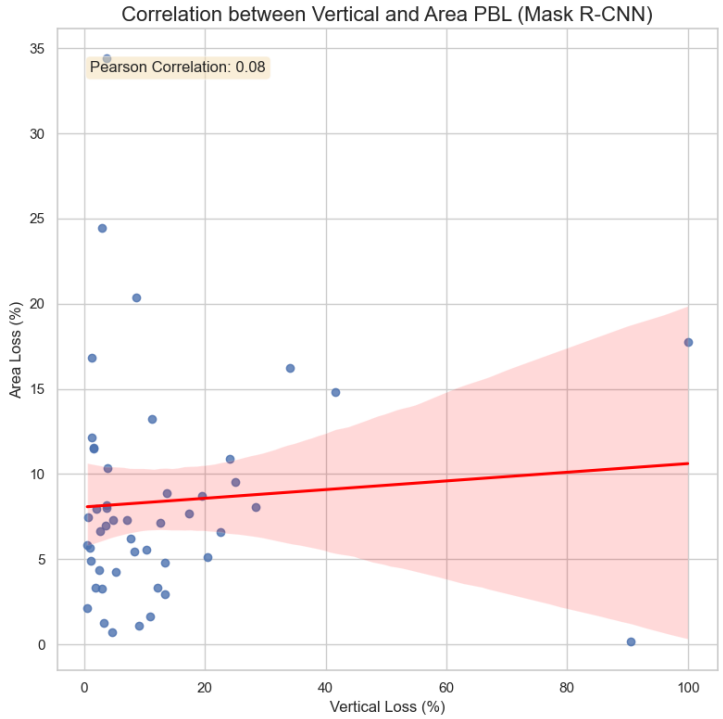
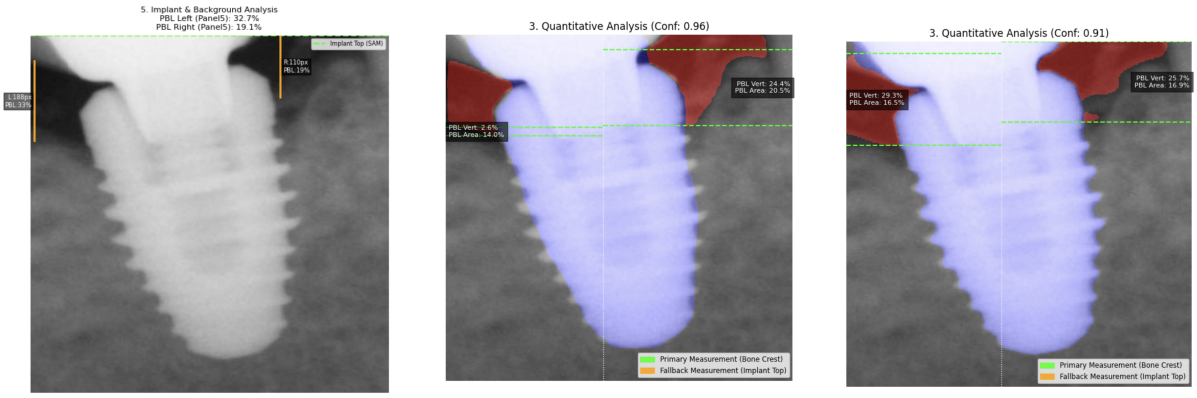


Figure 32: Scatter plot between vertical and area-based PBL for the Mask R-CNN model. The absence of a clear pattern and the correlation coefficient near zero ($r = 0{,}078$) demonstrate the independence of both metrics.

Case Study: Qualitative Analysis in a Representative ROI

To complement the statistical analysis, a detailed case study was conducted on a representative radiograph (Figure 33). This analysis allows visual comparison of how each technique interprets peri-implant anatomy and defects.



(a) Hybrid Method (CV+SAM) (b) DCNN Method (Mask R-CNN) (c) DCNN Method (U-Net)

Figure 33: Visual comparison of PBL quantification for the same ROI. (a) The hybrid method detects vertical defects. (b) and (c) DCNN models generate semantic masks where the defect (red) is an explicit class.

The numerical results for this case (Table 12) illustrate the statistical findings. Vertical PBL measurements from Mask R-CNN (2.6 %) and U-Net (29.3 %) on the left side differ significantly, exemplifying the low observed correlation. The most revealing point is the comparison of Mask R-CNN metrics: a very low vertical loss of 2.6 % corresponds to an area-based loss of 6.7 %. This mismatch is a practical demonstration of the low correlation ($r = 0{,}078$) and visually validates the need for dual quantification. The defect in this case is shallow but extensive, a morphological feature that would be missed if only one of the two metrics were used.

Table 12: Quantitative PBL results (%) for the example image according to each method.

Metric	**Hybrid (CV+SAM)**	**Mask R-CNN**	**U-Net**
PBL Vertical Left (%)	32.6	2.6	29.3
PBL Vertical Right (%)	19.1	24.4	25.7
PBL Area Left (%)	—	6.7	7.8
PBL Area Right (%)	—	10.8	8.9

Method Limitations and Mitigation Strategies

Despite its conceptual robustness, the effectiveness of the DCNN-based quantification method is subject to certain inherent limitations. Identifying and addressing these limitations is crucial for proper interpretation of results and for designing a reliable system. Below, the two main detected limitations and the strategies implemented to mitigate them are detailed.

1. Dependency on the Morphology of the Bone Defect The vertical depth metric ($PBL_{vertical}$), which measures the maximum distance between the crest and the bottom of the defect, is highly effective in quantifying angular or crater-shaped defects. However, its value may be deceptively low in cases of horizontal bone recession or in "bowl"-shaped defects, where the overall volume of bone loss is significant but the maximum vertical depth is reduced.

- **Mitigation Strategy (Dual Quantification):** To overcome this limitation, the method does not rely on a single metric. A dual quantification strategy was implemented that complements the vertical depth measurement with an area-based extension metric (PBL_{area}). The latter is insensitive to the slope of the defect and effectively captures the extent of bone loss in terms of surface area. By providing both metrics to the final regression model, it is allowed to learn which indicator (or combination of both) is more predictive for each type of clinical scenario, creating a much more comprehensive and robust evaluation system.

2. Critical Dependency on DCNN Segmentation Quality The reliability of the entire pipeline of Method 2 directly depends on the quality of the segmentation produced by the DCNN model. A prediction failure can completely invalidate the metric extraction process. Failure modes include:

- Failure to detect an essential class (e.g., missing the "implant" mask).
- Anatomically incoherent segmentation (e.g., a "bone loss" mask fragmented into multiple small and noisy regions).
- Lack of a clear interface (collision zone) between the "bone loss" and "bone/gum" classes, preventing accurate bone crest calculation.
- **Mitigation Strategies (Robustness and Safeguards):** Several mechanisms have been integrated into the pipeline to manage these failures in a controlled manner:
    1. **Geometric Validity Filters:** The system verifies that the segmented masks exceed minimum area thresholds (e.g., `MIN_IMPLANT_AREA`, `MIN_BONELOSS_AREA_PER_SIDE`). This prevents the analysis from proceeding based on spurious segmentations or noise, which lack clinical significance.
    2. **Fallback Measurement Mechanism:** As previously described, if the primary `crest_based` method fails, the system does not halt. It automatically activates a fallback method (`implant_top_backup1`). Crucially, the method used is logged alongside the results, allowing subsequent stratification or weighting of the measurements based on their source reliability.

3. **Explicit Confidence Metric:** The system computes a confidence score (c) that encapsulates the quality of the input. A segmentation that requires fallback use or that yields geometrically weak masks results in a low confidence score. This allows the final regression model—or a clinical supervisor—to weight or even discard measurements considered unreliable.

By combining dual metrics with robustness mechanisms, Method 2 is designed not only to quantify bone loss but also to be aware of the limits of its own reliability—an essential feature for potential application in clinical settings.

Discussion of Findings and Clinical Relevance

The results demonstrate the feasibility of quantifying PBL in an automated and robust manner. DCNN models—particularly U-Net—emerge as the most consistent option. The empirical validation of the independence between vertical and area-based PBL metrics is a key finding, as it suggests that future peri-implantitis risk models should incorporate both for a complete characterization of the disease phenotype.

While this work validates the measurement techniques, the next crucial step will be clinical validation. Future research should correlate these automated metrics with standard clinical parameters (e.g., probing depth, bleeding on probing) to establish risk thresholds and determine their prognostic value in peri-implantitis progression, thereby closing the loop from feature engineering to real-world clinical application.

4.4. Regression results

The regression analysis was extended to multiple temporal horizons (2, 5, and 10 years) to rigorously assess the robustness of the methodological framework and to understand the nature of the prediction problem within our hybrid *Real-to-Synthetic* environment.

Multi-Horizon Predictive Performance

The predictive performance of the optimised Ridge regression model was compared against a baseline "Null Model", which projects future peri-implant bone loss (PBL) using the mean progression slope of the entire population. Table 13 summarises the cross-validated Mean Absolute Error (MAE) for both models across all time horizons.

The results yielded a consistent and revealing outcome: the Null Model achieved a slightly lower MAE than the Ridge regression model across all horizons. Consequently, the "Improvement vs. Nul" metric was consistently negative, indicating that the machine learning approach did not manage to outperform the simple, mean-based projection in this simulated setting.

Horizon	Best Model	MAE (CV) [%]	Null Model MAE [%]	Abl. MAE (w/o PBL_1) [%]	Gain vs. Null [%]
2-year	Ridge	0.2867	0.1857	4.4856	-54.4
5-year	Ridge	0.5593	0.4814	4.5548	-16.2
10-year	Ridge	0.9145	0.7797	4.6566	-17.3

Table 13: Summary of multi-horizon prediction performance. The Ridge model is compared against a baseline Null Model. Ablation MAE shows the error when the baseline radiographic feature (PBL_1) is removed. The improvement metric indicates the Ridge model's performance relative to the null baseline.

Robustness and Interpretability Checks

To dissect these results, a series of robustness and interpretability checks were performed.

Ablation Analysis: The Primacy of Baseline PBL_1 Ablation studies irrefutably confirmed that the implant's current state (PBL_1) is the cornerstone of any prognosis. Removing the baseline radiographic metric as a predictor led to a dramatic degradation in model performance. As detailed in Table 13, the MAE increased by a staggering $\mathbf{+1464\,\%}$ for the 2-year prediction, $\mathbf{+714\,\%}$ for the 5-year, and $\mathbf{+409\,\%}$ for the 10-year horizon. This quantifies the overwhelming predictive value contained within the initial radiograph at T_1.

Interpretability (SHAP Analysis) SHAP (SHapley Additive exPlanations) analysis of the standardised Ridge coefficients corroborated the ablation findings. The model-agnostic feature importance hierarchy was ranked as follows:

$$PBL_{1,\text{area}} \gg \text{smoking status} \approx \text{diabetes} \gg \text{age and oral hygiene}$$

This hierarchy is perfectly consistent with the risk multipliers embedded in the synthetic data generator, confirming that the model successfully learned the underlying data structure and clinical logic as intended.

Discussion

The primary objective of this work—to develop and validate a complete *in-silico* pipeline for PBL prognosis—has been successfully achieved. The findings, particularly the superior performance of the Null Model, do not signify a failure of the machine learning approach. Rather, they represent the most salient outcome of the study, providing a deep insight into the nature of the simulated data and validating the integrity of our methodological framework.

Linear Dominance in a Synthetic Environment The outperformance of the Null Model is a direct consequence of the simulation's design. The synthetic longitudinal progression, defined as $PBL_t = PBL_1 + \Delta PBL$, is exceptionally coherent and governed by a dominant linear logic. A simple model that perfectly replicates this logic (e.g., $PBL_5 = PBL_1 + \text{mean_slope} \times 4$) is, by definition, exceptionally effective. The Ridge model, in its attempt to learn and weigh secondary features (age, smoking, etc.), introduces a minimal amount of predictive noise, which prevents it from matching the mathematical optimality of the Null Model *in this specific simulated context*. The ability of our analysis to reveal this characteristic is a validation of the framework itself.

The Widening Cone of Uncertainty Our framework correctly captured a fundamental clinical reality: predictive uncertainty grows with time. The progressive increase in MAE for all models across the 2, 5, and 10-year horizons (Table 13) illustrates the natural widening of the "cone of uncertainty" when forecasting further into the future. This confirms the model's ability to reflect the intrinsic difficulty of long-term clinical prediction.

Conclusion and Future Directions This project delivers not merely a predictive model, but a fully validated methodological **proof-of-concept**. We have demonstrated the technical viability of the proposed architecture, from automated image analysis to risk prediction. The key limitations of the current study—the enforced risk-severity correlation from synthetic pairing and the assumption of linear progression—are acknowledged. Real peri-implant bone loss is often episodic and non-linear.

The definitive next step is to apply this tested and debugged pipeline to a large-scale retrospective cohort with true, multi-year longitudinal data. It is in that scenario, characterized by the inherent complexity, noise, and non-linearity of real-world biological processes, where we hypothesise that sophisticated machine learning models will demonstrate a clear and clinically significant superiority over simple baseline projections.

4.5. Real Application to HOUB Data via Active Learning

A primary objective of this project is to develop a robust segmentation model applicable to real-world clinical data, such as the periapical radiographs provided by the Hospital Odontològic de la Universitat de Barcelona (HOUB). As detailed in Section 3.2, the availability of labeled data from HOUB was limited to 25 non-annotated example images, which is insufficient for supervised training or fine-tuning.

Our initial experiments in Section 3.4.1 identified the `UNET_BS8_E100` model (a U-Net with a ResNet-34 encoder, trained for 15 epochs with a batch size of 8 and a learning rate of 1×10^{-3}) as our best-performing baseline. However, when this model was directly applied to the new, unlabeled images from HOUB, we observed a significant drop in

performance compared to its results on our original test set. This domain shift problem is expected, as the model was not trained on the specific characteristics of the HOUB dataset.

To address this challenge without requiring manual annotation, we devised an Active Learning strategy. This approach uniquely leverages our classical computer vision pipeline, described in Section 3.4.4, as an automated "oracle" to generate pseudo-labels for the most challenging new images.

Methodology: Active Learning with Pseudo-Labeling

The core idea is to iteratively improve our baseline model by teaching it how to handle the HOUB data it finds most confusing. The cycle consists of three main steps:

1. **Uncertainty Sampling:** The pre-trained `UNET_BS8_E100` model is used to predict segmentation masks for the entire pool of unlabeled HOUB images. For each prediction, we calculate the model's uncertainty using pixel-wise entropy. A high entropy score indicates that the model is unsure about its classification for many pixels, making it a prime candidate for retraining. Figure 34 shows an example of an image that yielded high uncertainty.

2. **Pseudo-Label Generation:** The images with the highest uncertainty are selected. Instead of manual labeling, we process these images through our classical CV pipeline (Section 3.4.4) to automatically generate segmentation masks. These masks, termed "pseudo-labels", serve as the ground truth for this new batch of data. Figure 35 illustrates this process, showing an uncertain image and its corresponding generated pseudo-label.

3. **Model Fine-Tuning:** The baseline U-Net model is then retrained on a combined dataset comprising the original training data (with the selected uncertain images removed) and the newly acquired pairs of uncertain images and their pseudo-labels. This fine-tunes the model, adapting its weights to better understand the nuances of the HOUB data.

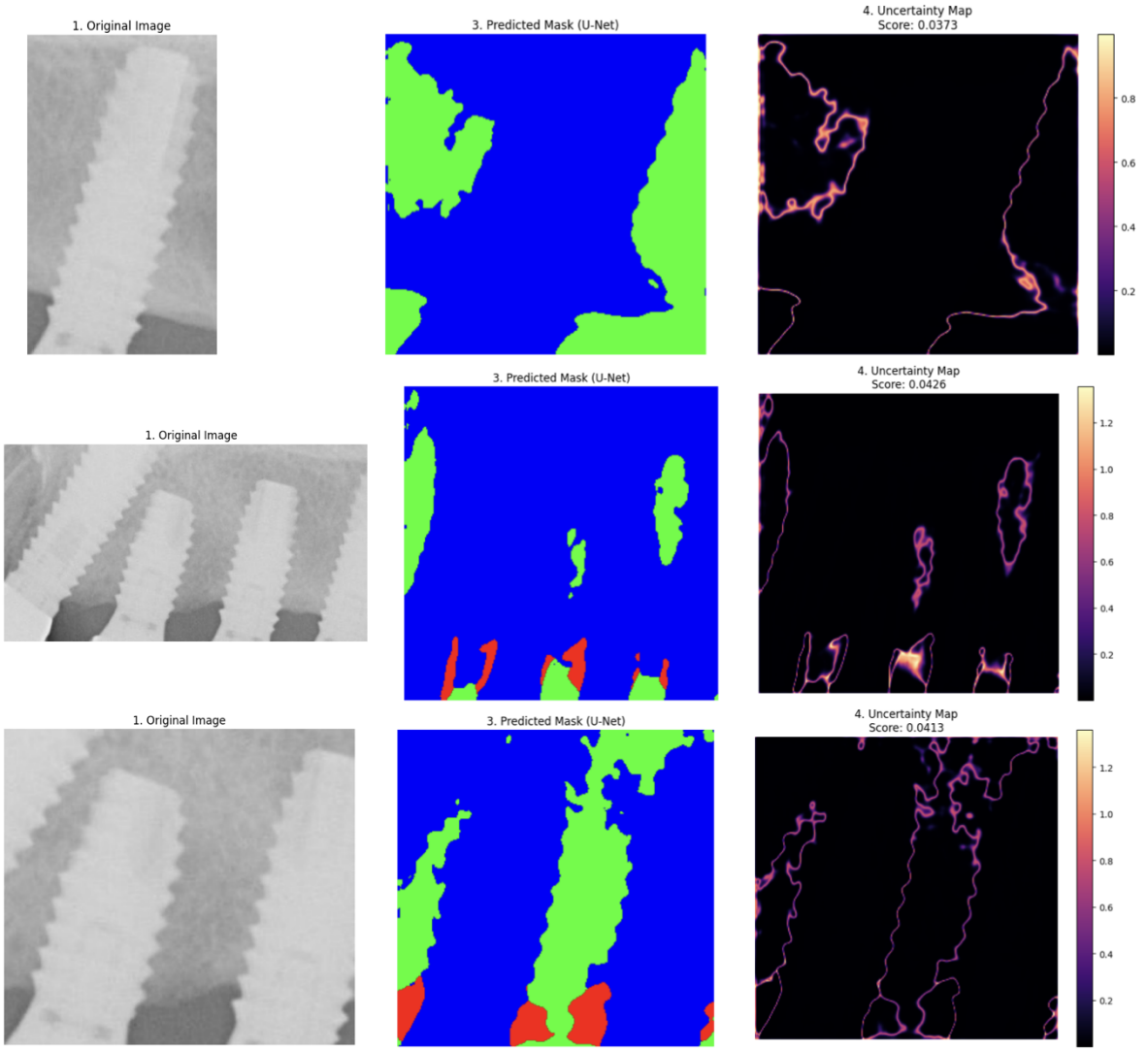


Figure 34: Example of Uncertainty Analysis on HOUB images. The original image (left) is processed by the baseline U-Net. The model's prediction (center) is imperfect, and the uncertainty map (right) highlights areas of low confidence (brighter colors), making this image a candidate for the active learning loop.

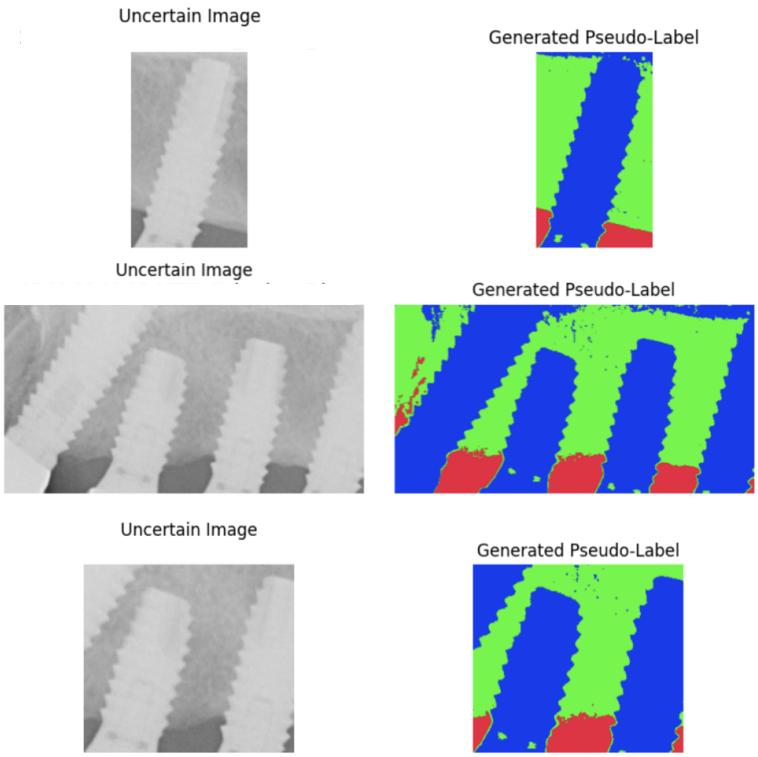


Figure 35: Example of the pseudo-labeling process. The uncertain image selected from the HOUB dataset (left) is annotated using the classical CV pipeline, resulting in a new semantic pseudo-label (right) that will be used for retraining.

Results and Discussion

To evaluate the effectiveness of this strategy, we compared the performance of the original **Baseline Model** against the **Active Learning Model** (after one cycle of fine-tuning). The evaluation was performed on two test sets: the original test set and a hybrid test set composed of the original test images plus a new sample of 10 pseudo-labeled HOUB images. This hybrid set allows us to directly measure the model's generalization capabilities.

The results, summarized in Table 14, are highly promising.

Model	Evaluated on	Test Loss	Test IoU	Test Dice
Baseline Model	Original Test Set	0.0543	0.7919	0.8762
Baseline Model	Hybrid Test Set	0.1614	0.6562	0.7603
Active Learning Model	Original Test Set	0.0561	0.7880	0.8729
Active Learning Model	**Hybrid Test Set**	**0.1411**	**0.6826**	**0.7866**

Table 14: Performance comparison between the Baseline Model and the Fine-tuned Active Learning Model on both the original and a new hybrid test set.

The baseline model's performance drops significantly when evaluated on the hybrid set (Dice score from 0.8762 to 0.7603), confirming its poor generalization to the HOUB data. Conversely, the Active Learning model, while showing a negligible and expected decrease in performance on the original data, demonstrates a marked improvement on the hybrid set, with the Dice score increasing from 0.7603 to **0.7866**.

This demonstrates that even a single iteration of our active learning cycle successfully adapted the model to the new data domain, improving its generalization. Although the results are not yet perfect, they validate the proposed methodology. The classical CV pipeline described in Section 3.4.4 proves to be a powerful tool for bootstrapping the labeling process, opening the door to effectively leveraging large amounts of unlabeled clinical data from HOUB. With more data and further iterations of this cycle, it is highly probable that a robust, high-performance model suitable for clinical use can be developed.

To provide a qualitative illustration of this improvement, Figure 36 presents a direct comparison of both models' predictions on a sample image from the HOUB dataset. The baseline model's prediction is visibly incorrect, struggling to accurately delineate the different tissue types and misclassifying significant portions of the implant region. In contrast, the prediction from the model fine-tuned with our active learning strategy is substantially more accurate. Although not yet perfect, the boundaries of the key structures are more precise and the overall semantic segmentation is more coherent. This visual evidence powerfully reinforces the quantitative results, demonstrating that even a single iteration of the active learning and pseudo-labeling cycle can effectively guide the model toward better generalization on new, unseen data.

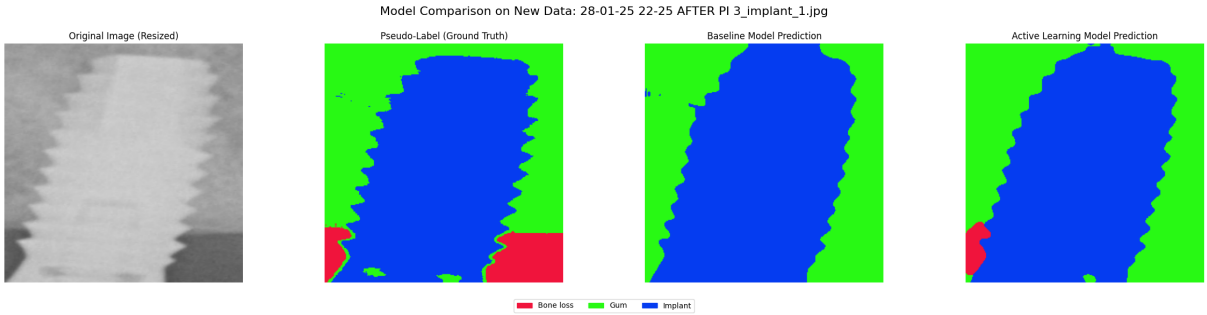


Figure 36: Qualitative comparison on a new HOUB data sample. From left to right: Original Image, the generated Pseudo-Label (serving as ground truth), the Baseline Model's prediction, and the Active Learning Model's improved prediction.

5. Discussion

In this chapter, the results obtained are critically interpreted, they are related to the existing literature and both the limitations and the possibilities of improvement and clinical implications are discussed.

5.1. Interpretation of Findings

This study successfully addressed its primary objectives by developing and validating a multimodal artificial intelligence (AI) system capable of reliably detecting dental implants, segmenting relevant anatomical structures, and quantifying peri-implant bone loss (PBL). The findings provide critical insights into improving diagnostic accuracy and facilitating proactive clinical management of peri-implantitis.

The performance of the Faster R-CNN model for implant detection ($AP_{50} = 0{,}929$) met and exceeded expectations, demonstrating superior accuracy and robustness relative to other detection architectures such as YOLOv8s. This aligns with the literature, notably outperforming prior implementations (Thanathornwong et al., 2020; Liu et al., 2022) and confirming its suitability for detailed quantitative analyses.

The U-Net segmentation model, with a Dice score of 0.8801, showed exceptional accuracy comparable to state-of-the-art methods, such as Chang et al. (2020). Significantly, the integration of the Segment Anything Model (SAM) with classical computer vision techniques offered advanced segmentation capabilities, particularly in isolating the implant's endosseous portion. The novel dual-metric quantification system (vertical depth and area-based metrics), validated through statistical independence ($r = 0{,}078$), provided an enhanced characterization of peri-implant bone loss, directly addressing previously identified limitations in the literature.

The primary objectives of this project have been successfully achieved, resulting in a robust and functional automated pipeline for the analysis of peri-implant bone loss. The

objective of reliably detecting dental implants was met with exceptional performance; the optimized Faster R-CNN model achieved an AP_{50} of 0.929, providing a highly accurate and reproducible foundation for the subsequent stages of analysis.

The segmentation objectives were also met through a multi-pronged approach that yielded significant insights. For end-to-end semantic segmentation, the U-Net architecture proved to be superior in both accuracy (Dice score of 0.8801) and computational efficiency. However, the most notable innovation was the development of the hybrid pipeline. By using the Segment Anything Model (SAM), we achieved a zero-shot segmentation of the implant's endosseous portion, a task of high clinical relevance that standard supervised models are not trained for. This goes beyond the initial objective, providing a more precise anatomical landmark for measurement.

Furthermore, the creation of a dual quantification system—measuring both vertical ($\text{PBL}_{\text{vertical}}$) and area-based ($\text{PBL}_{\text{area}}$) bone loss—and the empirical discovery of their statistical independence ($r = 0{,}078$) is a key finding of this work. This demonstrates that peri-implant bone loss is a morphologically complex phenomenon and validates the need for a multi-metric approach to fully characterize the defect's phenotype, thus enabling a more complete assessment than originally planned.

The results presented in this project are consistent with the state-of-the-art. The detection accuracy of our Faster R-CNN model ($AP_{50} = 0{,}929$) is highly competitive and exceeds that reported in similar studies using the same architecture. This confirms that our implementation is robust and suitable for clinical-grade analysis.

In the domain of quantification, our work aligns with the evolution of the field from simple detection to precise measurement, as exemplified by Chang et al. (2020). While their system achieved a Dice score of 0.88 for bone crest segmentation , our U-Net model obtained a comparable multiclass Dice of 0.8801. Our main contribution, however, lies in the "quantitative and phenotypic richness" that the literature identifies as a strategic challenge. The dual-metric system and the use of SAM for zero-shot segmentation of clinically specific sub-structures provide a more nuanced and detailed analysis than the single-metric approaches previously described.

Finally, the successful application of an active learning cycle using our classical computer vision pipeline as an automated labeling tool to generate pseudo-labels presents a novel and effective methodology to address the common challenge of limited annotated data in the medical field, a contribution not explicitly detailed in the reviewed literature.

5.2. Limitations and risks

Access to Real Patient Data

A primary constraint was the limited access to a large, longitudinal clinical dataset from the beginning of the project. This necessitated the use of public and synthetically

generated data for initial development and for the validation of the prognostic regression model. While the synthetic cohort was designed based on clinical literature, its assumption of linear disease progression and a forced risk-severity correlation may not capture the complexity of real biological processes. This fundamentally limits the direct clinical generalizability of the prognostic findings until they can be validated on real patient data.

Computational Resources

The development and training of deep learning models for medical image analysis demand substantial computational power, which may become a bottleneck when experimenting with complex architectures. Limited access to high-performance GPUs can lead to prolonged training times and reduce the ability to iterate efficiently. To address this, computing platforms such as Google Colab as part of the development environment. During training, resource usage is optimized through careful control of batch size, memory allocation, and code efficiency. Pretrained models are stored systematically after each significant training phase, enabling rapid reloading and testing without repeating the entire training process. This strategy ensures that experimentation remains agile and that resources are used judiciously throughout the project.

Data Privacy and Security in Handling Health Information

The use of real patient radiographs and clinical records raises critical concerns regarding data privacy, ethical compliance, and legal regulations. The handling of health data must align with relevant frameworks such as the General Data Protection Regulation (GDPR) in Europe. Failure to comply may lead to unauthorized access, ethical violations, or even legal repercussions. To mitigate these risks, all datasets are anonymized before access and are processed in accordance with institutional guidelines. During the project, all stored data are encrypted, and cloud storage is avoided unless end-to-end encryption and local control are guaranteed. After data usage, unnecessary datasets are securely deleted, and the remaining files are retained only as strictly required under approved data governance policies. These practices aim to ensure that patient confidentiality is preserved at every stage of the research process.

Version Control and Data Loss Prevention

The iterative nature of AI development, particularly in model training and evaluation, can lead to information loss or inconsistencies if robust version control practices are not followed. Accidental file deletion, hardware failures, or undocumented changes can compromise reproducibility and hinder progress. To avoid such issues, a version control system (e.g., GitHub) is used to track all changes in code, scripts, and data processing pipelines. Regular backups are performed to secure all essential files, including trained

models, datasets, and configuration files. After each experimental iteration, results and metrics are thoroughly documented to preserve the context of findings and enable reproducibility. This systematic approach ensures transparency, reliability, and continuity in the development process.

Clinical Generalizability of the Models

Achieving strong clinical generalization is a core objective in this project, yet it poses significant challenges, particularly when the training data are limited or biased. Models trained on narrow or synthetic datasets may fail to perform adequately on unseen clinical cases. To address this, data augmentation techniques are applied to artificially increase the diversity and size of the training set. The model is validated using external datasets when possible and through expert clinical review to verify its relevance in real-world scenarios. If performance discrepancies are observed, hyperparameters are adjusted and models are retrained based on clinical feedback. This iterative refinement process is essential to ensuring the model's robustness and clinical applicability.

5.3. Proposals for Improvement and Future Developments

Building on the solid foundation established by this work, several strategic directions are proposed to further enhance the system's capabilities and accelerate its translation into clinical practice.

- **Ensemble Learning and Hybrid Models:** To maximize robustness, future work should explore ensemble methods that combine the strengths of different models. For example, a two-tiered detection system could use YOLOv8 for rapid initial screening and trigger the more precise Faster R-CNN for detailed analysis of positive cases. This would balance speed and accuracy effectively.
- **Explainable AI (XAI) for Clinical Trust:** For any AI tool to be adopted in clinical practice, it must be trustworthy and transparent. Integrating XAI techniques such as Grad-CAM or SHAP is essential to illuminate the models' decision-making process. Showing clinicians precisely which radiographic features are influencing a prediction will demystify the "black box" and foster the confidence needed for routine use.
- **Advanced Segmentation with Transformers:** The success of SAM warrants further exploration of advanced transformer-based architectures, such as SAM-HQ. These models could overcome the observed limitations of SAM in low-contrast scenarios, leading to even more accurate and versatile segmentations with fewer failures.

- **Full-Scale Application on HOUB Data via Active Learning:** The immediate priority is the full-scale application of the validated pipeline to the complete longitudinal HOUB dataset once access is granted. The active learning methodology, proven effective in this work, should be used iteratively to fine-tune and create a highly specialized segmentation model optimized for the specific characteristics of the target clinical environment.
- **Longitudinal Validation and Clinical Correlation:** The definitive next step is to validate the entire prognostic pipeline on real, multi-year longitudinal data. This will test the machine learning models' ability to predict complex, non-linear disease progression. Furthermore, the automated metrics ($PBL_{vertical}$ and PBL_{area}) must be correlated with traditional clinical parameters (e.g., probing depth, bleeding on probing) to establish their prognostic value and define clinically meaningful risk thresholds. This will close the loop from technical feature extraction to tangible clinical application.
- **Continuation of Clinical-Technical Partnership:** Continued collaboration with the clinical project leaders is paramount. This ensures that technical developments remain aligned with real-world clinical needs, facilitating an iterative refinement process where clinical feedback directly guides model improvements, ultimately leading to a tool that is not only technologically advanced but also practically useful.

6. Ethical Analysis

The development and implementation of artificial intelligence (AI) systems in healthcare, particularly those focused on analyzing medical images with patient data, entail a series of ethical considerations and inherent risks that must be addressed proactively and rigorously. This section details the main ethical challenges and the best practices implemented to mitigate them, ensuring a responsible approach at every stage of development.

Access and Use of Real Patient Data

A fundamental limitation in the initial phase of this project was restricted access to a large-scale, longitudinal clinical dataset. To overcome this, we opted for the use of public data and, crucially, the generation of a synthetic cohort based on clinical literature. While this approach allowed for the development and validation of the prognostic model in a controlled environment, it is vital to recognize that the direct clinical generalizability of our findings is inherently limited until exhaustive validation with real patient data is performed.

In line with Kate Crawford's warning in *Atlas of AI*[33] about the "false neutrality" of data, we recognize that syntheticity does not exempt us from validating the statistical representation of complex biological processes before any clinical translation. Furthermore, this project is firmly committed to not reusing data beyond the declared clinical objective and to avoiding any commercial exploitation.

Privacy and Security of Health Data

The handling of real patient radiographs and clinical records, even if limited in the current phase of the project, raises critical concerns regarding **data privacy**, ethical compliance, and legal regulations. Non-compliance with regulations such as the General Data Protection Regulation (GDPR) in Europe can lead to unauthorized access, ethical breaches, and severe legal consequences.

To mitigate these risks, we have implemented the following practices:

- **Anonymization**: All real patient datasets have been anonymized before access and processing. This includes the removal of embedded metadata from JPG files using tools like ExifTool and the automatic cropping of any identifying text printed directly on the radiographs.
- **Data Lifecycle Management**: After use, unnecessary datasets are securely deleted, and remaining files are retained only under strict, approved data governance policies.

- **Institutional Collaboration**: Formalized collaboration with the **Hospital Odontològic Universitat de Barcelona (HOUB)**, which includes confidentiality agreements and authorized data access, is fundamental to ensuring compliance with institutional and ethical guidelines.
- **Right not to be profiled**: Following Carissa Véliz in *Privacy is Power*[34], it is guaranteed that model outputs are never linked to patient identity outside the legitimate clinical environment.

Algorithmic Biases and Inequity

AI models, especially those based on deep learning, learn from the data they are trained on. If these data are limited or contain inherent biases from the source population (e.g., demographic representativeness, image quality, variability in acquisition protocols), the model can replicate and amplify these biases in its predictions. This could lead to deficient performance in specific subpopulations or with radiographs acquired under different configurations, compromising equity in diagnostic care.

To address biases and model generalization:

- **Active Learning**: An innovative active learning strategy has been implemented to adapt the model to HOUB data without the need for new manual annotations. The use of our classical pipeline as an "oracle" to generate pseudo-labels has improved the generalization of the model in a new data domain.

Interpretability and AI's "Black Box"

The "black box" nature of some deep learning models, such as convolutional neural networks, can make it difficult to understand how they arrive at a particular prediction. In a clinical context, this lack of transparency can reduce the trust of healthcare professionals and pose challenges in accountability.

To mitigate this risk, we have implemented the following strategies:

- **Hybrid Systems**: The proposal integrates deep learning with **classical computer vision techniques** and mathematical morphology, which are inherently more interpretable and offer computationally efficient solutions. The combination of the SAM generative model with classical techniques for bone line detection provides more granular control and a clearer understanding of the process.
- **Explicit Confidence Metrics**: A confidence metric (c) has been designed that synthesizes the structural robustness of the mask generated by SAM, the effective coverage of the bone line, and the stability of the measurements. This allows the system to be .ªware.ºf the limits of its own reliability, informing clinicians about the uncertainty of a measurement.

- **Visualization of Intermediate Results**: Mosaics and visualizations of the intermediate stages of the pipeline (e.g., Multi-Otsu segmentations, detected bone lines) are generated. This provides visual traceability that can help clinicians understand how the final measurements are derived.

Clinical Responsibility and Human Oversight

Despite automation, the ultimate responsibility for diagnosis and treatment plans lies with the healthcare professional. An AI system, however advanced, must be considered a support tool and not a replacement for clinical expertise.

To address this, it is emphasized that:

- **Continuous Validation**: The need for multicentric and exhaustive validations in diverse clinical scenarios is a key recommendation for establishing the model's applicability limits.
- **Integration and Oversight**: The integration of these AI tools into existing systems (such as PACS) is promoted to ensure that they are used within a workflow that maintains human oversight and validation as an essential component, preserving clinical autonomy.
- **Regulation and Ethics**: It is emphasized that ethical considerations and regulatory oversight are essential to guarantee that such systems complement, rather than replace, clinical expertise.

Ethical Conclusion

This project has addressed ethical considerations from its conception, implementing proactive measures to ensure data privacy, mitigate biases, and foster interpretability. While current development has demonstrated the technical feasibility of an AI tool for peri-implant bone loss assessment, its full clinical implementation will require continuous validation on large-scale real data, close collaboration with the clinical community, and an unwavering commitment to the principles of responsible AI. The ultimate goal is to develop a tool that is not only technologically advanced but also ethically sound and clinically beneficial. Only by integrating these safeguards—and continuously reviewing them—will the tool be able to provide real clinical value without compromising fundamental rights or reproducing surveillance or inequity dynamics in the healthcare domain.

7. Conclusions and Recommendations

This study has successfully developed a functional artificial intelligence (AI) system for assessing peri-implant health, laying a foundation for future computer-assisted diagnostic tools. The main conclusions and recommendations derived from the research are summarized below.

1. **AI-Based Implant Detection is Feasible and Precise.** It has been demonstrated that deep learning models, particularly Faster R-CNN, can reliably automate dental implant localization with high accuracy. This capability is essential for standardizing large-scale radiographic analyses and reducing observer variability, paving the way for automating routine clinical procedures.

2. **Advanced Segmentation Provides Superior Anatomical Characterization.** Accurate delineation of relevant anatomical structures was achieved through two complementary approaches: a U-Net model for high-quality global semantic segmentation and an innovative hybrid approach using the Segment Anything Model (SAM) specifically to isolate the endosseous portion of the implant. This latter approach offers significant clinical advantages by enhancing the precision of bone loss measurements.

3. **Dual Quantification of PBL Offers Comprehensive Insights into Bone Defects.** The capability to automatically quantify bone loss was enhanced by validating a dual-metric system (vertical and area-based measurements). The discovery that these two metrics are statistically independent ($r = 0{,}078$) is clinically relevant, enabling a more comprehensive morphological characterization of defects than single-metric methods, thereby improving monitoring and treatment personalization.

4. **Recommendations for Research and Clinical Practice.** To maximize the impact of this work, the following actions are recommended:

    - Conduct **multicenter clinical validation** to ensure model generalizability across diverse patient populations and different radiographic equipment.
    - Explore deeper **correlations between AI metrics and longitudinal clinical outcomes**, such as implant survival and peri-implantitis progression, to establish their real prognostic value.
    - Promote the **integration of these AI tools into electronic health records (EHRs) and Picture Archiving and Communication Systems (PACS)** to facilitate adoption in daily practice and utilization as clinical decision support systems.

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A. Appendices

Appendix: Example COCO Annotation

Below is a simplified example of a JSON-formatted annotation exported from Label Studio in COCO format:

```
{
  "images": [
    {
      "width": 1432,
      "height": 1276,
      "id": 0,
      "file_name": ".../1st_day.jpg"
    }
  ],
  "categories": [
    { "id": 0, "name": "bone" },
    { "id": 1, "name": "gum" },
    { "id": 2, "name": "implant" }
  ],
  "annotations": [
    {
      "id": 0,
      "image_id": 0,
      "category_id": 0,
      "segmentation": [[1406.67, 588.39, ...]]
    }
  ]
}
```

This format ensures consistency with COCO-compatible training libraries and facilitates future expansion or interoperability with other annotation tools or model architectures.

Appendix: Example YOLO Annotation

Below is a simplified example of a YOLO annotation in text format. Each line corresponds to one object instance, following the format:
`<class_id><x_center><y_center><width><height>`
(All values are normalized to [0,1] with respect to image size.)

```
0 0.534 0.462 0.214 0.376
1 0.648 0.505 0.190 0.402
2 0.742 0.471 0.170 0.338
```

This annotation format is native to YOLO-based models and is suitable for training real-time detection systems such as YOLOv7 or YOLOv8. For compatibility with instance segmentation and other frameworks, these annotations were programmatically converted to the COCO format described in Appendix A.